

REFERENCE NOTE FOR PDE ON A HALF SPACE

Sungbae Jeong

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1 Weighted Sobolev Spaces with Weights

By $\mathcal{D}(\mathbb{R}_+^d)$ we denote the space of all distributions on $\mathbb{R}_+^d$.

Definition 1.0.1 Let us fix $\zeta \in C_0^\infty(\mathbb{R}_+)$ such that for every $x \in \mathbb{R}$,

$$\sum_{n=-\infty}^{\infty} \zeta^p(e^{x-n}) \geq 1. \quad (1)$$

For $\gamma, \theta \in \mathbb{R}$ and $p \in (1, \infty)$ we define a space $H_{p,\theta}^\gamma$ by all distributions u such that

$$\|u\|_{\gamma,p,\theta}^p := \sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p < \infty. \quad (2)$$

This new space is a Banach space, but the proof is slightly computational.

Theorem 1.0.2 The space $H_{p,\theta}^\gamma$ is a Banach space.

proof. Define two maps $S_{p,\theta} : H_{p,\theta}^\gamma \rightarrow l_p^\theta(\mathbb{Z}, H_p^\gamma)$ and $R_{p,\theta} : l_p^\theta(\mathbb{Z}, H_p^\gamma) \rightarrow H_{p,\theta}^\gamma$ by

$$S_{p,\theta} u = (u(e^n \cdot) \zeta)_n, \quad R_{p,\theta} (u_n)_n = \zeta^{-1} \sum_{n=-\infty}^{\infty} u_n(e^{-n} \cdot) \zeta^{p-1}(e^{-n} \cdot),$$

where $\zeta = \sum_n \zeta^p(e^n \cdot) \geq 1$, and $l_p^\theta(\mathbb{Z}, H_p^\gamma)$ is the space of H_p^γ -valued sequences such that

$$\|f\|_{\gamma,p,\theta}^p = \sum_{n=-\infty}^{\infty} e^{n\theta} \|f_n\|_{\gamma,p}^p < \infty.$$

The map $S_{p,\theta}$ is well-defined but we should check that $R_{p,\theta}$ is also well-defined. Fix $(u_n)_n \in l_p^\theta(\mathbb{Z}, H_p^\gamma)$. Since $\zeta \in C_0^\infty(\mathbb{R}_+)$, there exists a positive integer M (observe that this M only depends on ζ) such that for every $n \in \mathbb{Z}$,

$$\sum_{m=-\infty}^{\infty} u_m(e^{n-m} \cdot) \zeta^{p-1}(e^{n-m} \cdot) \zeta = \sum_{m=n-M}^{n+M} u_m(e^{n-m} \cdot) \zeta^{p-1}(e^{n-m} \cdot) \zeta.$$

This gives

$$\begin{aligned} \left\| \zeta^{-1} \sum_{n=-\infty}^{\infty} u_n(e^{-n} \cdot) \zeta^{p-1}(e^{-n} \cdot) \right\|_{\gamma,p,\theta}^p &= N \sum_{n=-\infty}^{\infty} e^{n\theta} \left\| \sum_{m=-\infty}^{\infty} u_m(e^{n-m} \cdot) \zeta^{p-1}(e^{n-m} \cdot) \zeta \right\|_{\gamma,p}^p \\ &= N \sum_{n=-\infty}^{\infty} e^{n\theta} \left\| \sum_{m=n-M}^{n+M} u_m(e^{n-m} \cdot) \zeta^{p-1}(e^{n-m} \cdot) \zeta \right\|_{\gamma,p}^p \\ &= N \sum_{n=-\infty}^{\infty} e^{n\theta} \left\| \sum_{m=-M}^M u_{n+m}(e^{-m} \cdot) \zeta^{p-1}(e^{-m} \cdot) \zeta \right\|_{\gamma,p}^p \\ &\leq N \sum_{m=-M}^M \sum_{n=-\infty}^{\infty} e^{n\theta} \|\zeta\|_{B|\gamma|+\delta}^{p^2} \|u_{m+n}\|_{\gamma,p}^p \end{aligned}$$

$$= N \sum_{m=-M}^M \sum_{n=-\infty}^{\infty} e^{(n-m)\theta} \|u_n\|_{\gamma,p}^p$$

$$< \infty,$$

where $\delta = \delta(\gamma) \in (0, 1)$ is some appropriate constant. Observe that $R_{p,\theta} S_{p,\theta} = \text{id}_{H_{p,\theta}^\gamma}$, and thus $H_{p,\theta}^\gamma$ and $S_{p,\theta}[H_{p,\theta}^\gamma]$ are pseudo-isometric.⁽ⁱ⁾ Since $l_p^\theta(\mathbb{Z}, H_p^\gamma)$ is a reflexive Banach space by Lemma 8.1.1, we need to show that the space $S_{p,\theta}[H_{p,\theta}^\gamma]$ is closed to finish the proof.

Notice that if we define $P_{p,\theta} = S_{p,\theta} R_{p,\theta}$, then

$$P_{p,\theta}^2 = (S_{p,\theta} R_{p,\theta})(S_{p,\theta} R_{p,\theta}) = S_{p,\theta}(R_{p,\theta} S_{p,\theta}) R_{p,\theta} = S_{p,\theta} R_{p,\theta} = P_{p,\theta}.$$

Hence, by the well-known property of a projection map, $P_{p,\theta}[l_p^\theta(\mathbb{Z}, H_p^\gamma)] = \ker(I - P_{p,\theta})$, hence closed because $P_{p,\theta}$ is also bounded. Since $R_{p,\theta}$ is surjective,

$$S_{p,\theta}[H_{p,\theta}^\gamma] = S_{p,\theta}[R_{p,\theta}[l_p^\theta(\mathbb{Z}, H_p^\gamma)]] = P_{p,\theta}[l_p^\theta(\mathbb{Z}, H_p^\gamma)]$$

holds, hence it is closed.

Remark 1.0.3 The space $H_{p,\theta}^\gamma$ is separable. Indeed, the space $l_p^\theta(\mathbb{Z}, H_p^\gamma)$ is separable. Since $H_{p,\theta}^\gamma$ and $S_{p,\theta}[H_{p,\theta}^\gamma]$ are isometric and the latter space is indeed separable because it is closed subspace of a separable space, $H_{p,\theta}^\gamma$ is, therefore, separable.

Remark 1.0.4 Observe that

$$\sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_p^p = \sum_{n=-\infty}^{\infty} e^{n\theta} \int_{\mathbb{R}_+^d} \zeta^p(x) |u|^p(e^n x) dx = \int_{\mathbb{R}_+^d} \eta(x) |u|^p(x) dx,$$

where

$$\eta(x) = \sum_{n=-\infty}^{\infty} e^{n(\theta-d)} \zeta^p(e^{-n} x).$$

Next, the function

$$\xi(t) = \sum_{n=-\infty}^{\infty} e^{(n-t)(\theta-d)} \zeta^p(e^{t-n})$$

is periodic with period 1, bounded, and bounded away from zero. Since $\eta(x) = \xi(\ln x^1)(x^1)^{\theta-d}$, we get that $(x^1)^{\theta-d} \leq N\eta(x) \leq N(x^1)^{\theta-d}$.

In this section, the change of variables in spaces H_p^γ is heavily used. One can see this theorem at section 4.3.2 of [Tri92].

Theorem 1.0.5 Let $k \in \mathbb{Z}_+$, and we say that the 1-1 mapping ψ of $\mathbb{R}^d$ onto itself is a k -diffeomorphism if $\psi \in C^k(\mathbb{R}^d)$ and $|\det \psi_*(x)| \geq c > 0$ for some c and all $x \in \mathbb{R}^d$. Here ψ_* stands for the Jacobian matrix.

Let $s \in \mathbb{R}$, $0 < p < \infty$, and let ψ be a k -diffeomorphism where k is sufficiently large (in dependence of γ, p) then

$$\|f \circ \psi\|_{\gamma,p} \leq N \|f\|_{\gamma,p} \quad \forall f \in H_p^\gamma,$$

where N depends only on d, γ, p , and ψ .

Following lemma shows that norms of $H_{p,\theta}^\gamma$ are equivalent for different ζ .

⁽ⁱ⁾Two Banach spaces X and Y are pseudo-isometric if there exists two constants $c, C > 0$ such that $c\|\cdot\|_X \leq \|\cdot\|_Y \leq C\|\cdot\|_X$.

Lemma 1.0.6 Take ζ from Definition 1.0.1 and let $\eta \in C_0^\infty(\mathbb{R}_+)$. Then for any $u \in \mathcal{D}(\mathbb{R}_+^d)$ and γ, θ, p we have

$$\sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \eta\|_{\gamma,p}^p \leq N \sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p$$

where N depends only on $d, \gamma, \theta, p, \eta$, and ζ .

proof. As η has a compact support, there is a finite integer m such that the function

$$\xi(x) := \eta(x) \left[\sum_{|n| \leq m} \zeta(e^n x) \right]^{-1}$$

belongs to $C_0^\infty(\mathbb{R}_+)$. Also remember that the $C_0^\infty(\mathbb{R}_+)$ -functions are pointwise multipliers in any H_p^γ (see, for instance, Corollary 4.2.2 of [Tri92], or Lemma 5.2 of [Kry99a]) and that by Theorem 1.0.5, we have

$$\begin{aligned} \|u(e^n \cdot) \eta\|_{\gamma,p}^p &= \left\| u(e^n \cdot) \xi \sum_{|k| \leq m} \zeta(e^k \cdot) \right\|_{\gamma,p}^p \\ &\leq N \left\| u(e^n \cdot) \sum_{|k| \leq m} \zeta(e^k \cdot) \right\|_{\gamma,p}^p \\ &\leq N \sum_{|k| \leq m} \|u(e^n \cdot) \zeta(e^k \cdot)\|_{\gamma,p}^p \\ &\leq N \sum_{|k| \leq m} \|u(e^{n-k} \cdot) \zeta\|_{\gamma,p}^p. \end{aligned}$$

Hence,

$$\begin{aligned} \sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \eta\|_{\gamma,p}^p &\leq N \sum_{n=-\infty}^{\infty} \sum_{|k| \leq m} e^{n\theta} \|u(e^{n-k} \cdot) \zeta\|_{\gamma,p}^p \\ &= N \sum_{|k| \leq m} e^{k\theta} \sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p \\ &= N \sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p. \end{aligned}$$

This proves the lemma.

Remark 1.0.7 The norm in $H_{p,\theta}^\gamma$ depends on ζ , so that, constants entering various estimates of $H_{p,\theta}^\gamma$ -norms also depends on ζ . However, we will not mention the dependence on ζ for the sake of brevity and because ζ has been fixed.

Corollary 1.0.8 Let $\alpha \in \mathbb{R}$. We have $u \in H_{p,\theta}^\gamma$ if and only if $u = M^\alpha v$ with $v \in H_{p,\theta+\alpha p}^\gamma$. Hence,

$$M^\alpha H_{p,\theta+\alpha p}^\gamma = H_{p,\theta}^\gamma.$$

In addition,

$$\|u\|_{\gamma,p,\theta} \leq N \|M^{-\alpha} u\|_{\gamma,p,\theta+\alpha p} \leq N \|u\|_{\gamma,p,\theta},$$

where N are independent of u .

proof. Notice that $M^\alpha \zeta \in C_0^\infty(\mathbb{R}_+)$. Thus by Lemma 1.0.6,

$$\begin{aligned}
\|u\|_{\gamma,p,\theta} &= \sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p \\
&= \sum_{n=-\infty}^{\infty} e^{n\theta} \|(M^\alpha v)(e^n \cdot) \zeta\|_{\gamma,p}^p \\
&= \sum_{n=-\infty}^{\infty} e^{n(\theta+\alpha p)} \|v(e^n \cdot) M^\alpha \zeta\|_{\gamma,p}^p \\
&\leq N \sum_{n=-\infty}^{\infty} e^{n(\theta+\alpha p)} \|v(e^n \cdot) \zeta\|_{\gamma,p}^p \\
&= N \|v\|_{\gamma,p,\theta+\alpha p}^p.
\end{aligned}$$

Similarly one can prove the other side. This time, remember that $M^{-\alpha} \zeta \in C_0^\infty(\mathbb{R}_+)$.

Corollary 1.0.9 For any $a \in [1, e]$,

$$\|u(a \cdot)\|_{\gamma,p,\theta} \leq N(\gamma, p, d) \|u\|_{\gamma,p,\theta} \leq N(\gamma, p, d) \|u(a \cdot)\|_{\gamma,p,\theta}.$$

proof. By Theorem 1.0.5, for any $a \in [1, e]$,

$$\|u(e^n a \cdot) \zeta\|_{\gamma,p} \leq N \|u(e^n \cdot) \zeta(a^{-1} \cdot)\|_{\gamma,p} \leq N \|u(e^n \cdot) \zeta\|_{\gamma,p},$$

where N depends only on d, γ , and p . Now applying Lemma 1.0.6, we obtain the main result of this corollary.

Corollary 1.0.10 For any $a > 0$,

$$\|u(a \cdot)\|_{\gamma,p,\theta}^p \leq a^{-\theta} N_1(\gamma, p, d) \|u\|_{\gamma,p,\theta}^p \leq N_2(\gamma, p, d) \|u\|_{\gamma,p,\theta}^p. \quad (3)$$

proof. Obviously,

$$\|u(e^k \cdot)\|_{\gamma,p,\theta}^p = \sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^{n+k} \cdot) \zeta\|_{\gamma,p}^p = \sum_{n=-\infty}^{\infty} e^{(n-k)\theta} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p = e^{-k\theta} \|u\|_{\gamma,p,\theta}^p.$$

Using this and Corollary 1.0.9 and representing $a = e^k b$, where $b \in [1, e]$, we easily get the first inequality. The second one is obtained by substituting $u(x) = v(a^{-1}x)$ into the first one.

The core inequality of Lemma 1.0.11 is the following. For any $\gamma \in \mathbb{R}$ and $p \in (1, \infty)$, there exist $\kappa > 0$ such that, for any invertible constant $d \times d$ matrix σ and $u \in H_p^\gamma$,

$$\|u(\sigma \cdot)\|_{\gamma,p} \leq N(d, \gamma, p)(\lambda^{-\kappa}(\sigma) + \Lambda^\kappa(\sigma)) \|u\|_{\gamma,p}, \quad (4)$$

where $\lambda(\sigma)$ and $\Lambda(\sigma)$ are the smallest and the largest eigenvalues of $\sigma \sigma^*$.

However, the inequality does not follow from subsection 4.3.2 of [Tri92] where [Kry99b] said. The proof of eq. (4) is in the appendix.

Lemma 1.0.11 Let σ be a nondegenerate $d \times d$ -matrix such that $\sigma : \mathbb{R}_+^d \rightarrow \mathbb{R}_+^d$. Denote by $\lambda(\sigma)$ and $\Lambda(\sigma)$ the smallest and largest eigenvalues of $\sigma \sigma^*$. Then for any γ, p, θ there exists $\kappa > 0$ such that for any $u \in H_{p,\theta}^\gamma$ we have

$$\|u(\sigma \cdot)\|_{\gamma,p,\theta} \leq N(\lambda^{-\kappa}(\sigma) + \Lambda^\kappa(\sigma)) \|u\|_{\gamma,p,\theta}, \quad (5)$$

where the constant N is independent of σ and u .

proof. We use eq. (4) which is proved in the appendix. In particular,

$$\|u(e^n \sigma \cdot) \zeta\|_{\gamma,p} \leq N(\lambda^{-\kappa}(\sigma) + \Lambda^\kappa(\sigma)) \|u(e^n \cdot) \zeta(\sigma^{-1} \cdot)\|_{\gamma,p}. \quad (6)$$

Also notice that if $\sigma : \mathbb{R}_+^d \rightarrow \mathbb{R}_+^d$, then $(\sigma x)^1 = ax^1$, where $a := \sigma^{11} > 0$, so that $\zeta(\sigma^{-1}x) = \zeta(a^{-1}x^1)$. Applying eq. (4) to $u(e^n ax) \zeta(x)$ with σ defined as multiplying a^{-1} , we get

$$\|u(e^n \cdot) \zeta(\sigma^{-1} \cdot)\|_{\gamma,p} \leq N(a^\kappa + a^{-\kappa}) \|u(e^n a \cdot) \zeta\|_{\gamma,p}.$$

Bearing in mind the possibility of multiplying $e^{n\theta}$ and summation on n and also using Corollary 1.0.10, the proof is finished.

To prove one more unusual property of spaces $H_{p,\theta}^\gamma$ define an operator

$$A : x \mapsto Ax = (1/x^1, x'/(x^1)^2), \quad A : u \mapsto Au(x) = u(Ax).$$

Theorem 1.0.12 We have $u \in H_{p,\theta}^\gamma$ if and only if $Au \in H_{p,-\theta}^\gamma$. Moreover, the operator $A : H_{p,\theta}^\gamma \rightarrow H_{p,-\theta}^\gamma$ and its inverse are bounded.

proof. As easy to see $A = A^{-1}$ and $A(cx) = c^{-1}Ax$ for any $c > 0$. Furthermore, as

$$\frac{\partial Ax}{\partial x} = \begin{pmatrix} -\frac{1}{(x^1)^2} & 0 & 0 & \cdots & 0 \\ -2\frac{x^1}{(x^1)^3} & \frac{1}{(x^1)^2} & 0 & \cdots & 0 \\ -2\frac{x^2}{(x^1)^3} & 0 & \frac{1}{(x^1)^2} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -2\frac{x^d}{(x^1)^3} & 0 & 0 & \cdots & \frac{1}{(x^1)^2} \end{pmatrix},$$

$\det(DAx) = -(x^1)^{-2d}$, which is bounded and bounded away from zero on the support of ζ . Therefore, again by the above mentioned results from [Tri92], (and remember $A = A^{-1}$),

$$\|(Au)(e^n \cdot) \zeta\|_{\gamma,p} = \|u(e^{-n} A \cdot) \zeta\|_{\gamma,p} \leq N \|u(e^{-n} \cdot) \zeta(A \cdot)\|_{\gamma,p}.$$

Upon multiplying this by $e^{n\theta}$, summing up, and using Lemma 1.0.6, we get that A is bounded operator from $H_{p,\theta}^\gamma$ to $H_{p,-\theta}^\gamma$. The same conclusion is true for $A^{-1}(=A)$. The theorem is proved.

Theorem 1.0.13 If $\gamma \leq \alpha \leq \beta$ and $\gamma \neq \beta$, then

$$\|u\|_{\alpha,p,\theta} \leq N(\alpha, \beta, \gamma) \|u\|_{\beta,p,\theta}^{(\alpha-\gamma)/(\beta-\gamma)} \|u\|_{\gamma,p,\theta}^{(\beta-\alpha)/(\beta-\gamma)}.$$

proof. The spaces $H_{p,\theta}^\gamma$ enjoy the same interpolation properties as H_p^γ . By Theorem 14.2 of [KZiPSi76] for $\gamma \leq \alpha \leq \beta$, $\gamma \neq \beta$, we have

$$\|u\|_{\alpha,p} \leq N(\alpha, \beta, \gamma) \|u\|_{\beta,p}^{(\alpha-\gamma)/(\beta-\gamma)} \|u\|_{\gamma,p}^{(\beta-\alpha)/(\beta-\gamma)}.$$

Put $\phi = (\alpha - \gamma)/(\beta - \gamma)$ and $\psi = (\beta - \alpha)/(\beta - \gamma)$. By the Hölder's inequality,

$$\begin{aligned} \|u\|_{\alpha,p,\theta} &= \left(\sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\alpha,p}^p \right)^{1/p} \\ &\leq N(\alpha, \beta, \gamma) \left(\sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\beta,p}^{p\phi} \|u(e^n \cdot) \zeta\|_{\gamma,p}^{p\psi} \right)^{1/p} \\ &= N(\alpha, \beta, \gamma) \left(\sum_{n=-\infty}^{\infty} (e^{n\theta} \|u(e^n \cdot) \zeta\|_{\beta,p}^p)^\phi (e^{n\theta} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p)^\psi \right)^{1/p} \\ &\leq N(\alpha, \beta, \gamma) \left(\sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\beta,p}^p \right)^{\phi/p} \left(\sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p \right)^{\psi/p} \\ &= N(\alpha, \beta, \gamma) \|u\|_{\beta,p,\theta}^\phi \|u\|_{\gamma,p,\theta}^\psi. \end{aligned}$$

Remark 1.0.14 We will also use to a great extent other properties of spaces H_p^γ . In this connection notice that if a distribution u is supported in the strip $a < x^1 < b$ for some $0 < a < b$, then $u \in H_{p,\theta}^\gamma$ if and only if $u \in H_p^\gamma$. Also in this case $\|u\|_{\gamma,p} \leq N\|u\|_{\gamma,p,\theta} \leq N\|u\|_{\gamma,p}$, where the constant N depends only on p, d, γ, θ, a , and b .

Indeed, for distributions supported in the strip $a < x^1 < b$, the number of nonzero terms on the right-hand side of eq. (2) is finite and can be estimated by a constant N depending only on a, b , and ζ . Therefore, our assertion follows from Section 4.3.2 in [Tri92] dealing with the change of variables, Corollary 4.2.2 in [Tri92] on pointwise multipliers, and from [Kry94].

Remark 1.0.15 Speaking about elements of H_p^γ supported in the strip $a < x^1 < b$ let us also mention a well-known fact which will be later. It turns out that for such functions u we have $\|u\|_{\gamma,p} \leq N\|u_x\|_{\gamma-1,p}$, where N depends only on d, γ, p, a , and b .

Indeed, it suffices to assume that $u \in C_0^\infty(\mathbb{R}_+^d)$ because of the density argument. If $\eta \in C_0^\infty(\mathbb{R})$ is such that $\eta = 1$ on (a, b) and η vanishes outside an interval (s, r) , then

$$u(x) = -\eta(x) \int_0^{r-s} u_{x^1}(x^1 + t, x') dt.$$

Hence,

$$\|u\|_{\gamma-1,p} \leq N \left\| \int_0^{r-s} u_{x^1}(\cdot + t, \cdot) dt \right\|_{\gamma-1,p} \leq N \int_0^{r-s} \|u_{x^1}(\cdot + t, \cdot)\|_{\gamma-1,p} dt \leq N\|u_{x^1}\|_{\gamma-1,p}.$$

It only remains to notice that $\|u\|_{\gamma,p} \leq N\|u_x\|_{\gamma-1,p} + N\|u\|_{\gamma-1,p}$.

The following theorem deals with the interpolation with respect to θ . It turns out to be very important in applications to SPDEs.

Theorem 1.0.16 Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a measure space, $\theta_1 < \theta_2$, and X, Y be separable Banach spaces. Define $\mathbb{L}_{p,\theta}(X)$ as the space of all X -valued functions $f(\omega, x)$ on $\Omega \times \mathbb{R}_+^d$ measurable with respect to $\mathcal{F} \otimes \mathfrak{B}(\mathbb{R}_+^d)$, and such that

$$\|f\|_{\mathbb{L}_{p,\theta}(X)}^p := \int_{\Omega \times \mathbb{R}_+^d} (x^1)^{\theta-d} |f(\omega, x)|_X^p \mathbb{P}(d\omega) dx = \mathbb{E} \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |f(x)|_X^p dx < \infty.$$

Let T be a linear operator with values on $\mathbb{L}_{p,\theta_1}(Y) \cap \mathbb{L}_{p,\theta_2}(Y)$ defined on all finite linear combinations of the set

$$\mathfrak{A} := \{\zeta(e^n x) f(\omega, x) : n \in \mathbb{Z}, f \in \mathbb{L}_{p,d}(X)\}.$$

Moreover, assume that, for any $\zeta(e^n \cdot) f \in \mathfrak{A}$ and $i = 1, 2$, we have

$$\|T(\zeta(e^n \cdot) f)\|_{\mathbb{L}_{p,\theta_i}(Y)}^p \leq N_i^p e^{-n(\theta_i-d)} \|f\|_{\mathbb{L}_{p,d}(X)}^p, \quad (7)$$

where N_i are some constants independent of f and n . Then for any $\theta \in (\theta_1, \theta_2)$ the operator T is extendible to a bounded linear operator acting from $\mathbb{L}_{p,\theta}(X)$ to $\mathbb{L}_{p,\theta}(Y)$ and if we keep the same notation T for the extension, then for any $f \in \mathbb{L}_{p,\theta}(X)$

$$\|Tf\|_{\mathbb{L}_{p,\theta}(Y)} \leq NN_1^\alpha N_2^\beta \|f\|_{\mathbb{L}_{p,\theta}(X)}, \quad (8)$$

where $\alpha := (\theta_2 - \theta)/(\theta_2 - \theta_1)$, $\beta = 1 - \alpha$, and N is independent of f and N_i .

proof. First, for any $f \in \mathbb{L}_{p,\theta}(X)$ and n , $\zeta(e^n \cdot) f \in \mathfrak{A}$. Indeed, take $\eta \in C_0^\infty(\mathbb{R}_+)$ such that $\eta = 1$ on the support of $\zeta(e^n \cdot)$. Then we can easily checked that $f\eta \in \mathbb{L}_{p,d}(X)$ because

$$\mathbb{E} \int_{\mathbb{R}_+^d} |\eta f(x)|_X^p dx = \mathbb{E} \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |f(x)|_X^p (M^{d-\theta} \eta^p)(x) dx \leq N \mathbb{E} \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |f(x)|_X^p dx.$$

Thus we obtain $\zeta(e^n \cdot) f = \zeta(e^n \cdot) f\eta \in \mathfrak{A}$, and that $T(\zeta(e^n \cdot) f)$ belongs to $\mathbb{L}_{p,\theta_1}(Y) \cap \mathbb{L}_{p,\theta_2}(Y)$. The inequality $(x^1)^{\theta-d} \leq (x^1)^{\theta_1-d} + (x^1)^{\theta_2-d}$ implies $T(\zeta(e^n \cdot) f) \in \mathbb{L}_{p,\theta}(Y)$.⁽ⁱⁱ⁾ Furthermore, if we define $\xi = \sum_n \zeta(e^n \cdot)$, for

⁽ⁱⁱ⁾ $\|f\|_{\mathbb{L}_{p,\theta}(Y)}^p \leq \|f\|_{\mathbb{L}_{p,\theta_1}(Y)}^p + \|f\|_{\mathbb{L}_{p,\theta_2}(Y)}^p$.

any $g \in \mathbb{L}_{p,\theta}(X)$ can be represented as ξf with comparable $\mathbb{L}_{p,\theta}(X)$ -norms of g and f . Indeed, first of all, $1 \leq \sum_n \zeta^p(e^n \cdot) \leq \|\zeta\|_{L_\infty}^{p-1} \sum_n \zeta(e^n \cdot)$ implies $f := \xi^{-1}g$ is well-defined and $\|f\|_{\mathbb{L}_{p,\theta}(X)} \leq N \|g\|_{\mathbb{L}_{p,\theta}(X)}$. Now consider an interval $(a, b) \subset \mathbb{R}_+$ which contains the support of ζ . Then one can easily obtain that for every $x \in \mathbb{R}_+^d$,

$$\#\{n : x^1 \in (e^{-n}a, e^{-n}b)\} \leq \log b - \log a + 3.$$

This implies $\xi(x) \leq (\log b - \log a + 3) \|\zeta\|_{L_\infty}$ for every $x \in \mathbb{R}_+$, hence, $\|g\|_{\mathbb{L}_{p,\theta}(X)} \leq N \|f\|_{\mathbb{L}_{p,\theta}(X)}$ because ξ is bounded.

Above observations gives that, to prove the theorem, it suffices to prove that for any integer $r > 1$ and $f \in \mathbb{L}_{p,\theta}(X)$, we have

$$\left\| T \left(\sum_{n=-r}^r \zeta(e^n \cdot) f \right) \right\|_{\mathbb{L}_{p,\theta}(Y)}^p \leq N N_1^{\alpha p} N_2^{\beta p} \|f\|_{\mathbb{L}_{p,\theta}(X)}^p, \quad (9)$$

where N is independent of f , r , and N_i .

Denote $u_n = T(\zeta(e^n \cdot) f)$ and observe that by definition the left-hand side of eq. (9) equals

$$\begin{aligned} \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} \left| \sum_{|n| \leq r} u_n(x) \right|_{L_p(\Omega, Y)}^p dx &\leq 2^p \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} \left| \sum_{|n| \leq r} \mathbb{1}_{x^1 e^n \leq s} u_n(x) \right|_{L_p(\Omega, Y)}^p dx \\ &\quad + 2^p \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} \left| \sum_{|n| \leq r} \mathbb{1}_{x^1 e^n \geq s} u_n(x) \right|_{L_p(\Omega, Y)}^p dx =: 2^p I_1 + 2^p I_2, \end{aligned}$$

where s is a constant to be specified later.

To estimate I_1 , define $\delta = (\theta - \theta_1)/p$, $q = p/(p-1)$ and use Hölder's inequality to find

$$\begin{aligned} \left| \sum_{n=-r}^r \mathbb{1}_{x^1 e^n \leq s} u_n \right|_Y^p &\leq \left(\sum_{n=-r}^r |\mathbb{1}_{x^1 e^n \leq s} u_n|_Y \right)^p \\ &= \left(\sum_{n=-r}^r |e^{n\delta} e^{-n\delta} \mathbb{1}_{x^1 e^n \leq s} u_n|_Y \right)^p \\ &\leq \left(\sum_{n=-r}^r |e^{n\delta} \mathbb{1}_{x^1 e^n \leq s}|_Y^q \right)^{p-1} \left(\sum_{n=-r}^r |e^{-n\delta} \mathbb{1}_{x^1 e^n \leq s} u_n|_Y^p \right) \\ &= \left(\sum_{n=-r}^r e^{nq\delta} \mathbb{1}_{x^1 e^n \leq s} \right)^{p-1} \sum_{n=-r}^r e^{-np\delta} \mathbb{1}_{x^1 e^n \leq s} |u_n|_Y^p. \end{aligned}$$

As easy to see, the first factor on the right is less than $N(x^1/s)^{-\delta q(p-1)} = N s^{\theta-\theta_1} (x^1)^{\theta_1-\theta}$. This and assumption eq. (7) imply that

$$I_1 \leq N s^{\theta-\theta_1} \sum_n e^{-np\delta} \|\mathbb{1}_{e^n \cdot \leq s} T(\zeta(e^n \cdot) f)\|_{\mathbb{L}_{p,\theta_1}(Y)}^p \leq N N_1^p s^{\theta-\theta_1} \sum_n e^{-np\delta-n(\theta_1-d)} \|\eta(e^n \cdot) f\|_{\mathbb{L}_{p,d}(X)}^p, \quad (10)$$

where η is any function from $C_0^\infty(\mathbb{R}_+)$ such that $\eta = 1$ on the support of ζ . Upon noticing that $\delta p + \theta_1 - d = \theta - d$ and (cf. Remark 1.0.4)

$$\sum_n e^{-n(\theta-d)} \eta(e^n x) \leq N (x^1)^{\theta-d}, \quad (11)$$

we obtain

$$I_1 \leq N N_1^p s^{\theta-\theta_1} \|f\|_{\mathbb{L}_{p,\theta}(X)}^p.$$

Similarly, for $\epsilon = (\theta_2 - \theta)/p$ we get

$$\left| \sum_{n=-r}^r \mathbb{1}_{x^1 e^n \geq s} u_n \right|_Y^p \leq \left(\sum_{n=-r}^r e^{-nq\epsilon} \mathbb{1}_{x^1 e^n \geq s} \right)^{p-1} \sum_{n=-r}^r e^{np\epsilon} \mathbb{1}_{x^1 e^n \geq s} |u_n|_Y^p,$$

where the first factor on the right is less than $N(x^1/s)^{\epsilon q(p-1)}$, which equals $N s^{\theta-\theta_2} (x^1)^{\theta_2-\theta}$. This implies

$$I_2 \leq N s^{\theta-\theta_2} \sum_n e^{np\epsilon} \|\mathbb{1}_{e^n \cdot \geq s} T(\zeta(e^n \cdot) f)\|_{\mathbb{L}_{p,\theta_2}(Y)}^p \leq N N_2^p s^{\theta-\theta_2} \sum_n e^{-np\epsilon - n(\theta_2-d)} \|\eta(e^n \cdot) f\|_{\mathbb{L}_{p,d}(X)}^p. \quad (12)$$

Since $np\epsilon - n(\theta_2 - d) = -n(\theta - d)$ and we have eq. (11),

$$I_2 \leq N N_2^p s^{\theta-\theta_2} \|f\|_{\mathbb{L}_{p,\theta}(X)}^p.$$

By combining the above estimates we see that the left-hand side of eq. (9) is less than

$$N(N_1^p s^{\theta-\theta_1} + N_2^p s^{\theta-\theta_2}) \|f\|_{\mathbb{L}_{p,\theta}(X)}^p.$$

It only remains to take $s = (N_2 N_1^{-1})^{p/(\theta_2-\theta_1)}$. The theorem is proved.

Remark 1.0.17 Condition eq. (7) is satisfied if, for instance, T is a bounded operator from $\mathbb{L}_{p,\theta_i}(X)$ to $\mathbb{L}_{p,\theta_i}(Y)$ for $i = 1, 2$.

Indeed, $\zeta(e^n x) \leq N(e^n x^1)^{(d-\theta_i)/p}$ and

$$\|\zeta(e^n \cdot) f\|_{\mathbb{L}_{p,\theta_i}(X)}^p \leq N e^{n(d-\theta_i)} \|f\|_{\mathbb{L}_{p,d}(X)}^p.$$

Remark 1.0.18 It follows from eq. (10) and eq. (12) with $s = 1$, that if instead of eq. (7) we have, for all n and $\zeta(e^n \cdot) f \in \mathfrak{A}$,

$$\begin{aligned} \|\mathbb{1}_{e^n \cdot \leq 1} T(\zeta(e^n \cdot) f)\|_{\mathbb{L}_{p,\theta_1}(Y)}^p &\leq N_0^p e^{-n(\theta_1-d)} \|f\|_{\mathbb{L}_{p,d}(X)}^p, \\ \|\mathbb{1}_{e^n \cdot \geq 1} T(\zeta(e^n \cdot) f)\|_{\mathbb{L}_{p,\theta_2}(Y)}^p &\leq N_0^p e^{-n(\theta_2-d)} \|f\|_{\mathbb{L}_{p,d}(X)}^p, \end{aligned} \quad (13)$$

where N_0 is independent of n and f , then instead if eq. (8) we get

$$\|Tf\|_{\mathbb{L}_{p,\theta}(Y)} \leq N N_0 \|f\|_{\mathbb{L}_{p,\theta}(X)}, \quad (14)$$

where N is independent of f and N_0 .

Remark 1.0.19 It turns out that eq. (14) holds under slightly less restrictive assumption than eq. (13). We mean

$$\begin{aligned} \|\zeta(e^m \cdot) T(\zeta(e^n \cdot) f)\|_{\mathbb{L}_{p,d}(Y)}^p &\leq N_0^p e^{(m-n)(\theta_1-d)} \|f\|_{\mathbb{L}_{p,d}(X)}^p \quad \text{if } m \geq n, \\ \|\zeta(e^m \cdot) T(\zeta(e^n \cdot) f)\|_{\mathbb{L}_{p,d}(Y)}^p &\leq N_0^p e^{(m-n)(\theta_2-d)} \|f\|_{\mathbb{L}_{p,d}(X)}^p, \quad \text{if } m \leq n. \end{aligned} \quad (15)$$

Indeed, first assume that eq. (15) holds.

[Fill here!!]

Corollary 1.0.20 Let γ be a nonnegative integer, $\theta_1 < \theta_2$, and let T be a bounded operator from L_{p,θ_i} to H_{p,θ_i}^γ for $i = 1, 2$. Then T is a bounded operator from $L_{p,\theta}$ to $H_{p,\theta}^\gamma$ for any $\theta \in [\theta_1, \theta_2]$.

Theorem 1.0.21 The space $C_0^\infty(\mathbb{R}_+^d)$ is dense in $H_{p,\theta}^\gamma$.

proof. The idea of the proof of this theorem is the following: First, approximate $u \in H_{p,\theta}^\gamma$ by an element $v \in V_{p,\theta}^\gamma$, a collection of all $H_{p,\theta}^\gamma$ which vanishing for x^1 outside of a closed bounded subset of $\mathbb{R}_+$. By Remark 1.0.14, $H_{p,\theta}^\gamma$ -norm of v and H_p^γ -norm of which are comparable, so approximate v with elements of $C_0^\infty(\mathbb{R}_+^d)$ on H_p^γ .

Fix $\epsilon > 0$ and $v \in V_{p,\theta}^\gamma$. By Remark 1.0.14, $v \in H_p^\gamma$. First of all, take $\xi \in C_0^\infty(\mathbb{R}^d)$ such that $\|\xi v - v\|_{\gamma,p} < \epsilon$. As $C_0^\infty(\mathbb{R}^d)$ is dense in H_p^γ , take $\phi \in C_0^\infty(\mathbb{R}^d)$ such that $\|v - \phi\|_{\gamma,p} < \epsilon$. Take any $\xi \in C_0^\infty(\mathbb{R}^d)$ such that, $0 \leq \xi \leq 1$,

$\xi = 1$ if $|x| \leq 1$, and define $\xi_R(x) = \xi(x/R)$ for $R \geq 1$. Then one can take R large enough so that $\xi_R = 1$ on the support of ϕ . Then

$$\begin{aligned} \|\xi_R v - v\|_{\gamma,p} &= \|\xi_R v - v - \phi + \phi\|_{\gamma,p} \\ &= \|\xi_R v - v - \xi_R \phi + \phi\|_{\gamma,p} \\ &\leq \|\xi_R(v - \phi)\|_{\gamma,p} + \|v - \phi\|_{\gamma,p} \\ &\leq \|\xi_R\|_{B^{|\gamma|+\alpha}} \|v - \phi\|_{\gamma,p} \\ &\leq \|\xi_R\|_{B^{|\gamma|+\alpha}} \epsilon \\ &\leq N \|\xi\|_{B^{|\gamma|+\alpha}} \epsilon, \end{aligned}$$

where $\alpha \in [0, 1)$ such that $\alpha = 0$ if γ is an integer; otherwise $\alpha > 0$ and such that $|\gamma| + \alpha$ is not an integer. The space $B^{|\gamma|+\alpha}$ is defined in [Kry99a]. In addition, the constant N is independent of R . This is because we chose $R \geq 1$ and by the definition of the space $B^{|\gamma|+\alpha}$.

Observe that the support of $\xi_R v$ lie in $\mathbb{R}_+^d$. Therefore, by the mollifying argument gives a smooth function with compact support on $\mathbb{R}_+^d$ which is closed enough to $\xi_R v$ as we want in H_p^γ -norm sense. This proves the first claim.

Now take a function $\eta \in C_0^\infty(\mathbb{R})$ such that $\eta(r) = 1$ for $r \notin (-2, 2)$ and $\eta(r) = 0$ for $|r| \leq 1$. Then for any $k \geq 1$ the function $\eta_k(x) := \eta(k^{-1} \log x^1)$ satisfies $\eta_k(x^1) = 0$ for $x^1 \in (e^{-k}, e^k)$ and $\eta_k(x^1) = 1$ if $x^1 \in (0, e^{-2k})$ or $x^1 \geq e^{2k}$. Furthermore, take a function $u \in H_{p,\theta}^\gamma$ and write

$$\|\eta_k u\|_{\gamma,p,\theta}^p = \sum_{n=-\infty}^{\infty} e^{n\theta} \|\eta_k(e^n \cdot) u(e^n \cdot) \zeta\|_{\gamma,p}^p.$$

For any given n the function $\eta_k(e^n x)$ vanishes on the support of ζ if k is large enough. Therefore, each term in the above sum tends to zero as $k \rightarrow \infty$. In addition, since every derivative of $\eta_k(e^n x) = \eta(nk^{-1} + k^{-1} \log x^1)$ is bounded on the support of ζ uniformly with respect to $k \geq 1$ and n , we have

$$\|\eta_k(e^n \cdot) u(e^n \cdot) \zeta\|_{\gamma,p}^p \leq N \|u(e^n \cdot) \zeta\|_{\gamma,p}^p,$$

where N is independent of k, n (and u). Hence by DCT on summation, $\|\eta_k u\|_{\gamma,p,\theta} \rightarrow 0$ as $k \rightarrow \infty$, so that $\|(1 - \eta_k)u - u\|_{\gamma,p,\theta} \rightarrow 0$. Here, $(1 - \eta_k)u \in V_{p,\theta}^\gamma$, thus the theorem is proved.

Corollary 1.0.22 Define $\zeta_k = 1 - \eta_k$, then $\zeta_k(x^1)$ vanishes near zero and infinity and for any $u \in H_{p,\theta}^\gamma$

$$\|\zeta_k u\|_{\gamma,p,\theta} \leq \|u\|_{\gamma,p,\theta}, \quad \|\zeta_k u - u\|_{\gamma,p,\theta} \rightarrow 0$$

as $k \rightarrow \infty$, where N is independent of k and u .

Remark 1.0.23 Define $\zeta_k = 1 - \eta_k$, then $\zeta_k(x^1)$ vanishes near zero and infinity and for any $u \in H_{p,\theta}^\gamma$

$$\|\zeta_k u\|_{\gamma,p,\theta} \leq N \|u\|_{\gamma,p,\theta}, \quad \|\zeta_k u - u\|_{\gamma,p,\theta} \rightarrow 0$$

as $k \rightarrow \infty$, where N is independent of k and u .

2 Characterizations of $H_{p,\theta}^\gamma$

Recall that by M we denote the operator of multiplying by x^1 .

Theorem 2.0.1 The following conditions are equivalent

- (i) $u \in H_{p,\theta}^\gamma$,
- (ii) $u \in H_{p,\theta}^{\gamma-1}$ and $Mu_x \in H_{p,\theta}^{\gamma-1}$,
- (iii) $u \in H_{p,\theta}^{\gamma-1}$ and $(Mu)_x \in H_{p,\theta}^{\gamma-1}$.

In addition, under either of these three conditions for some constants $N = N(\gamma, p, d)$ we have

$$\|u\|_{\gamma,p,\theta} \leq N\|u\|_{\gamma-1,p,\theta} + N\|Mu_x\|_{\gamma-1,p,\theta} \leq N\|u\|_{\gamma,p,\theta}, \quad (16)$$

$$\|u\|_{\gamma,p,\theta} \leq N\|u\|_{\gamma-1,p,\theta} + N\|(Mu)_x\|_{\gamma-1,p,\theta} \leq N\|u\|_{\gamma,p,\theta}. \quad (17)$$

proof. Equivalence of (ii) and (iii) is trivial since $Mu_{x^i} = (Mu)_{x^i} - u\delta^{1i}$. By the same reason eq. (16) and eq. (17) are equivalent.

Under the assumption that (ii) holds, notice that

$$\begin{aligned} \|u(e^n \cdot) \zeta\|_{\gamma,p}^p &\leq N\|u(e^n \cdot) \zeta\|_{\gamma-1,p}^p + N\|(u(e^n \cdot) \zeta)_x\|_{\gamma-1,p}^p \\ &\leq N\|u(e^n \cdot) \zeta\|_{\gamma-1,p}^p + N\|u(e^n \cdot) \zeta_x\|_{\gamma-1,p}^p + N\|e^n u_x(e^n \cdot) \zeta\|_{\gamma-1,p}^p. \end{aligned}$$

Upon observing that $e^n u_x(e^n \cdot) \zeta = (Mu_x)(e^n \cdot) M^{-1} \zeta$ and using Lemma 1.0.6, we get that (i) holds and also get the left estimate in eq. (16).

On the other hand, if $u \in H_{p,\theta}^\gamma$, we have, of course, $u \in H_{p,\theta}^{\gamma-1}$. Also

$$\zeta(x) e^n u_x(e^n x) = (\zeta(x) u(e^n x))_x - \zeta_x(x) u(e^n x),$$

$$\|(Mu_x)(e^n \cdot) \zeta\|_{\gamma-1,p}^p = \|M(u(e^n \cdot) \zeta)_x - u(e^n \cdot) M \zeta_x\|_{\gamma-1,p}^p \leq N\|f(u(e^n \cdot) \zeta)_x\|_{\gamma-1,p}^p + \|u(e^n \cdot) M \zeta_x\|_{\gamma,p}^p,$$

where f is any function of class $C_0^\infty(\mathbb{R}_+)$ such that $f(x^1) = 1$ if $\zeta(x^1) \neq 0$.⁽ⁱⁱⁱ⁾ By using Lemma 1.0.6 and the fact that

$$\|f(u(e^n \cdot) \zeta)_x\|_{\gamma-1,p}^p \leq N\|(u(e^n \cdot) \zeta)_x\|_{\gamma-1,p}^p \leq N\|u(e^n \cdot) \zeta\|_{\gamma,p}^p,$$

we conclude the proof.

Remark 2.0.2 Theorem 2.0.1 implies, in particular, that the operators MD_i and $D_i M$ are *bounded* operators from $H_{p,\theta}^\gamma$ to $H_{p,\theta}^{\gamma-1}$.

Theorem 2.0.1 and Remark 1.0.4 or Corollary 1.0.8 allows us to get the following characterization of spaces $H_{p,\theta}^\gamma$ for integral γ .

Corollary 2.0.3 If γ is nonnegative integer, then the space $H_{p,\theta}^\gamma$ is

$$\{f : f, x^1 f_{x^1}, \dots, (x^1)^{|\alpha|} D^\alpha f \in L_{p,\theta} \quad \forall \alpha : |\alpha| \leq \gamma\}$$

with natural norm.

proof (Corollary 1.0.20). It suffices to apply Theorem 1.0.16 along with Remark 1.0.17 and Corollary 2.0.3 to the operator

$$\tilde{T}f = \{Tf, x^1 T f_{x^1}, \dots, (x^1)^{|\alpha|} D^\alpha T f : |\alpha| \leq \gamma\}$$

taking Ω as a singleton, $X = \mathbb{R}$, and $Y = \mathbb{R}^m$ with sufficiently large m .^(iv)

From Corollary 2.0.3 we also get the following.

⁽ⁱⁱⁱ⁾To detour the operator M into C_0^∞ function as $\zeta \in C_0^\infty(\mathbb{R}_+)$, and apply the well-known inequality for multiplier.

^(iv)Regard $H_{p,\theta}^\gamma$ as $\mathbb{L}_{p,\theta}(\mathbb{R}^m)$ for sufficiently large m where $\mathbb{L}_{p,\theta}(X)$ is defined in Theorem 1.0.16. Noticed that $L_{p,\theta}$ is a collection of real-valued functions, thus $L_{p,\theta} = \mathbb{L}_{p,\theta}(\mathbb{R})$.

Corollary 2.0.4 Let γ be a nonnegative integer and $a(x)$ be a real-valued function on $\mathbb{R}^d$ satisfying

$$|(x^1)^{|\alpha|} D^\alpha(x)| \leq K \quad \forall |\alpha| \leq \gamma,$$

where K is a constant. Then, for any $f \in H_{p,\theta}^\gamma$, we have $\|af\|_{\gamma,p,\theta} \leq N\|f\|_{\gamma,p,\theta}$, where N is independent of f and a .

The following operators will play an important role in the future. For $b, c \in \mathbb{R}$ define

$$\mathcal{L}_{b,c}u = M^2\Delta u + bMD_1u - cu.$$

By using Remark 2.0.2 and the formulas

$$\begin{aligned} M^2\Delta &= \sum_{i=1}^d (MD_i)^2 - MD_1, \\ M^{-1}\mathcal{L}_{b,c}Mu &= MD_iD_iMu + bD_1Mu - cu \end{aligned} \quad (18)$$

we get the following.

Corollary 2.0.5 The operators $\mathcal{L}_{b,c}$ and $M^{-1}\mathcal{L}_{b,c}M$ are bounded operators from $H_{p,\theta}^\gamma$ to $H_{p,\theta}^{\gamma-2}$.

Theorem 2.0.6 We have $M^{-1}u \in H_{p,\theta}^\gamma$ if and only if

$$u_x \in H_{p,\theta}^{\gamma-1}, \quad M^{-1}u \in \bigcup_{\mu} H_{p,\theta}^\mu. \quad (19)$$

Moreover, there exists constants $N = N(d, \gamma, \mu, \theta, p)$ such that, for any $\mu \leq \gamma$ and any θ, p , and $M^{-1}u \in H_{p,\theta}^\gamma$,

$$\|M^{-1}u\|_{\gamma,p,\theta} \leq N\|u_x\|_{\gamma-1,p,\theta} + N\|M^{-1}u\|_{\mu,p,\theta} \leq N\|M^{-1}u\|_{\gamma,p,\theta}. \quad (20)$$

proof. Observe that

$$MD_i(M^{-1}u) = u_{x^i} - \delta^{1i}M^{-1}u. \quad (21)$$

By Theorem 2.0.1 and eq. (21), we can obtain the following inequalities.

$$\begin{aligned} \|M^{-1}u\|_{\gamma,p,\theta} &\leq N\|M^{-1}u\|_{\gamma-1,p,\theta} + N\|M(M^{-1}u)_x\|_{\gamma-1,p,\theta} \leq N\|M^{-1}u\|_{\gamma-1,p,\theta} + N\|u_x\|_{\gamma-1,p,\theta}, \\ \|M^{-1}u\|_{\gamma-1,p,\theta} + \|u_x\|_{\gamma-1,p,\theta} &\leq N\|M^{-1}u\|_{\gamma-1,p,\theta} + \|M(M^{-1}u)_x\|_{\gamma-1,p,\theta} \leq N\|M^{-1}u\|_{\gamma,p,\theta}. \end{aligned}$$

We can apply the above inequalities with $\gamma - n$ instead of γ where n is any positive integer. Thus inductively, we obtain

$$\begin{aligned} \|M^{-1}u\|_{\gamma,p,\theta} &\leq N\|u_x\|_{\gamma-1,p,\theta} + N\|M^{-1}u\|_{\gamma-1,p,\theta} \\ &\leq N\|u_x\|_{\gamma-1,p,\theta} + N\|u_x\|_{\gamma-2,p,\theta} + N\|M^{-1}u\|_{\gamma-2,p,\theta} \\ &\leq \dots \\ &\leq N\sum_{j=1}^n \|u_x\|_{\gamma-j,p,\theta} + N\|M^{-1}u\|_{\gamma-n,p,\theta} \\ &\leq N\|u_x\|_{\gamma-1,p,\theta} + N\|M^{-1}u\|_{\gamma-n,p,\theta}, \\ \|M^{-1}u\|_{\gamma-n,p,\theta} + \|u_x\|_{\gamma-1,p,\theta} &\leq \|M^{-1}u\|_{\gamma-1,p,\theta} + \|u_x\|_{\gamma-1,p,\theta} \leq N\|M^{-1}u\|_{\gamma,p,\theta}. \end{aligned}$$

Thus we proved eq. (20) when $\mu = \gamma - n$ for every positive integer n , and thus covers the case of all such $\mu \leq \gamma$. Obviously, eq. (20) implies eq. (19), and thus the theorem is proved.

Next, we need several results allowing us to move along the scale of spaces $H_{p,\theta}^\gamma$. Also, these results characterize $H_{p,\theta}^\gamma$ as an image of $H_{p,\theta}^{\gamma-2}$.

Theorem 2.0.7 For any b, γ, p , and θ there exists a constant $c_0 > 0$ such that for any $c \geq c_0$ and $u \in H_{p,\theta}^\gamma$,

$$\|u\|_{\gamma,p,\theta} \leq N \|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}, \quad c \|u\|_{\gamma-2,p,\theta} \leq N \|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}, \quad (22)$$

where the constant N are independent of u and c .

proof. First, notice that, by Remark 2.0.2 and Theorem 1.0.13

$$\|MD_i u\|_{\gamma-2,p,\theta} \leq N \|u\|_{\gamma-1,p,\theta} \leq N \|u\|_{\gamma,p,\theta}^{1/2} \|u\|_{\gamma-2,p,\theta}^{1/2}. \quad (23)$$

Next, we have

$$I := \|u\|_{\gamma,p,\theta}^p = \sum_{n=-\infty}^{\infty} e^{n\theta} \|(1-\Delta)(u(e^n \cdot)\zeta)\|_{\gamma-2,p}^p \leq N \|u\|_{\gamma-2,p,\theta}^p + \sum_{n=-\infty}^{\infty} e^{n\theta} \|\Delta(u(e^n \cdot)\zeta)\|_{\gamma-2,p}^p. \quad (24)$$

Here

$$\Delta(u(e^n \cdot)\zeta) = u(e^n \cdot)\Delta\zeta + (M^2\Delta u)(e^n \cdot)M^{-2}\zeta + 2(MD_i u)(e^n \cdot)M^{-1}D_i\zeta,$$

so that by Lemma 1.0.6 the last sum in eq. (24) is less than

$$N \|u\|_{\gamma-2,p,\theta}^p + N \|MD_i u\|_{\gamma-2,p,\theta}^p + N \|\mathcal{L}_{b,c}u - bMD_1 u + cu\|_{\gamma-2,p,\theta}^p.$$

Owing to eq. (23) this expression is less than

$$N(1+c) \|u\|_{\gamma-2,p,\theta}^p + NI^{1/2} \|u\|_{\gamma-2,p,\theta}^{p/2} + N \|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}^p.$$

Now we are going to use Young's inequality^(v) provided that the second inequality of eq. (22) holds. For every $\epsilon > 0$ and $c \geq c_0$,

$$\begin{aligned} I &\leq N(1+c) \|u\|_{\gamma-2,p,\theta}^p + NI^{1/2} \|u\|_{\gamma-2,p,\theta}^{p/2} + N \|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}^p \\ &\leq N \|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}^p + N \left(\frac{\epsilon}{c} I^{1/2}\right) \left(\frac{c}{\epsilon} \|u\|_{\gamma-2,p,\theta}^{p/2}\right) \\ &\leq N \|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}^p + N \frac{\epsilon^2}{c^2} I + N \frac{c^2}{\epsilon^2} \|u\|_{\gamma-2,p,\theta}^p. \end{aligned}$$

This, therefore, gives that

$$\left(1 - N \frac{\epsilon^2}{c^2}\right) I \leq \left(N + N \frac{c^{2-p}}{\epsilon^2}\right) \|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}^p.$$

By letting $\epsilon > 0$ small enough, one can make the coefficient of the left side be positive, thus to prove the first one of eq. (22) it now suffices to prove the second one.

To prove the second inequality of eq. (22), first of all, assume that $u \in C_0^\infty(\mathbb{R}_+^d)$ because of Theorem 1.0.21, put $f = -\mathcal{L}_{b,c}u$, and for any $x \in \mathbb{R}_+^d$ denote by $x_t(x)$ a unique solution of the stochastic equation

$$dx_t = bx_t^1 e_1 dt + \sqrt{2} x_t^1 dw_t$$

with the initial data $x_0 = x$ where $w_t = (w_t^1, w_t')$ is a d -dimensional Wiener process and $e_1 = (1, 0, \dots, 0)$. Notice that the above stochastic equation corresponds to the Kolmogorov equation

$$\mathcal{L}_{b,c}u = -f.$$

Observe that $x_t(x) = \sigma_t x$, where

$$\sigma_t = \begin{pmatrix} e^{\xi_t} & 0 & \dots & 0 \\ | & & & \\ \eta_t & & I_{d-1} & \\ | & & & \end{pmatrix}, \quad \xi_t = w_t^1 \sqrt{2} + (b-1)t, \quad \eta_t = \sqrt{2} \int_0^t e^{\xi_s} dw_s'.$$

Indeed, we can split the stochastic equation by two parts:

^(v)If $p, q > 0$ and $p^{-1} + q^{-1} = 1$, then for every $a, b > 0$, $ab \leq p^{-1}a^p + q^{-1}b^q$.

$$dx_t^1 = bx_t^1 dt + \sqrt{2}x_t^1 dw_t^1, \quad dx_t' = \sqrt{2}x_t^1 dw_t^1.$$

By Lemma IV.3.1 on [Kry95], we have $x_t^1(x) = (\sigma_t x)^1 = \exp(\xi_t)x^1$. Then $x_t' = x' + \sqrt{2} \int_0^t x_t^1 dw_t^1$, thus we obtain the above result.

By Itô's formula applied to $u(x_t(x)) \exp(-ct)$ we get

$$du(x_t) = u_{x_i}(x_t)dx_t^i + \frac{1}{2}u_{x_i x_j}(x_t)(dx_t^i)(dx_t^j) = (M^2 \Delta u + bMD_1 u)(x_t)dt + \sqrt{2}(MDu)(x_t)dw_t,$$

$$d[u(x_t) \exp(-ct)] = -e^{-ct}f(x_t)dt + \sqrt{2}e^{-ct}(MDu)(x_t)dw_t.$$

By taking the expectation, we have

$$\mathbb{E}u(x_t)e^{-ct} = u(x) - \mathbb{E} \int_0^t e^{-cs}f(x_s)ds + \sqrt{2}\mathbb{E} \int_0^t e^{-cs}(MDu)(x_s)dw_s.$$

Since $u \in C_0^\infty(\mathbb{R}_+^d)$, the integrand of the stochastic integral lies in $L_2((0, \infty), \mathcal{P})$, expectation of the stochastic integral vanishes. In addition, letting $t \rightarrow \infty$, $\mathbb{E}u(x_t)e^{-ct} \rightarrow 0$ because $u \in C_0^\infty(\mathbb{R}_+^d)$. Hence, we get

$$u(x) = \mathbb{E} \int_0^\infty e^{-ct}f(x_t(x))dt = \mathbb{E} \int_0^\infty e^{-ct}f(\sigma_t x)dt, \quad (25)$$

and^(vi)

$$\|u\|_{\gamma-2,p,\theta} \leq \mathbb{E} \int_0^\infty e^{-ct}\|f(\sigma_t \cdot)\|_{\gamma-2,p,\theta}dt.$$

Next, we want to apply Lemma 1.0.11. It follows that by definition that

$$|\sigma_t^* x|^2 = |e^{\xi_t} x^1 + \eta_t \cdot x'|^2 + |x'|^2, \quad \det \sigma_t \sigma_t^* = (\det \sigma_t)^2 = e^{2\xi_t}.$$

Using this, we find

$$\begin{aligned} \Lambda(\sigma_t) &= \sup_{|x|=1} |\sigma_t^* x|^2 \leq N(e^{\xi_t} + |\eta_t|)^2 \leq N(e^{2\xi_t} + |\eta_t|^2), \\ \lambda(\sigma_t) &\geq (\Lambda(\sigma_t))^{-d+1} \det \sigma_t \sigma_t^* \geq N^{-1}e^{2\xi_t}(e^{2\xi_t} + |\eta_t|^2)^{-d+1}, \\ \lambda^{-\kappa}(\sigma_t) + \Lambda^\kappa(\sigma_t) &\leq Ne^{N|\xi_t|} + N|\eta_t|^N + N. \end{aligned}$$

Now we are going to estimate the expectation of the last expression. It is easy to check that $e^{|x|} \leq e^x + e^{-x}$, and simple calculation about $\mathbb{E}e^{aw_t}$,^(vii)

$$\mathbb{E}e^{N|\xi_t|} \leq \mathbb{E}e^{N\xi_t} + \mathbb{E}e^{-N\xi_t} \leq e^{N(b-1)t+Nt} + e^{-N(b-1)t+Nt},$$

where N are constants independent of t .

To estimate $\mathbb{E}|\eta_t|^N$, we use Burkholder-Davis-Gundy inequality. We find $\langle \eta \rangle_t = \int_0^t 2e^{2\xi_s} ds$, and

$$\mathbb{E}|\eta_t|^N \leq N\mathbb{E}\langle \eta \rangle_t^{N/2} = \mathbb{E} \left(\int_0^t 2e^{2\xi_s} ds \right)^{N/2} =: I.$$

One can take N large enough so we assume that $N \geq 2$. In this case, the Minkowski's inequality gives

$$I \leq \left(\int_0^t \mathbb{E}(2e^{2\xi_s})^{N/2} ds \right)^{N/2} = N \left(\int_0^t \mathbb{E}e^{N\xi_s} ds \right)^{N/2} = Ne^{N(b-1)t+Nt}.$$

Notice that N are constants independent of t .

Hence, they are less than $Ne^{c_0 t/2}$ for an appropriate $c_0 > 0$. This proves the second inequality in eq. (22) and the theorem.

^(vi) Consider $\Phi : \Omega \times (0, \infty) \rightarrow H_{p,\theta}^{\gamma-2}$ by $\Phi(t) = e^{-ct}f(\sigma_t \cdot)$. Since $H_{p,\theta}^{\gamma-2}$ is separable, by the Pettis theorem, Φ is strongly measurable. If $\mu = \mathbb{P} \times \ell$, then $u = \int_X \Phi d\mu$ where $X = \Omega \times (0, \infty)$ and the integration is understood by Bochner integral sense. Hence, by the Bochner integral theory,

$$\|u\|_{\gamma-2,p,\theta} \leq \int_X \|\Phi\|_{\gamma-2,p,\theta} d\mu = \mathbb{E} \int_0^\infty e^{-ct}\|f(\sigma_t \cdot)\|_{\gamma-2,p,\theta} dt.$$

^(vii) $\mathbb{E}e^{aw_t} = e^{a^2 t/2}$

Theorem 2.0.8 Take b, γ, p, θ , and c_0 from Theorem 2.0.7. Then there exists a constant c_1 depending only on d and b such that for any $c \geq c_0 \vee c_1$ the operator $\mathcal{L}_{b,c}$ is a bounded 1-1 operator from the space $H_{p,\theta}^\gamma$ onto $H_{p,\theta}^{\gamma-2}$ and its inverse is also bounded.

proof. Owing to Theorem 2.0.7 it suffices to prove that, for large c , the range of $\mathcal{L}_{b,c}$ is dense in $H_{p,\theta}^{\gamma-2}$. By virtue of Theorem 1.0.21 we have only to prove that given $f \in C_0^\infty(\mathbb{R}_+^d)$, there exists a function $u \in H_{p,\theta}^\gamma$ such that $\mathcal{L}_{b,c}u = -f$.

Define u by eq. (25). Then by well-known results about Kolmogorov's equations (see, for instance, Theorem V.8.5 of [Kry95]) there exists a constant c_1 depending only on d and b such that if $c \geq c_1$, then u is twice continuously differentiable and $\mathcal{L}_{b,c}u = -f$ (even in $\mathbb{R}^d$). In the proof of Theorem 2.0.7 we have seen that if in addition $c \geq c_0$, then

$$\|u\|_{\gamma,p,\theta} \leq \mathbb{E} \int_0^\infty e^{-ct} \|f(\sigma_t \cdot)\|_{\gamma,p,\theta} dt \leq N \int_0^\infty e^{(c_0-c)t} dt < \infty,$$

hence, $u \in H_{p,\theta}^\gamma$.

Remark 2.0.9 In eq. (22), the first inequality is proved provided that $u \in H_{p,\theta}^\gamma$. Hence, one cannot use the first one directly in the proof of Theorem 2.0.8. However, the second inequality is proved without the condition $u \in H_{p,\theta}^\gamma$.

Corollary 2.0.10 For every $u \in H_{p,\theta}^\gamma$ and $\mu \geq 1$ there exists functions $u^i \in H_{p,\theta}^{\gamma+1}$, $i = 1, \dots, d$, and $u^0 \in H_{p,\theta}^{\gamma+\mu}$ such that

$$u = u^0 + MD_i u^i \quad (26)$$

and

$$\|u^0\|_{\gamma+\mu,p,\theta} + \sum_{i=1}^d \|u^i\|_{\gamma+1,p,\theta} \leq N \|u\|_{\gamma,p,\theta} \leq N \left(\|u^0\|_{\gamma+\mu,p,\theta} + \sum_{i=1}^d \|u^i\|_{\gamma+1,p,\theta} \right) \quad (27)$$

with constants N independent of u .

proof. Notice that given eq. (26), the estimate on the right in eq. (27) follows from Remark 2.0.2. Our assertion become stronger if μ becomes larger. Therefore, we may concentrate on integral μ . In this case one can use the induction on μ . Indeed, if we have our assertions for $\mu = 1$, then representing u^0 as another sum like eq. (26) with a new first term from $H_{p,\theta}^{\gamma+2}$, we will have our assertion for $\mu = 2$, and so on.^(viii) This shows that it only remains to consider $\mu = 1$.

By Theorem 2.0.7 and 2.0.8 we have $u = M^2 \Delta v - cv$ and $\|v\|_{\gamma+2,p,\theta} \leq N \|u\|_{\gamma,p,\theta}$. This, formulas eq. (18), and Remark 2.0.2 finish the proof.

Remark 2.0.11 Generally the term u^0 in eq. (27) cannot be dropped. Indeed, for $d = 1$, $u(x) := I_{x \geq 2}(\ln x)^{-1}$, and $\theta = \gamma = 0$, we have $u \in H_{p,\theta}^\gamma = L_p(\mathbb{R}, x^{-1} dx)$. However, there is no solution $v \in L_p(\mathbb{R}, x^{-1} dx)$ of the equation $xv'(x) = u(x)$.

The above results admit stronger forms needed in the future in the particular case $\theta \neq d - 1$ and $\theta \neq p + d - 1$.

Theorem 2.0.12

(i) Let $\theta \neq d - 1$ and b be such that

$$(\theta - d + 1)^2 < (\theta - d + 1)(b - 1). \quad (28)$$

Then for any γ and p the operator $\mathcal{L}_{b,0}$ is a bounded 1-1 operator from the space $H_{p,\theta}^\gamma$ onto $H_{p,\theta}^{\gamma-2}$ and its inverse is also bounded.

^(viii) Also remember that, obviously, $H_{p,\theta}^{\gamma+1} \subset H_{p,\theta}^\gamma$.

(ii) Let $\theta \neq p + d - 1$ and b such that

$$(\theta - p - d + 1)^2 < (\theta - p - d + 1)(b - 1).$$

Then for any γ and p the operator $M^{-1}\mathcal{L}_{b,0}M = MD_iD_iM + bD_1M$ is a bounded 1-1 operator from the space $H_{p,\theta}^\gamma$ onto $H_{p,\theta}^{\gamma-2}$ and its inverse is also bounded.

proof.

(i) Owing to Theorem 2.0.8 and the method of continuity, we only need to show that for all $c \geq 0$ and $u \in H_{p,\theta}^\gamma$ we have

$$\|u\|_{\gamma,p,\theta} \leq N\|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}, \quad (29)$$

where N is independent of u and c . Indeed, for fixed $f \in H_{p,\theta}^{\gamma-2}$, consider a collection $C \subset [0, \infty)$ of c such that $\mathcal{L}_{b,c}$ is bijective and satisfies eq. (29). Fix $c^* \in C$, $c \in [0, \infty)$ and define an operator $Q_c : H_{p,\theta}^\gamma \rightarrow H_{p,\theta}^\gamma$ such that for any $u \in H_{p,\theta}^\gamma$,

$$\mathcal{L}_{b,c^*}Q_cu + (c^* - c)u = f.$$

Notice that for any $u_1, u_2 \in H_{p,\theta}^\gamma$,

$$\|Q_cu_1 - Q_cu_2\|_{\gamma,p,\theta} \leq N\|\mathcal{L}_{b,c^*}Q_cu_1 - \mathcal{L}_{b,c^*}Q_cu_2\|_{\gamma-2,p,\theta} = N|c - c^*|\|u_1 - u_2\|_{\gamma,p,\theta},$$

where N is independent of c and c^* . Thus one can make Q_c a contraction if c is close enough to c^* and the distance of which does not depends on c and c^* , say $\delta > 0$.

What we proved is that there exists $\delta > 0$ such that for any $c^* \in C$ and $|c - c^*| < \delta$, then $c \in C$. Since C is not empty because of Theorem 2.0.8, we therefore obtain $C = [0, \infty)$, which proves (i) of this theorem.

We know from Theorem 2.0.7 that this estimate holds for large c , say $c \geq c^* = c^*(\gamma) = c^*(b, \gamma)$. For $c \in [0, c^*(\gamma)]$ we have

$$\begin{aligned} \|u\|_{\gamma,p,\theta} &\leq N\|\mathcal{L}_{b,c^*(\gamma)}u\|_{\gamma-2,p,\theta} \\ &\leq N\|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta} + |c^*(\gamma) - c|N\|u\|_{\gamma-2,p,\theta} \\ &\leq N\|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta} + |c^*(\gamma) - c|N\|\mathcal{L}_{b,c^*(\gamma-2)}u\|_{\gamma-4,p,\theta} \\ &\leq N\|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta} + |c^*(\gamma) - c|N\|\mathcal{L}_{b,c}u\|_{\gamma-4,p,\theta} + |c^*(\gamma) - c||c^*(\gamma - 2) - c|N\|u\|_{\gamma-4,p,\theta} \\ &\leq (N + N|c^*(\gamma) - c|)\|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta} + (N|c^*(\gamma) - c|^2 + N|c^*(\gamma) - c|)\|u\|_{\gamma-4,p,\theta} \end{aligned}$$

because

$$|c^*(\gamma - 2) - c| \leq |c^*(\gamma - 2) - c^*(\gamma)| + |c^*(\gamma) - c|.$$

Continuing this process and since $c^*(\gamma) - c \leq c^*(\gamma)$, we have $\leq N\|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta} + N\|u\|_{\gamma-2n,p,\theta}$. Here n is any nonnegative integer and N are independent of u and $c \in [0, c^*(\gamma)]$. It follows that it suffices to prove that for any γ there exists $n \geq 1$ and $N \in (0, \infty)$ such that, for any $u \in H_{p,\theta}^\gamma$ and $c \geq 0$,

$$\|u\|_{\gamma-2n,p,\theta} \leq N\|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta}. \quad (30)$$

Of course, we may confine ourselves to the case $u \in C_0^\infty(\mathbb{R}_+^d)$.

First, let $\gamma \geq 2$. Then take n such that $\gamma - 2n \leq 0$ and notice that, for $f := -\mathcal{L}_{b,c}u$ by formula eq. (25) and Remark 1.0.4, we have^(ix)

$$\begin{aligned} \|u\|_{\gamma-2n,p,\theta} &\leq N\|u\|_{0,p,\theta} \\ &\leq N\mathbb{E} \int_0^\infty e^{-ct} \|f(\sigma_t \cdot)\|_{0,p,\theta} dt \end{aligned}$$

^(ix)In the third equality, we used the following fact.

$$\mathbb{E} \exp\left(w_t^1 - \frac{t}{2}\right) = \frac{1}{\sqrt{2\pi t}} \int_{\mathbb{R}} e^{x-t/2} e^{-x^2/(2t)} dx = \frac{1}{\sqrt{2\pi t}} \int_{\mathbb{R}} e^{-(x-t)^2/(2t)} dx = 1.$$

Similar method also proves that $\mathbb{E} \exp(\alpha w_t^1 + \beta t) = \exp(\alpha^2 t/2 + \beta t)$. Now remember that $\xi_t = w_t^1 \sqrt{2} + (b-1)t$.

$$\begin{aligned}
 &= N \mathbb{E} \int_0^\infty e^{-ct} e^{-\xi_t(\theta-d+1)/p} \|f(\sigma_t \cdot)\|_{0,p,\theta} dt \\
 &= N \|\mathcal{L}_{b,c} u\|_{0,p,\theta} \int_0^\infty e^{-ct - (\theta-d+1)(b-1)t/p + (\theta-d+1)^2 t/p^2} dt \\
 &\leq N \|\mathcal{L}_{b,c} u\|_{0,p,\theta} \int_0^\infty e^{\frac{-(\theta-d+1)(b-1)t + (\theta-d+1)^2 t}{p}} dt \\
 &= N \|\mathcal{L}_{b,c} u\|_{0,p,\theta} \\
 &\leq N \|\mathcal{L}_{b,c} u\|_{\gamma-2,p,\theta},
 \end{aligned}$$

where N is independent of u and c . In this way, we have also proved eq. (29) for $\gamma \geq 2$.

If $\gamma < 2$, take $n = 1$ and an integer $m \geq 2$ such that $\gamma + 2(m-1) < 4$ and $\gamma + 2m \geq 4$ and define

$$\bar{c} = c^*(\gamma-2) \vee c^*(\gamma) \vee \dots \vee c^*(\gamma+2m-2).$$

Then

$$\|u\|_{\gamma-2n,p,\theta} = N \|u\|_{\gamma-2,p,\theta} \leq N \|\mathcal{L}_{b,\bar{c}}^{-1} u\|_{\gamma,p,\theta} \leq \dots \leq N \|\mathcal{L}_{b,\bar{c}}^{-m} u\|_{\gamma+2m-2,p,\theta}.$$

In addition, by the above result ($\gamma + 2m - 2 \geq 2$) (and eq. (29) for $\gamma \geq 2$ since we already proved)

$$\begin{aligned}
 \|\mathcal{L}_{b,\bar{c}}^{-m} u\|_{\gamma+2m-2,p,\theta} &\leq N \|\mathcal{L}_{b,c} \mathcal{L}_{b,\bar{c}}^{-m} u\|_{\gamma+2m-4,p,\theta} = N \|\mathcal{L}_{b,\bar{c}}^{-m} \mathcal{L}_{b,c} u\|_{\gamma+2m-4,p,\theta} \\
 &\leq N \|\mathcal{L}_{b,c} u\|_{\gamma-4,p,\theta} \leq N \|\mathcal{L}_{b,c} u\|_{\gamma-2,p,\theta}
 \end{aligned}$$

with the constant N independent of u and c . This proves the assertion (i).

- (ii) This follows from (i) and Corollary 1.0.8. Indeed, the equation $M^{-1} \mathcal{L}_{b,0} M u = f$ is equivalent to $\mathcal{L}_{b,0} v = g$ with $v = M u$ and $g = M f$. In addition, $u \in H_{p,\theta}^\gamma$ and $f \in H_{p,\theta}^{\gamma-2}$ if and only if $v \in H_{p,\theta-p}^\gamma$ and $g \in H_{p,\theta-p}^{\gamma-2}$.

The theorem is proved.

Corollary 2.0.13

- (i) If $\theta \neq d-1$ and $u \in H_{p,\theta}^\gamma$, then

$$\|u\|_{\gamma,p,\theta} \leq N(d, \gamma, p, \theta) \|M D u\|_{\gamma-1,p,\theta}. \quad (31)$$

In particular, in this case the term $N \|u\|_{\gamma-1,p,\theta}$ in eq. (16) can be dropped. Furthermore, in Corollary 2.0.10 one can take $u_0 = 0$.

- (ii) If $\theta \neq d+p-1$ and $u \in H_{p,\theta}^\gamma$, then

$$\|u\|_{\gamma,p,\theta} \leq N(d, \gamma, p, \theta) \|D M u\|_{\gamma-1,p,\theta}.$$

In particular, in this case the terms $N \|u\|_{\gamma-1,p,\theta}$ in eq. (17) and $\|M^{-1} u\|_{\mu,p,\theta}$ in eq. (20) can be dropped.

proof.

- (i) Observe that $\mathcal{L}_{b,0} u = M D_i M D_i u + (b-1) M D_1 u$.^(x) Hence, for an appropriate b ,

$$\|u\|_{\gamma,p,\theta} \leq N \|\mathcal{L}_{b,0} u\|_{\gamma-2,p,\theta} \leq N \|M D u\|_{\gamma-1,p,\theta} + N \|M D_1 u\|_{\gamma-2,p,\theta},$$

and eq. (31) follows.

The second assertion in (i) follows from the fact that $u = \mathcal{L}_{b,0} v = M D_i v^i + (b-1) M D_1 v$, where $v := \mathcal{L}_{b,0}^{-1} u$, $v^i := M D_i v$, and

^(x) Notice that $M^2 \Delta u = \sum_1^d M D_j M D_j u - D_1 u$.

$$\|v\|_{\gamma+1,p,\theta} + \|v^i\|_{\gamma+1,p,\theta} \leq \|v\|_{\gamma+1,p,\theta} + N\|v\|_{\gamma+2,p,\theta} \leq N\|v\|_{\gamma+2,p,\theta} \leq N\|u\|_{\gamma,p,\theta}.$$

Indeed, first inequality in eq. (16) comes immediately from eq. (31). The second part is obvious. Now in Corollary 2.0.10, we have

$$u = MD_i u^i$$

where $u^1 = (b-1)v + v^1$ and $u^i = v^i$ for $i = 2, 3, \dots, d$.

(ii) Apply Corollary 1.0.8 and the result in (i) to get

$$\|u\|_{\gamma,p,\theta} \leq N\|Mu\|_{\gamma,p,\theta-p} \leq N\|MDMu\|_{\gamma-1,p,\theta-p} \leq N\|DMu\|_{\gamma-1,p,\theta}.$$

The same result can also be obtained from Theorem 2.0.12 (ii) if one notices that $M^{-1}\mathcal{L}_{b,0}Mu = MD_i D_i Mu + bD_1 Mu$.

Indeed, it suffices to show the second assertion. First of all,

$$\|Du\|_{\gamma-1,p,\theta} = \|DMM^{-1}u\|_{\gamma-1,p,\theta} \leq N\|M^{-1}u\|_{\gamma,p,\theta}.$$

On the other hand, assume that $Du \in H_{p,\theta}^{\gamma-1}$, and let $v = M^{-1}\mathcal{L}_{b,0}u$. Since

$$\begin{aligned} \|v\|_{\gamma-2,p,\theta} &= \|M^{-1}\mathcal{L}_{b,0}u\|_{\gamma-2,p,\theta} \\ &\leq \|MD_i D_i u\|_{\gamma-2,p,\theta} + N\|D_1 u\|_{\gamma-2,p,\theta} \\ &\leq N\|Du\|_{\gamma-1,p,\theta} + N\|Du\|_{\gamma-2,p,\theta} \\ &\leq N\|Du\|_{\gamma-1,p,\theta}, \end{aligned}$$

we have $v \in H_{p,\theta}^{\gamma-2}$. Thus by Theorem 2.0.12 (ii), there exists $w \in H_{p,\theta}^{\gamma}$ such that $M^{-1}\mathcal{L}_{b,0}Mw = v$. Since $M^{-1}\mathcal{L}_{b,0}M$ is bijective, $w = M^{-1}u$, hence $M^{-1}u \in H_{p,\theta}^{\gamma}$ and

$$\|M^{-1}u\|_{\gamma,p,\theta} \leq N\|v\|_{\gamma-2,p,\theta} \leq N\|Du\|_{\gamma-1,p,\theta}.$$

The corollary is proved.

Remark 2.0.14 The way we arrived at Corollary 2.0.13 is much more complicated than in one-dimensional case (see [KL99]), where we had similar assertions almost for free.

Definition 2.0.15 Let $\theta \neq d-1$. Fix a b such that eq. (28) holds and for any $u \in H_{p,\theta}^{\gamma}$ define $Qu = (Q_1 u, \dots, Q_d u)$, where

$$Q_1 u = MD_1 \mathcal{L}_{b,0}^{-1} u + (b-1)\mathcal{L}_{b,0}^{-1} u, \quad Q_i u = MD_i \mathcal{L}_{b,0}^{-1} u, \quad i = 2, \dots, d.$$

Remark 2.0.16 By what we have seen in the proof of Corollary 2.0.13, Q is a bounded operator from $H_{p,\theta}^{\gamma}$ to $H_{p,\theta}^{\gamma+1}$ (or better say to $(H_{p,\theta}^{\gamma+1})^d$) and $u = MD_i Q_i u$.

If $\theta > d-1$, then condition eq. (28) means that $b > 1$ and $\theta < (b-1) + d-1$. It turns out that, for $\theta > d-1$, assertion (i) of Theorem 2.0.12 holds under less restrictive conditions.

Theorem 2.0.17 Let $b, c \in \mathbb{R}$ and

$$d-1 < \theta < d-1 + p(b-1) + p^2 c(\theta-d+1)^{-1}.$$

Then for any γ the operator $\mathcal{L}_{b,c}$ is a bounded 1-1 operator from $H_{p,\theta}^{\gamma}$ onto $H_{p,\theta}^{\gamma-2}$ and its inverse is also bounded.

To prove this theorem we need the following standard result, which is proved in Appendix.

Lemma 2.0.18 Let $(a, b) \subset \mathbb{R}$ and $u \in W_p^2(a, b)$. Then

- (i) the functions $|u|^{p/2}$ and $|u|^{p-2}uu_x$ are absolutely continuous on (a, b) ,
- (ii) almost everywhere on (a, b) ($\infty \cdot 0 := 0$ if $1 < p < 2$)

$$(|u|^{p/2})_x = \frac{p}{2}|u|^{p/2-2}uu_x, \quad (|u|^{p-2}uu_x)_x = |u|^{p-2}uu_{xx} + (p-1)|u|^{p-2}(u_x)^2. \quad (32)$$

proof (Theorem 2.0.17). As in the proof of Theorem 2.0.12 it suffices to prove estimate eq. (30) with $\gamma \geq 2$ for $u \in C_0^\infty(\mathbb{R}_+^d)$.^(xi) This case can be further reduced into by proving the following inequality

$$\|u\|_{0,p,\theta} \leq N\|\mathcal{L}_{b,c}u\|_{0,p,\theta}.$$

Indeed then for sufficiently large n ,

$$\|u\|_{\gamma-2n,p,\theta} \leq \|u\|_{0,p,\theta} \leq N\|\mathcal{L}_{b,c}u\|_{0,p,\theta} \leq N\|\mathcal{L}_{b,c}u\|_{\gamma-2,p,\theta},$$

because $\gamma \geq 2$.

Define

$$f = (x^1)^2\Delta u + bx^1u_{x^1} - cu \quad (= \mathcal{L}_{b,c}u),$$

multiply both parts of this equality by $(x^1)^{\theta-d}|u|^{p-2}u$, and integration by parts, Lemma 2.0.18,^(xii) and Fubini's theorem (also remember that $u \in C_0^\infty(\mathbb{R}_+^d)$).^(xiii)

$$\begin{aligned} \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2}|u|^{p-2}u\Delta u dx &= - \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2}(p-1)|u|^{p-2}|u_x|^2 dx + \sum_{i=1}^d \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2}(|u|^{p-2}uu_{x^i})_{x^i} dx \\ &= - \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2}(p-1)|u|^{p-2}|u_x|^2 dx \\ &\quad + \sum_{i=1}^d (x^1)^{\theta-d+2}|u|^{p-2}uu_{x^i} \Big|_{\text{if } (i=1) \text{ } 0 \text{ else } -\infty}^\infty \\ &\quad - \int_{\mathbb{R}_+^d} (\theta-d+2)(x^1)^{\theta-d+1}|u|^{p-2}uu_{x^1} dx \\ &= - \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2}(p-1)|u|^{p-2}|u_x|^2 dx - \int_{\mathbb{R}_+^d} p^{-1}(\theta-d+2)(x^1)^{\theta-d+1}(|u|^p)_{x^1} dx, \\ \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+1}|u|^{p-2}uu_{x^1} dx &= (x^1)^{\theta-d+1}|u|^p \Big|_0^\infty - \int_{\mathbb{R}_+^d} p^{-1}(\theta-d+1)(x^1)^{\theta-d}|u|^p dx \\ &= - \int_{\mathbb{R}_+^d} p^{-1}(\theta-d+1)(x^1)^{\theta-d}|u|^p dx. \end{aligned}$$

Combining these, we have

$$\begin{aligned} &- \int_{\mathbb{R}_+^d} (x^1)^{\theta-d}|u|^{p-2}uf dx \\ &= (p-1) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2}|u|^{p-2}|u_x|^2 dx + \alpha \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+1}(|u|^p)_{x^1} dx + \beta \int_{\mathbb{R}_+^d} (x^1)^{\theta-d}|u|^p dx \end{aligned}$$

^(xi) Repeat the proof of Theorem 2.0.12 for the case $\gamma < 2$.

^(xii) Lemma 2.0.18 states for only one dimensional. However, as u has a compact support, we can apply Lemma 2.0.18 per coordinates.

^(xiii) If $1 < p < 2$, then $|u|^{p-2}u$ is nonsense because the case 0^{-1} can happen. However if one adopt $\infty \cdot 0 := 0$, on the set $\{u = 0\}$, every calculation also valid, thus we can just calculate integrals like the context.

In addition, Fubini's theorem is used in order to apply the fundamental theorem of calculus.

$$= (p-1) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |u|^{p-2} |u_x|^2 dx + (\beta - (\theta - d + 1)\alpha) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |u|^p dx$$

where $\alpha = p^{-1}(\theta - d + 2)$ and $\beta = p^{-1}(\theta - d + 1)b + c$, and in the middle term we again integrate by parts. On the left use Hölder's inequality.^(xiv) We also define $v = |u|^{p/2}$. This and Lemma 2.0.18 yields

$$\begin{aligned} \|f\|_{0,p,\theta} \|u\|_{0,p,\theta}^{p-1} &\geq \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |u|^{p-1} |f| dx \\ &\geq \left| \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |u|^{p-2} u f dx \right| \\ &\geq - \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |u|^{p-2} u f dx \\ &= (p-1) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |u|^{p-2} |u_x|^2 dx + \tau \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |u|^p dx \\ &\geq 4p^{-2}(p-1) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |v_x|^2 dx + \tau \|u\|_{0,p,\theta}^p \\ &\geq 4p^{-2}(p-1) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |v_{x^1}|^2 dx + \tau \|u\|_{0,p,\theta}^p, \end{aligned}$$

where $\tau = \beta - (\theta - d + 1)\alpha$. To transform the remaining integral we use Minkowski's inequality and again Lemma 2.0.18 to find^(xv)

$$\begin{aligned} \|u\|_{0,p,\theta}^p &= \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |v|^2 dx \\ &= \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} \left| x^1 \int_1^\infty v_{x^1}(tx^1, x') dt \right|^2 dx \\ &\leq \left(\int_1^\infty \left(\int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |v_{x^1}(tx^1, x')|^2 dx \right)^{1/2} dt \right)^2 \\ &= \left(\int_1^\infty \left(\int_{\mathbb{R}_+^d} (x^1/t)^{\theta-d+2} t^{-1} |v_{x^1}(x^1, x')|^2 dx \right)^{1/2} dt \right)^2 \\ &= \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |v_{x^1}|^2 dx \left(\int_1^\infty t^{-(\theta-d+3)/2} dt \right)^2, \end{aligned}$$

where the integral with respect to t converges and equals $2(\theta - d + 1)^{-1}$ since $\theta > d - 1$. Thus

$$\frac{(\theta - d + 1)^2}{4} \|u\|_{0,p,\theta}^p \leq \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |v_{x^1}|^2 dx, \quad \|f\|_{0,p,\theta} \|u\|_{0,p,\theta}^{p-1} \geq \kappa^{-1} \|u\|_{0,p,\theta}^p,$$

where

$$\kappa := \tau + (\theta - d + 1)^2 p^{-2}(p-1) = p^{-2}(\theta - d + 1)(p(b-1) + d - 1 + p^2 c(\theta - d + 1)^{-1} - \theta).$$

By assumption $\kappa > 0$, so that $\|u\|_{0,p,\theta} \leq \kappa^{-1} \|f\|_{0,p,\theta}$ and the theorem is proved.

3 Embedding Theorems for $H_{p,\theta}^\gamma$

The following theorem is based on Sobolev's theorem saying that $H_p^\gamma \subset \mathcal{C}^{\gamma-d/p}$ if $\gamma > d/p$.

^(xiv)with measure $(x^1)^{\theta-d} dx$.

^(xv)Since $v = |u|^{p/2}$ is absolutely continuous, we can apply the fundamental theorem of calculus to write (also $|v(x)| \rightarrow 0$ as $|x| \rightarrow \infty$)

$$v(x) = - \int_{x^1}^\infty v_{x^1}(t, x') dt = -x^1 \int_1^\infty v_{x^1}(tx^1, x') dt.$$

Theorem 3.0.1 Assume that $\gamma p > d$ and represent $\gamma - d/p$ as $k + \alpha$, where k is an integer and $\alpha \in (0, 1]$. Let i, j be multiindices such that $|i| \leq k$ and $|j| = k$. Then for any $u \in H_{p,\theta}^\gamma$, we have

$$M^{|i|+\theta/p} D^i u \in C(\mathbb{R}_+^d), \quad M^{k+\alpha+\theta/p} D^j u \in \mathcal{C}_{\text{loc}}^\alpha(\mathbb{R}_+^d), \quad (33)$$

$$\|M^{|i|+\theta/p} D^i u\|_{C(\mathbb{R}_+^d)} \leq N \|u\|_{\gamma,p,\theta}, \quad [M^{k+\alpha+\theta/p} D^j u]_{\mathcal{C}^\alpha(\mathbb{R}_+^d)} \leq N \|u\|_{\gamma,p,\theta}, \quad (34)$$

where the constants N depend only on d, γ, p, θ .

proof. First of all, we need to show that $M^{|i|+\theta/p} D^i u$ is continuous on $\mathbb{R}_+^d$ in order to say whether it belongs to $C(\mathbb{R}_+^d)$. By Theorem 2.0.1 and Corollary 1.0.8, $u \in H_{p,\theta}^\gamma$ implies $M^{|i|+\theta/p} D^i u \in H_{p,0}^{\gamma-|i|}$, so it suffices to observe the space $H_{p,0}^\gamma$ where $\gamma p > d$. For fixed $f \in H_{p,0}^\gamma$, we have

$$\|f\|_{\gamma,p,0}^p = \sum_{n=-\infty}^{\infty} \|f(e^n \cdot) \zeta\|_{\gamma,p}^p.$$

This implies for every $n \in \mathbb{Z}$, $f(e^n \cdot) \zeta \in H_p^\gamma$. By changing of variables on spaces H_p^γ (see [Tri92]), $f \zeta(e^{-n} \cdot) \in H_p^n \subset \mathcal{C}^{\gamma-d/p}(\mathbb{R}_+^d) \subset C(\mathbb{R}_+^d)$ for every $n \in \mathbb{Z}$. Since $\sum \zeta^p(e^{-n} x) \geq 1$, by observing locally on each support of $\zeta(e^{-n} \cdot)$, we have $f \in C(\mathbb{R}_+^d)$.

For the inequality, notice that

$$\begin{aligned} \sup_{x \in \mathbb{R}_+^d} |(x^1)^{|i|+\theta/p} D^i u(x)|^p &\leq \sup_{x \in \mathbb{R}_+^d} \sum_{n=-\infty}^{\infty} \zeta^p(e^{-n} x) |(x^1)^{|i|+\theta/p} D^i u(x)|^p \\ &\leq \sum_{n=-\infty}^{\infty} e^{n\theta} \sup_{x \in \mathbb{R}_+^d} [(e^n x^1)^{|i|} D^i u(e^n x)] [(x^1)^{\theta/p} \zeta(x)]^p =: I. \end{aligned}$$

Here we use the Sobolev's embedding theorem, which is $H_p^\gamma \subset \mathcal{C}^{\gamma-d/p}(\mathbb{R}_+^d) \subset C(\mathbb{R}_+^d)$. In addition, by Theorem 2.0.1 and the induction (or see Corollary 2.0.3), the operator $M^{|i|} D^i$ is a bounded operator from $H_{p,\theta}^\gamma$ into $H_{p,\theta}^{\gamma-|i|}$. Finally, as $|i| \leq k$, $\gamma - |i| - d/p \geq \alpha > 0$ and thanks to Lemma 1.0.6, we finally have

$$I \leq \sum_{n=-\infty}^{\infty} e^{n\theta} \|M^{|i|} D^i u(e^n \cdot) M^{\theta/p} \zeta\|_{\gamma-|i|,p}^p \leq N \sum_{n=-\infty}^{\infty} e^{n\theta} \|M^{|i|} D^i u(e^n \cdot) \zeta\|_{\gamma-|i|,p}^p = N \|M^{|i|} D^i u\|_{\gamma-|i|,p,\theta}^p \leq N \|u\|_{\gamma,p,\theta}^p.$$

Next, assume that $\alpha < 1$. It is convenient to take ζ so that

$$\sum_{n=-\infty}^{\infty} \zeta(e^{-n} x) \equiv 1 \quad \text{on } \mathbb{R}_+^d.$$

Notice that for any function v

$$[v]_{\mathcal{C}^\alpha(\mathbb{R}_+^d)}^p = [v]_{C^\alpha(\mathbb{R}_+^d)}^p = \sup_{x,y \in \mathbb{R}_+^d} \frac{|u(x) - u(y)|^p}{|x - y|^{p\alpha}} = \sup_{x,y \in \mathbb{R}_+^d} \left| \sum_{n=-\infty}^{\infty} \frac{[v(x)\zeta(e^{-n}x) - v(y)\zeta(e^{-n}y)]}{|x - y|^\alpha} \right|^p. \quad (35)$$

Here if both $\zeta(e^{-n}x)$ and $\zeta(e^{-n}y)$ are nonzero, then

$$n \in \Gamma(x, y) := \{m : ae^m \leq x, y \leq be^m\}.$$

If we consider

$$\Gamma(x) = \{m \in \mathbb{Z} : ae^m \leq x \leq be^m\},$$

where $(a, b) \subset \mathbb{R}_+$ is an interval containing the support of ζ . Furthermore, if $n \notin \Gamma(x, y)$ and at least one of $\zeta(e^{-n}x)$ or $\zeta(e^{-n}y)$ is not zero, then $|x - y| \geq \epsilon e^n$, where ϵ is the distance from points a, b to the support of ζ . Assume that $\zeta(e^{-n}x) \neq 0$. Since $n \notin \Gamma(x, y)$, we have $e^{-n}x \in \text{supp} \zeta$ and $e^{-n}y \notin (a, b)$, thus

$$|e^{-n}x - e^{-n}y| \geq \epsilon.$$

Furthermore, notice that $\Gamma(x, y) = \Gamma(x) \cap \Gamma(y)$, where

$$\Gamma(x) = \{m \in \mathbb{Z} : \log x - \log b \leq m \leq \log x - \log a\}.$$

This gives $|\Gamma(x)| \leq \log(b/a) + 2$, and we have $|\Gamma(x, y)| \leq |\Gamma(x)| \leq \log(b/a) + 2$. Finally we have the following inequality:

$$\sum_{n=-\infty}^{\infty} e^{-n\alpha} \zeta(e^{-n}x) = \sum_{n: x^1 \in [ae^n, be^n]} e^{-n\alpha} \zeta(e^{-n}x) \leq \sum_{n: x^1 \in [ae^n, be^n]} b^{\alpha}(x^1)^{-\alpha} \zeta(e^{-n}x) \leq b^{\alpha}(x^1)^{-\alpha}.$$

Therefore, the expression under sup in eq. (35) is less than

$$\begin{aligned} N \left| \sum_{n \in \Gamma(x, y)} \frac{[v(x)\zeta(e^{-n}x) - v(y)\zeta(e^{-n}y)]}{|x - y|^{\alpha}} \right|^p + N|v(x)|^p \left| \sum_{n=-\infty}^{\infty} e^{-n\alpha} \zeta(e^{-n}x) \right|^p + N|v(y)|^p \left| \sum_{n=-\infty}^{\infty} e^{-n\alpha} \zeta(e^{-n}y) \right|^p \\ \leq N \sum_{n \in \Gamma(x, y)} \left| \frac{[v(x)\zeta(e^{-n}x) - v(y)\zeta(e^{-n}y)]}{|x - y|^{\alpha}} \right|^p + N|(x^1)^{-\alpha}v(x)|^p + N|(y^1)^{-\alpha}v(y)|^p \\ \leq N \sum_{n=-\infty}^{\infty} \left| \frac{[v(x)\zeta(e^{-n}x) - v(y)\zeta(e^{-n}y)]}{|x - y|^{\alpha}} \right|^p + N|(x^1)^{-\alpha}v(x)|^p + N|(y^1)^{-\alpha}v(y)|^p. \end{aligned}$$

Thus

$$\begin{aligned} [v]_{\mathcal{C}^{\alpha}(\mathbb{R}_+^d)}^p &\leq \sup_{x \in \mathbb{R}_+^d} N|(x^1)^{-\alpha}v(x)|^p + N \sup_{x, y \in \mathbb{R}_+^d} \sum_{n=-\infty}^{\infty} \left| \frac{[v(x)\zeta(e^{-n}x) - v(y)\zeta(e^{-n}y)]}{|x - y|^{\alpha}} \right|^p \\ &\leq \sup_{x \in \mathbb{R}_+^d} N|(x^1)^{-\alpha}v(x)|^p + N \sum_{n=-\infty}^{\infty} e^{-np\alpha} \sup_{x, y \in \mathbb{R}_+^d} \left| \frac{[v(e^n x)\zeta(x) - v(e^n y)\zeta(y)]}{|x - y|^{\alpha}} \right|^p \end{aligned} \quad (36)$$

Now we apply this inequality for $v = M^{k+\alpha+\theta/p} D^j u$. Then first of all, $(x^1)^{-\alpha}v(x) = M^{k+\theta/p} D^j u$, so by the previous result, $\|M^{k+\theta/p} D^j u\|_{\mathcal{C}^{\alpha}(\mathbb{R}_+^d)} \leq N\|u\|_{\gamma, p, \theta}$. Next,

$$v(e^n x) = (e^n x^1)^{(k+\alpha+\theta/p)n} (D^j u)(e^n x) = e^{n\theta+np\alpha} e^{nk} (x^1)^{k+\alpha+\theta/p} (D^j u)(e^n x),$$

$$\begin{aligned} v(e^n x)\zeta(x) - v(e^n y)\zeta(y) &= e^{n\theta+np\alpha} [e^{nk} (x^1)^{k+\alpha+\theta/p} (D^j u)(e^n x)\zeta(x) - e^{nk} (y^1)^{k+\alpha+\theta/p} (D^j u)(e^n y)\zeta(y)] \\ &= e^{n\theta+np\alpha} [(M^k D^j u)(e^n x)(M^{\alpha+\theta/p} \zeta)(x) - (M^k D^j u)(e^n y)(M^{\alpha+\theta/p} \zeta)(y)], \end{aligned}$$

and thus we have

$$\begin{aligned} e^{-np\alpha} \sup_{x, y \in \mathbb{R}_+^d} \left| \frac{[v(e^n x)\zeta(x) - v(e^n y)\zeta(y)]}{|x - y|^{\alpha}} \right|^p \\ = e^{n\theta} \sup_{x, y \in \mathbb{R}_+^d} \left| \frac{[(M^k D^j u)(e^n x)(M^{\alpha+\theta/p} \zeta)(x) - (M^k D^j u)(e^n y)(M^{\alpha+\theta/p} \zeta)(y)]}{|x - y|^{\alpha}} \right|^p \\ = e^{n\theta} [(M^k D^j u)(e^n \cdot)(M^{\alpha+\theta/p} \zeta)]_{\mathcal{C}^{\alpha}(\mathbb{R}_+^d)}^p. \end{aligned}$$

Finally, by the Sobolev's embedding theorem gives and remembering that $\gamma = \alpha + k + d/p$, therefore that

$$\begin{aligned} [M^{k+\alpha+\theta/p} D^j u]_{\mathcal{C}^{\alpha}(\mathbb{R}_+^d)}^p &\leq N\|u\|_{\gamma, p, \theta}^p + N \sum_{n=-\infty}^{\infty} e^{n\theta} [(M^k D^j u)(e^n \cdot)(M^{\alpha+\theta/p} \zeta)]_{\mathcal{C}^{\alpha}(\mathbb{R}_+^d)}^p \\ &\leq N\|u\|_{\gamma, p, \theta}^p + N \sum_{n=-\infty}^{\infty} e^{n\theta} \|(M^k D^j u)(e^n \cdot)(M^{\alpha+\theta/p} \zeta)\|_{\alpha+d/p, p}^p \\ &\leq N\|u\|_{\gamma, p, \theta}^p + N \sum_{n=-\infty}^{\infty} e^{n\theta} \|(M^k D^j u)(e^n \cdot)\zeta\|_{\alpha+d/p, p}^p \\ &\leq N\|u\|_{\gamma, p, \theta}^p + N\|M^k D^j u\|_{\alpha+d/p, p, \theta}^p \\ &\leq N\|u\|_{\gamma, p, \theta}^p + N\|u\|_{k+\alpha+d/p, p, \theta}^p \\ &= N\|u\|_{\gamma, p, \theta}^p. \end{aligned}$$

Now we prove the theorem for the remaining case $\alpha = 1$. Notice that the definition of a seminorm of $\mathcal{C}^1(\mathbb{R}_+^d)$ is the following.

$$\begin{aligned} [v]_{\mathcal{C}^1(\mathbb{R}_+^d)}^p &:= \sup_{x, y \in \mathbb{R}_+^d} \left| \frac{v(x) - 2v((x+y)/2) + v(y)}{|x-y|} \right|^p \\ &= \sup_{x, y \in \mathbb{R}_+^d} \left| \sum_{n=-\infty}^{\infty} \frac{|v(x)\zeta(e^{-n}x) - 2v((x+y)/2)\zeta(e^{-n}(x+y)/2) + v(y)\zeta(e^{-n}y)|}{|x-y|} \right|^p \end{aligned}$$

We should apply similar argument for this. Notice that open intervals are convex. Hence the expression under sup is less than

$$\begin{aligned} &\left| \sum_{n \in \Gamma(x, y)} \frac{|v(x)\zeta(e^{-n}x) - 2v((x+y)/2)\zeta(e^{-n}(x+y)/2) + v(y)\zeta(e^{-n}y)|}{|x-y|} \right|^p \\ &+ \left| \sum_{n \notin \Gamma(x, y)} \frac{|v(x)\zeta(e^{-n}x) - 2v((x+y)/2)\zeta(e^{-n}(x+y)/2) + v(y)\zeta(e^{-n}y)|}{|x-y|} \right|^p \\ &\leq N \sum_{n \in \Gamma(x, y)} \left| \frac{|v(x)\zeta(e^{-n}x) - 2v((x+y)/2)\zeta(e^{-n}(x+y)/2) + v(y)\zeta(e^{-n}y)|}{|x-y|} \right|^p \\ &\quad + N|v(x)|^p \left| \sum_{n=-\infty}^{\infty} e^{-n}\zeta(e^{-n}x) \right|^p + N|v(y)|^p \left| \sum_{n=-\infty}^{\infty} e^{-n}\zeta(e^{-n}y) \right|^p \\ &\quad + N|v((x+y)/2)|^p \left| \sum_{n=-\infty}^{\infty} e^{-n}\zeta(e^{-n}(x+y)/2) \right|^p \\ &\leq N \sum_{n=-\infty}^{\infty} \left| \frac{|v(x)\zeta(e^{-n}x) - 2v((x+y)/2)\zeta(e^{-n}(x+y)/2) + v(y)\zeta(e^{-n}y)|}{|x-y|} \right|^p \\ &\quad + N|(x^1)^{-1}v(x)|^p + N|(y^1)^{-1}v(y)|^p + N|((x+y)^1)^{-1}v((x+y)/2)|^p \\ &\leq N|(x^1)^{-1}v(x)|^p + N|(y^1)^{-1}v(y)|^p + N|((x+y)^1)^{-1}v((x+y)/2)|^p + N \sum_{n=-\infty}^{\infty} e^{-np} [v(e^n \cdot) \zeta]^p_{\mathcal{C}^1(\mathbb{R}_+^d)}. \end{aligned}$$

Therefore we find

$$[v]_{\mathcal{C}^1(\mathbb{R}_+^d)}^p \leq N \sup_{x \in \mathbb{R}_+^d} |(x^1)^{-1}v(x)|^p + N \sum_{n=-\infty}^{\infty} e^{-np} [v(e^n \cdot) \zeta]^p_{\mathcal{C}^1(\mathbb{R}_+^d)}.$$

Now we apply this inequality to $v = M^{k+1+\theta/p} D^j u$. Because of the first estimate in eq. (34), Sobolev's embedding theorem, Theorem 2.0.1, we can estimate $[M^{\gamma+\theta/p} D^j u]_{\mathcal{C}^1(\mathbb{R}_+^d)}$ similarly as in above (also remember that $\gamma = k + 1 + d/p$ in this case) to obtain the final result for $\alpha = 1$. The theorem is proved.

Theorem 3.0.2 Assume $\gamma p > d$ and let $\mu < \gamma$ and $q (> p)$ be such that

$$\gamma - \frac{d}{p} = \mu - \frac{d}{q}.$$

Denote $\theta_1 = \theta q/p$ (so that $\theta_1/q = \theta/p$). Then for any $u \in H_{p, \theta}^\gamma$ we have

$$u \in H_{q, \theta_1}^\mu, \quad \|u\|_{\mu, q, \theta_1} \leq N(d, \gamma, \mu, p, \theta) \|u\|_{\gamma, p, \theta}.$$

proof. This is an easy consequence of Sobolev's embedding theorems. Indeed, by remembering that $\ell_p(A) \subset \ell_q(A)$ for $0 < p < q \leq \infty$ and $\|f\|_q \leq \|f\|_p$,

$$\|u\|_{\mu, q, \theta_1}^q = \sum_{n=-\infty}^{\infty} e^{n\theta_1} \|u(e^n \cdot) \zeta\|_{\mu, q}^q$$

$$\begin{aligned}
 &\leq \left(\sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot)\zeta\|_{\mu,q}^p \right)^{q/p} \\
 &\leq N \left(\sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^n \cdot)\zeta\|_{\mu,p}^p \right)^{q/p} \\
 &= N \|u\|_{\gamma,p,\theta}^q.
 \end{aligned}$$

By taking $\mu = 0$ we get the following corollary.

Corollary 3.0.3 Let $0 < \gamma p < d$ and $u \in H_{p,\theta}^\gamma$. Then $u \in L_{p',\theta'}$, where $p' = pd/(d - \gamma p)$ and $\theta' = \theta d/(d - \gamma p)$. Moreover,

$$\int_{\mathbb{R}_+^d} (x^1)^{\theta'-d} |u(x)|^{p'} dx \leq N(p, d, \gamma, \theta) \|u\|_{\gamma,p,\theta}^{p'}.$$

4 Elliptic Equations in $\mathbb{R}_+^d$

In the remaining part of the paper we assume that^(xvi)

$$d - 1 < \theta < d - 1 + p \quad (37)$$

(and as always, $p \in (1, \infty)$). In this section we are going to consider the equation

$$\Delta u - \lambda u = D_i f^i, \quad (38)$$

where λ is a nonnegative constant.

Given $f^i \in H_{p,\theta}^{\gamma-1}$, we look for solutions u on some appropriate space. Under these circumstances, as we know from Remark 2.0.2 and 2.0.16, equation eq. (38) is equivalent to the equation

$$\Delta u - \lambda u = M^{-1} f, \quad (39)$$

where $f \in H_{p,\theta}^{\gamma-2}$.

If $(f^1, \dots, f^d)$ for $f^i \in H_{p,\theta}^\gamma$ is given, put $f = MD_i f^i$. Then $f \in H_{p,\theta}^{\gamma-2}$ by Remark 2.0.2. Conversely, of $f \in H_{p,\theta}^{\gamma-2}$ is fixed, consider $f^i = Q_i f$ where Q_i is come from Definition 2.0.15. Then by Remark 2.0.16, one have $M^{-1} f = \sum_i D_i f^i$.

Here is the main result of this section.

Theorem 4.0.1 For any $f^i \in H_{p,\theta}^{\gamma-1}$ in the space $MH_{p,\theta}^\gamma$ there exists a unique solution u of equation eq. (38). For this solution we have

$$\lambda \|Mu\|_{\gamma-2,p,\theta} + \|M^{-1}u\|_{\gamma,p,\theta} \leq N(d, \gamma, p, \theta) \sum_i \|f^i\|_{\gamma-1,p,\theta}. \quad (40)$$

Remark 4.0.2 For equation eq. (39) we have the same result assuming that $f \in H_{p,\theta}^{\gamma-2}$ and replacing the sum on the right in eq. (40) with $\|f\|_{\gamma-2,p,\theta}$.

Remark 4.0.3 If $\theta \geq p + d - 1$ or $\theta \leq d - 1$, then the statement of Theorem 4.0.1 is false in general.

It suffices to show that there is no constant N depends only on d, γ, p , and θ such that

$$\|M^{-1}u\|_{0,p,\theta} \leq N \|M\Delta u\|_{0,p,\theta}. \quad (41)$$

This is because by considering $\Delta u = M^{-1}f$ with $\gamma = 2$, eq. (40) becomes

^(xvi) By the way, this condition gives $(x^1)^{\theta-d} \in A_p$, where A_p is the Muckenhoupt class.

$$\|M^{-1}u\|_{2,p,\theta} \leq N\|f\|_{0,p,\theta} = N\|M\Delta u\|_{0,p,\theta}.$$

Since it is clear that $\|\cdot\|_{0,p,\theta} \leq \|\cdot\|_{2,p,\theta}$, we automatically get eq. (41) if Theorem 4.0.1 is true. In addition, we first assume $d \geq 2$.

Consider any nonnegative $g \in C_0^\infty(\mathbb{R}^{d-1})$ with $g(x) = 1$ on $|x| \leq 1$, and define $\psi(x) = g * P(x^1, \cdot)(x')$, where $P(t, x)$, $t > 0$, $x \in \mathbb{R}^{d-1}$ is a Poisson kernel. Finally, take a smooth function ϕ on $\mathbb{R}$ such that $0 \leq \phi \leq 1$, $\phi(x) = 1$ for $x \leq 1$, and $\phi(x) = 0$ for $x \geq 2$.

Assume that $\theta \leq d - 1$, and consider $u(x) = \phi(x^1)\psi(x)$. Then since ψ is harmonic on $\mathbb{R}_+^d$, we have

$$\Delta u(x) = \phi''(x)\psi(x) + 2\phi'(x)\psi_{x^1}(x).$$

This, Fubini's theorem, $\phi'(x) = \phi''(x) = 0$ outside of $(1, 2)$, and the Young's inequality^(xvii) shows that

$$\begin{aligned} \|M\Delta u\|_{0,p,\theta}^p &\leq N \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+p} |\phi''(x)\psi(x)|^p dx + N \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+p} |\phi'(x)\psi_{x^1}(x)|^p dx \\ &\leq N \int_1^2 \int_{\mathbb{R}^{d-1}} (x^1)^{\theta-d+p} |\phi''(x)\psi(x)|^p dx + N \int_1^2 \int_{\mathbb{R}^{d-1}} (x^1)^{\theta-d+p} |\phi'(x)\psi_{x^1}(x)|^p dx \\ &= N \int_1^2 (x^1)^{\theta-d+p} |\phi''(x^1)|^p \|\psi(x^1, \cdot)\|_p^p dx^1 + N \int_1^2 (x^1)^{\theta-d+p} |\phi'(x^1)|^p \|\psi_{x^1}(x^1, \cdot)\|_p^p dx^1 \\ &\leq N \|g\|_p^p \|\phi''\|_\infty \int_1^2 (x^1)^{\theta-d+p} dx^1 + N \|g\|_p^p \|\phi'\|_\infty \int_1^2 (x^1)^{\theta-d} dx^1 \\ &< \infty. \end{aligned}$$

On the other hand, for any $|x'| \leq 1$, since

$$|x' - y| \leq 2 \quad \forall |y| \leq 1,$$

we find

$$\begin{aligned} \psi(x) &= \int_{\mathbb{R}^{d-1}} g(y) P(x^1, x' - y) dy \geq \int_{|y| \leq 1} g(y) P(x^1, x' - y) dy \\ &= \int_{|y| \leq 1} P(x^1, x' - y) dy \geq N(d) \int_{|y| \leq 1} \frac{x^1}{((x^1)^2 + 4)^{d/2}} dy = N(d) \frac{x^1}{((x^1)^2 + 4)^{d/2}}. \end{aligned}$$

Hence,

$$\begin{aligned} \|M^{-1}u\|_{0,p,\theta}^p &= \int_{\mathbb{R}_+^d} (x^1)^{\theta-d-p} |\phi(x)\psi(x)|^p dx \\ &= \int_0^2 \int_{\mathbb{R}^{d-1}} (x^1)^{\theta-d-p} |\phi(x)\psi(x)|^p dx \\ &\geq \int_0^1 \int_{\mathbb{R}^{d-1}} (x^1)^{\theta-d-p} |\phi(x)\psi(x)|^p dx \\ &= \int_0^1 \int_{\mathbb{R}^{d-1}} (x^1)^{\theta-d-p} |\psi(x)|^p dx \\ &\geq \int_0^1 \int_{|x'| \leq 1} (x^1)^{\theta-d-p} |\psi(x)|^p dx \\ &\geq N(d) \int_0^1 \frac{(x^1)^{\theta-d}}{((x^1)^2 + 4)^{dp/2}} dx \\ &\geq N(d, p) \int_0^1 \frac{(x^1)^{\theta-d}}{((x^1)^{2p} \vee 2^{2p})} dx \\ &= N(d, p) \int_0^1 (x^1)^{\theta-d} dx \end{aligned}$$

^(xvii) See the appendix for the property of Poisson kernel.

$$= \infty$$

because $\theta - d \leq -1$. This proves that eq. (41) fails to hold if $d \geq 2$ and $\theta \leq d - 1$.

Now we are going to observe the case when $\theta \geq p - d + 1$ (also remember $d \geq 2$). Define $g_n(x) = g(x/n)$, $\psi_n(x) = g_n * P(x^1, \cdot)(x')$. Since

$$P(t, nx) = \frac{2t}{\omega_d(|nx|^2 + t^2)^{d/2}} = \frac{2n(t/n)}{\omega_d(|x|^2 + (t/n)^2)^{d/2} n^d} = n^{1-d} P(t/n, x),$$

one can show that

$$\psi_n(x) = n^{d-1} \int_{\mathbb{R}^{d-1}} g(y) P(x^1, x' - ny) dy = n^{d-1} (g * P(x^1, n \cdot))(x') = \psi(x/n), \quad \psi_{nx^1}(x) = n^{-1} \psi_{x^1}(x/n).$$

Thus, by Young's inequality,

$$\|\psi_n(x^1, \cdot)\|_p \leq N(d, p) n^{(d-1)/p} \|g\|_1, \quad \|\psi_{nx^1}(x^1, \cdot)\|_p \leq N(d, p) n^{(d-1)/p} (x^1)^{-1} \|g\|_1.$$

Furthermore, since $\psi(x) \rightarrow g(x')$ in L_p as $x^1 \downarrow 0$, there exists $\delta > 0$ such that $\|\psi(x^1, \cdot)\|_p \geq \|g\|_p/2$ for every $|x^1| \leq \delta$. We now only consider large n , meaning that n should satisfy $n\delta > 2$.

Now define $v_n(x) = (1 - \phi(x^1))\psi_n(x) =: \bar{\phi}(x^1)\psi_n(x)$. Since $\bar{\phi}'$ and $\bar{\phi}''$ vanishes outside of $(1, 2)$, and ψ_n is a harmonic function on $\mathbb{R}_+^d$, we have

$$\begin{aligned} \|M\Delta v_n\|_{0,p,\theta}^p &= \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+p} |\bar{\phi}''\psi_n + 2\bar{\phi}'\psi_{nx^1}|^p(x) dx \\ &\leq N \int_1^2 (x^1)^{\theta-d+p} \|\psi_n(x^1, \cdot)\|_p^p dx^1 + N \int_1^2 (x^1)^{\theta-d+p} \|\psi_{nx^1}(x^1, \cdot)\|_p^p dx^1 \\ &\leq N n^{d-1} \|g\|_1^p \int_1^2 (x^1)^{\theta-d+p} + (x^1)^{\theta-d} dx^1 \\ &= N(d, p, \theta, g) n^{d-1}. \end{aligned}$$

On the other hand, by our chosen n , we find

$$\begin{aligned} \|M^{-1}v_n\|_{0,p,\theta}^p &= \int_{\mathbb{R}_+^d} (x^1)^{\theta-d-p} |\bar{\phi}\psi_n|^p(x) dx \\ &= \int_2^\infty (x^1)^{\theta-d-p} \|\psi_n(x^1, \cdot)\|_p^p dx^1 \\ &= \int_2^\infty (x^1)^{\theta-d-p} \|\psi(x^1/n, \cdot/n)\|_p^p dx^1 \\ &= n^{d-1} \int_2^\infty (x^1)^{\theta-d-p} \|\psi(x^1/n, \cdot)\|_p^p dx^1 \\ &\geq n^{d-1} \int_2^{n\delta} (x^1)^{\theta-d-p} \|\psi(x^1/n, \cdot)\|_p^p dx^1 \\ &\geq 2^{-p} n^{d-1} \|g\|_p^p \int_2^{n\delta} (x^1)^{\theta-d-p} dx^1. \end{aligned}$$

Indeed for the last inequality, $x^1 \in [2, n\delta]$ implies that $|x^1/n| \leq \delta$, thus $\|\psi(x^1/n, \cdot)\|_p \geq \|g\|_p/2$.

If eq. (41) were true, we should have

$$2^{-p} n^{d-1} \|g\|_p^p \int_2^{n\delta} (x^1)^{\theta-d-p} dx^1 \leq \|M^{-1}v_n\|_{0,p,\theta}^p \leq N \|M\Delta v_n\|_{0,p,\theta}^p \leq N n^{d-1},$$

where N is independent of n . By our chosen g , $\|g\|_p > 0$, thus

$$\int_2^{n\delta} (x^1)^{\theta-d-p} dx^1 \leq N < \infty$$

where N is a constant independent of n . However as we assumed that $\theta \geq p + d - 1$, letting $n \rightarrow \infty$ gives the left hand side integral tends to infinity, which is a contradiction.

For the case $d = 1$, counterexamples are much simpler. If $\theta \leq 0$, then just consider ϕ as we discussed in above. Indeed, it is easy to show that $\|M\Delta\phi\|_{0,p,\theta} < \infty$ because $\Delta\phi = \phi''$ and it is supported in $(0, 3)$. In addition, $\|M^{-1}\phi\|_{0,p,\theta}^p \geq \int_0^1 x^{\theta-1-p} dx = \infty$.

If $\theta \geq p$, then consider $v(x) = x(1 - \phi(x))$. By the similar way, $\|M\Delta v\|_{0,p,\theta} < \infty$ because $\Delta v = v''$ and it is also supported in $(0, 3)$. Furthermore, $\|M^{-1}u\|_{0,p,\theta}^p \geq \int_2^\infty x^{\theta-1} dx = \infty$.

Remark 4.0.4 Lemma 1.0.11 bearing on change of coordinates shows that Theorem 4.0.1 also holds for the equation

$$a^{ij}u_{x^i x^j} - \lambda u = D_i f^i, \quad (42)$$

where $f^i \in H_{p,\theta}^{\gamma-1}$ and a^{ij} is a constant symmetric matrix satisfying

$$\delta_0 |\xi|^2 \leq a^{ij} \xi^i \xi^j \leq \delta_0^{-1} |\xi|^2 \quad \forall \xi \in \mathbb{R}^d, \quad (43)$$

with a constant $\delta_0 > 0$. In this situation the constant N in eq. (40) also depends on δ_0 . As above, eq. (42) is equivalent to the equation

$$a^{ij}u_{x^i x^j} - \lambda u = M^{-1}f,$$

where $f \in H_{p,\theta}^{\gamma-2}$.

Remark 4.0.5 If $\lambda = 0$, then the assertion of Theorem 4.0.1 follows from Theorem 2.0.17. Indeed, after substituting $u = Mv$, equation eq. (39) becomes $\mathcal{L}_{2,0}v = f$.

Remark 4.0.6 Observe that, if $\lambda > 0$, then $(\Delta - \lambda)MH_{p,\theta}^\gamma \neq M^{-1}H_{p,\theta}^{\gamma-2}$ because $\Delta MH_{p,\theta}^\gamma = M^{-1}H_{p,\theta}^{\gamma-2}$ (Remark 4.0.5) and $MH_{p,\theta}^{\gamma-1} \not\subseteq M^{-1}H_{p,\theta}^{\gamma-2}$. On the other hand, as we know

$$\begin{aligned} \|M(\Delta u - \lambda u)\|_{\gamma-2,p,\theta} &\leq \|MD_i D_i u\|_{\gamma-2,p,\theta} + \lambda \|Mu\|_{\gamma-2,p,\theta} \\ &\leq N(\|D_i u\|_{\gamma-1,p,\theta} + \|Mu\|_{\gamma-2,p,\theta}) \\ &\leq N(\|M^{-1}u\|_{\gamma,p,\theta} + \|Mu\|_{\gamma-2,p,\theta}) \\ &= N\|u\|_{MH_{p,\theta}^\gamma \cap M^{-1}H_{p,\theta}^{\gamma-2}}. \end{aligned}$$

Thus, for $\lambda > 0$, Theorem 4.0.1 implies that the operator $\Delta - \lambda$ is a bounded 1-1 operator from $MH_{p,\theta}^\gamma \cap M^{-1}H_{p,\theta}^{\gamma-2}$ onto $M^{-1}H_{p,\theta}^{\gamma-2}$ and its inverse is bounded.

The proof of this theorem is based on two lemmas.

Lemma 4.0.7 Let $\theta_1 \in \mathbb{R}$, $\mu \leq \gamma$, and let $u \in MH_{p,\theta_1}^\mu \cap MH_{p,\theta}^\gamma$ ^(xviii) be a solution of eq. (38) with $f^i \in H_{p,\theta}^{\gamma-1}$. Then $u \in MH_{p,\theta}^\gamma$ and

$$\lambda \|Mu\|_{\gamma-2,p,\theta} + \|M^{-1}u\|_{\gamma,p,\theta} \leq N \left(\sum_i \|f^i\|_{\gamma-1,p,\theta} + \|M^{-1}u\|_{\mu,p,\theta} \right), \quad (44)$$

where $N = N(d, \gamma, p, \theta)$. ^(xix)

proof. Since $\|\cdot\|_{\nu,p,\theta} \leq \|\cdot\|_{\mu,p,\theta}$ for $\nu \leq \mu$, it suffices to prove eq. (44) with some $\nu \leq \mu$ in place of μ . This shows that we may assume that $\gamma - \mu$ is an integer. Furthermore, inductively, for every positive integer n , we have

$$\lambda \|Mu\|_{\gamma-2,p,\theta} + \|M^{-1}u\|_{\gamma,p,\theta} \leq N \left(\sum_i \|f^i\|_{\gamma-1,p,\theta} + \|M^{-1}u\|_{\gamma-1,p,\theta} \right)$$

^(xviii) u can be a solution to eq. (38) on the space MH_{p,θ_1}^γ for different $\theta_1 \neq \theta$.

^(xix) Again, one can replace $\sum_i \|f^i\|_{\gamma-1,p,\theta}^p$ into $\|f\|_{\gamma-2,p,\theta}^p$ for $f \in H_{p,\theta}^{\gamma-2}$.

$$\begin{aligned}
 &\leq N \left(\sum_i \|f^i\|_{\gamma-1,p,\theta} + \sum_i \|f^i\|_{\gamma-2,p,\theta} + \|M^{-1}u\|_{\gamma-2,p,\theta} \right) \\
 &\leq \dots \\
 &\leq N \left(\sum_i \|f^i\|_{\gamma-1,p,\theta} + \dots + \sum_i \|f^i\|_{\gamma-n,p,\theta} + \|M^{-1}u\|_{\gamma-n,p,\theta} \right) \\
 &\leq N \left(\sum_i \|f^i\|_{\gamma-1,p,\theta} + \|M^{-1}u\|_{\gamma-n,p,\theta} \right).
 \end{aligned}$$

Because of such process, we reduce the general situation to the one with $\mu = \gamma - 1$.

Take a function $\zeta \in C_0^\infty(\mathbb{R}_+)$ satisfying eq. (1). Obviously, the functions $u_n(x) := u(e^n x)$ satisfy the equation

$$\Delta u_n - e^{2n} \lambda u_n = e^{2n} f_n,$$

where $f_n(x) = (D_i f^i)(e^n x)$. In addition, notice that

$$\Delta(\zeta u_n) = \zeta'' u_n + \zeta(\Delta u_n) + 2\zeta' u_{nx^1},$$

hence

$$\Delta(\zeta u_n) - e^{2n} \lambda \zeta u_n = \bar{f}_n, \quad (45)$$

where $\bar{f}_n = \zeta e^{2n} f_n - 2\zeta' u_{nx^1} - u_n \zeta''$. Remember that $\eta u_n = H_p^\mu = H_p^{\gamma-1}$ for any $\eta \in C_0^\infty(\mathbb{R}_+)$. It follows that $\bar{f}_n \in H_p^{\gamma-2}$. Also notice that eq. (45) holds in the entire space $\mathbb{R}^d$. We now prove that $\zeta u_n \in H_p^\gamma$ and

$$\lambda e^{2n} \|\zeta u_n\|_{\gamma-2,p} + \|(\zeta u_n)_{xx}\|_{\gamma-2,p} \leq N \|\bar{f}_n\|_{\gamma-2,p}. \quad (46)$$

By the classical result (see, for instance, Section 9.4 of [GT01] or Theorem 4.3.7 of [Kry08]), it follows that $\|v_{xx}\|_p \leq N \|\Delta v\|_p$ for $v \in H_p^2$.^(xx) Replacing v with $(1 - \Delta)^{(\gamma-2)/2} v$ we get

$$\begin{aligned}
 \|v_{xx}\|_{\gamma-2,p} &= \|(1 - \Delta)^{(\gamma-2)/2} \Delta^2 v\|_p \\
 &= \|D^2 (1 - \Delta)^{(\gamma-2)/2} v\|_p \\
 &\leq N \|\Delta (1 - \Delta)^{(\gamma-2)/2} v\|_p \\
 &\leq N \|(\Delta - \lambda)(1 - \Delta)^{(\gamma-2)/2} v\|_p + N \lambda \|(1 - \Delta)^{(\gamma-2)/2} v\|_p \\
 &= N \|\Delta v - \lambda v\|_{\gamma-2,p} + N \lambda \|v\|_{\gamma-2,p}.
 \end{aligned}$$

To get eq. (46), it only remains to notice that $v = (\Delta - \lambda)^{-1}[(\Delta - \lambda)v]$, and $\|(\Delta - \lambda)^{-1}\|_{H_p^\gamma \rightarrow H_p^{\gamma+2}} = \lambda^{-1}$.^(xxi) Put $e^{2n} \lambda$ instead of λ , therefore, gives

$$\begin{aligned}
 \lambda e^{2n} \|\zeta u_n\|_{\gamma-2,p} + \|(\zeta u_n)_{xx}\|_{\gamma-2,p} &\leq N \|\Delta(\zeta u_n) - \lambda e^{2n} \zeta u_n\|_{\gamma-2,p} + N \lambda e^{2n} \|\zeta u_n\|_{\gamma-2,p} \\
 &\leq N \|\Delta(\zeta u_n) - \lambda e^{2n} \zeta u_n\|_{\gamma-2,p} \\
 &= N \|\bar{f}_n\|_{\gamma-2,p}.
 \end{aligned}$$

Furthermore, by Remark 1.0.15, we have

$$N \|(\zeta u_n)_{xx}\|_{\gamma-2,p} \geq \|(\zeta u_n)_x\|_{\gamma-1,p} \geq \|\zeta u_{nx}\|_{\gamma-1,p} - \|u_n \zeta'\|_{\gamma-1,p}.$$

Also we use identities like

$$u_{nx}(x) = e^n u_x(e^n x), \quad e^{-n} u_n \zeta_x(x) = (M^{-1}u)(e^n x) M \zeta'(x),$$

and from eq. (46) we get

$$\begin{aligned}
 &\lambda \|(Mu)(e^n \cdot) M^{-1} \zeta\|_{\gamma-2,p} + \|u_x(e^n \cdot) \zeta\|_{\gamma-1,p} \\
 &= \lambda \|e^n u_n \zeta\|_{\gamma-2,p} + \|e^{-n} u_{nx} \zeta\|_{\gamma-1,p} \\
 &\leq \lambda \|e^n u_n \zeta\|_{\gamma-2,p} + \|e^{-n} u_n \zeta'\|_{\gamma-1,p} + N \|e^{-n} (\zeta u_n)_{xx}\|_{\gamma-2,p}
 \end{aligned}$$

^(xx) Actually, [GT01] only proves those with compactly supported. However we can drop that condition because $C_0^\infty(\mathbb{R}^d)$ is dense in H_p^2 .

^(xxi) The proof of which is presented in appendix.

$$\begin{aligned}
 &\leq N \|e^{-n} \tilde{f}_n\|_{\gamma-2,p} + \|e^{-n} u_n \zeta'\|_{\gamma-1,p} \\
 &\leq N (\|e^{-n} u_n \zeta'\|_{\gamma-1,p} + \|e^n f_n \zeta\|_{\gamma-2,p} + \|e^{-n} u_{nx^1} \zeta'\|_{\gamma-2,p} + \|e^{-n} u_n \zeta''\|_{\gamma-2,p}) \\
 &= N (\|(M^{-1}u)(e^n \cdot) M \zeta'\|_{\gamma-1,p} + \|(MD_i f^i)(e^n \cdot) M^{-1} \zeta\|_{\gamma-2,p} \\
 &\quad + \|u_{x^1}(e^n \cdot) \zeta'\|_{\gamma-2,p} + \|(M^{-1}u)(e^n \cdot) M \zeta''\|_{\gamma-2,p}).
 \end{aligned}$$

After taking this inequality to the p th power then multiplying through by $e^{n\theta}$, summing up over n , and using Lemma 1.0.6, we conclude first that

$$\lambda^p \sum_{n=-\infty}^{\infty} e^{n\theta} \|(Mu)(e^n \cdot) M^{-1} \zeta\|_{\gamma-2,p}^p + \|u_x\|_{\gamma-1,p,\theta}^p \leq N (\|MD_i f^i\|_{\gamma-2,p,\theta}^p + \|u_x\|_{\gamma-2,p,\theta}^p + \|M^{-1}u\|_{\gamma-1,p,\theta}^p).$$

Observe that in order to prove Lemma 1.0.6, the sufficient condition for ζ is

$$\sum_{n=-\infty}^{\infty} \zeta^p(e^{x-n}) > 0.$$

Notice that $M^{-1}\zeta$ also satisfies that condition, so we can apply Lemma 1.0.6 for ζ and $M^{-1}\zeta$ in place of η and ζ to obtain^(xxii)

$$\lambda \|Mu\|_{\gamma-2,p,\theta} + \|u_x\|_{\gamma-1,p,\theta} \leq N (\|MD_i f^i\|_{\gamma-2,p,\theta} + \|u_x\|_{\gamma-2,p,\theta} + \|M^{-1}u\|_{\gamma-1,p,\theta}).$$

It only remains to remember that MD_i are bounded operators from $H_{p,\theta}^{\gamma-1}$ to $H_{p,\theta}^{\gamma-2}$ and that the norms $\|u_x\|_{\gamma,p,\theta}$ and $\|M^{-1}u\|_{\gamma+1,p,\theta}$ are equivalent. The lemma is proved.

Lemma 4.0.8 Theorem 4.0.1 holds if $\gamma = 2$.

proof. By Remark 4.0.5, we may concentrate on the case $\lambda > 0$. First we prove the a priori estimate: for any $u \in MH_{p,\theta}^2$ we have

$$\lambda \|Mu\|_{0,p,\theta} + \|M^{-1}u\|_{2,p,\theta} \leq N(d, p, \theta) \|M(\Delta u - \lambda u)\|_{0,p,\theta}. \quad (47)$$

Of course, we only need to prove eq. (47) assuming that $\Delta u - \lambda u = M^{-1}f$ with $f \in L_{p,\theta}$ (otherwise the right-hand side is infinite).

In this case, by Lemma 4.0.7 we have $u \in MH_{p,\theta}^2 \cap M^{-1}L_{p,\theta}$. Observe that, by Corollary 1.0.8, we have $MH_{p,\theta}^2 = H_{p,\theta-p}^2$ and (obviously) $M^{-1}L_{p,\theta} = L_{p,\theta+p}$. By Remark 1.0.23 the set $C_0^\infty(\mathbb{R}_+^d)$ is everywhere dense in $MH_{p,\theta}^2 \cap M^{-1}L_{p,\theta}$. Since $\Delta - \lambda$ is bounded operator from the latter space to $M^{-1}L_{p,\theta}$ (Remark 4.0.6), we conclude that we only need to prove eq. (47) for $u \in C_0^\infty(\mathbb{R}_+^d)$.

In this case consider $v := M^{-1}u \in C_0^\infty(\mathbb{R}_+^d)$. Since

$$\begin{aligned}
 D_{x^i} v &= M^{-1}u_{x^i} - \delta^{1i} M^{-2}u; \\
 D_{x^i x^j} v &= M^{-1}u_{x^i x^j} - \delta^{1i} M^{-2}u_{x^j} - \delta^{1j} M^{-2}u_{x^i} + 2\delta^{1i} \delta^{1j} M^{-3}u,
 \end{aligned}$$

we have

$$\mathcal{L}_{2,0} v = (M^2 \Delta + 2MD_1)v = M\Delta u, \quad \mathcal{L}_{2,0} v - \lambda M^2 v = f.$$

In the same way as in the proof of Theorem 2.0.17 we get $\|v\|_{0,p,\theta} \leq \kappa^{-1} \|f\|_{0,p,\theta}$ with $\kappa := p^{-2}(\theta-d+1)(p+d-1-\theta)$. Indeed, similarly one can obtain with $w := |v|^{p/2}$ and

$$\frac{(\theta-d+1)^2}{4} \|v\|_{0,p,\theta}^p \leq \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |w_{x^1}|^2 dx,$$

we have

$$\|f\|_{0,p,\theta} \|v\|_{0,p,\theta}^{p-1} \geq \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |v|^{p-1} |f| dx$$

^(xxii) Additionally, we used

$$\sum_1^n a_j^p \leq \left(\sum_1^n a_j \right)^p \leq n^p \sum_1^n a_j^p$$

where $p \in [1, \infty)$ and $a_j \geq 0$.

$$\begin{aligned}
 &\geq - \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |v|^{p-2} v f dx \\
 &\geq 4p^{-2}(p-1) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |w_x|^2 dx \\
 &\quad + p^{-1}(\theta-d+1)(d-\theta) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |v|^p dx + \lambda \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |v|^p dx \\
 &\geq 4p^{-2}(p-1) \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2} |w_x|^2 dx + p^{-1}(\theta-d+1)(d-\theta) \|v\|_{0,p,\theta}^p \\
 &\geq p^{-1}(\theta-d+1)(p^{-1}(p-1)(\theta-d+1) + d-\theta) \|v\|_{0,p,\theta}^p \\
 &= \kappa \|v\|_{0,p,\theta}^p.
 \end{aligned}$$

Upon remembering that $v = M^{-1}u$ and using Lemma 4.0.7 with $\mu = 0$, we arrive at eq. (47).

Now we are going to prove the existence and the uniqueness of eq. (39). Of course, estimate eq. (47) implies the uniqueness. Owing to eq. (47), to prove the existence, we may assume that in eq. (39) we have $f \in C_0^\infty(\mathbb{R}_+^d)$. Indeed, for $f \in L_{p,\theta}$, take a sequence $(f_n)_n$ in $C_0^\infty(\mathbb{R}_+^d)$ such that $f_n \rightarrow f$. Take $u_n \in MH_{p,\theta}^2 \cap M^{-1}L_{p,\theta}$ solves

$$\Delta u_n - \lambda u_n = M^{-1}f_n.$$

Then eq. (47) implies that $(u_n)_n$ is Cauchy sequence in $MH_{p,\theta}^2 \cap M^{-1}L_{p,\theta}$, take the limit point $u \in MH_{p,\theta}^2 \cap M^{-1}L_{p,\theta}$. By Remark 4.0.6, we therefore have

$$M^{-1}f = \lim_{n \rightarrow \infty} M^{-1}f_n = \lim_{n \rightarrow \infty} (\Delta - \lambda)u_n = (\Delta - \lambda)u,$$

which proves the existence for the general case.

If $f \in C_0^\infty(\mathbb{R}_+^d)$, continue f to $\mathbb{R}^d$ to be an odd function with respect to x^1 and solve eq. (39) in $\mathbb{R}^d$ to get a bounded infinitely differentiable solution u . By uniqueness for equation eq. (39) in $\mathbb{R}^d$,^(xxiii) the function u is odd with respect to x^1 . Hence $u(0, x') = 0$.

Also u itself and all derivatives of u are known to be bounded and tends to zero as $|x| \rightarrow \infty$ exponentially fast ($\lambda > 0$).^(xxiv)

Now we are going to prove that $M^{-1}u \in H_{p,\theta}^2$. To prove this, we need a following inequality which can be proved by the fundamental theorem of calculus:

$$\left| \frac{u(x)}{x^1} \right| \leq \int_0^1 |u_{x^1}(tx^1, x')| dt \leq \sup_{0 \leq y \leq 1} |u_x(y, x')| \quad \forall x^1 \in (0, 1).$$

Notice that $2|x| = (|x^1|^2 + |x'|^2)^{1/2} \geq |x^1| + |x'|$. This and the calculation as in section 8.4 gives

$$|u_x(y, x')| + |u_{xx}(y, x')| \leq Ne^{-\mu|(y, x')|} \leq Ne^{-\mu(|y|+|x'|)/2}, \quad \sup_{x^1 \in (0,1)} \left| \frac{u(x)}{x^1} \right| \leq Ne^{-\mu|x'|/2}.$$

Therefore, since $d-1 < \theta < p+d-1$, and $\theta-d+p = (\theta-d+1) + (p-1) > p-1 > 0$, we obtain

$$\begin{aligned}
 \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |u(x)/x^1|^p dx &= \int_{x^1 < 1} (x^1)^{\theta-d} |u(x)/x^1|^p dx + \int_{x^1 \geq 1} (x^1)^{\theta-d} |u(x)/x^1|^p dx \\
 &= \int_{x^1 < 1} (x^1)^{\theta-d} |u(x)/x^1|^p dx + \int_{x^1 \geq 1} (x^1)^{\theta-d-p} |u(x)|^p dx \\
 &\leq N \int_0^1 \int_{\mathbb{R}^{d-1}} t^{\theta-d} e^{-\mu|y|/2} dy dt + N \int_1^\infty \int_{\mathbb{R}^{d-1}} t^{\theta-d-p} e^{-\mu(t+|y|)/2} dy dt \\
 &< \infty, \\
 \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |u_x(x)|^p dx &\leq N \int_0^1 \int_{\mathbb{R}^{d-1}} t^{\theta-d} e^{-\mu|y|/2} dy dt + N \int_1^\infty \int_{\mathbb{R}^{d-1}} t^{\theta-d-p} e^{-\mu(t+|y|)/2} dy dt < \infty.
 \end{aligned}$$

(xxiii) As $\lambda > 0$, we know that $(\lambda - \Delta)^{-1}$ is well-defined on $\mathcal{S}$, a collection of all Schwartz functions.

(xxiv) We will prove this in section 8.4. Beware that we cannot argue that u is a Schwartz function. Although every Schwartz function decreases to zero at infinity, its speed is fast than all polynomials. However this does not mean that the speed is exponential.

$$\int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |x^1 u_{xx}(x)|^p dx \leq N \int_0^\infty \int_{\mathbb{R}^{d-1}} t^{\theta-d+p} e^{-\mu(t+|y|)/2} dy dt < \infty.$$

This proves that $M^{-1}u$, u_x , $Mu_{xx} \in L_{p,\theta}$. Now we are going to use Corollary 2.0.3. Since

$$MD_i(M^{-1}u) = u_{xi} - \delta^{1i}M^{-1}u \in L_{p,\theta}, \quad M^2D_iD_j(M^{-1}u) = MD_{ij}u - \delta^{1i}D_ju - \delta^{1j}D_iu + 2\delta^{1i}\delta^{1j}M^{-1}u \in L_{p,\theta},$$

we, therefore, prove that $M^{-1}u \in H_{p,\theta}^2$. In conclusion, we have found the solution of eq. (39) in $MH_{p,\theta}^2$ and the lemma is proved.

proof (Theorem 4.0.1). We split two cases: $\gamma \geq 2$ and $\gamma < 2$.

Case $\gamma \geq 2$. The uniqueness follows from Lemma 4.0.8 and the fact that $H_{p,\theta}^\gamma \subset H_{p,\theta}^2$, which implies that the difference of two solutions belongs to $H_{p,\theta}^2$ and thus coincide because of the uniqueness when $\gamma = 2$. The existence and estimate eq. (40) follows from Lemma 4.0.8 and Lemma 4.0.7 (applied with $\mu = 2$) and the observation that by Lemma 4.0.8 we have

$$\|M^{-1}u\|_{2,p,\theta} \leq N \sum_i \|f^i\|_{1,p,\theta} \leq N \sum_i \|f^i\|_{\gamma-1,p,\theta}.$$

Case $\gamma < 2$. Denote by $\mathcal{R}_\lambda$ the operator which maps $(f^1, \dots, f^d)$ with $f^i \in H_{p,\theta}^{\gamma-1}$ into the solution $u \in MH_{p,\theta}^\gamma$ of equation eq. (38). So far we know that $\mathcal{R}$ is well-defined in spaces $(H_{p,\theta}^{\gamma-1})^d$ for $\gamma \geq 2$. We want to show that one can also defined $\mathcal{R}$ for $\gamma < 2$.

First, let $2 > \gamma \geq 1$. Take a b satisfying condition eq. (28) and observe that by Theorem 2.0.12 we have $\mathcal{L}_{b,0}^{-1}f^i \in H_{p,\theta}^{\gamma+1}$. Since $\gamma + 1 \geq 1$, by what we know in the case $\gamma \geq 2$, the function

$$v = \mathcal{R}_\lambda(\mathcal{L}_{b,0}^{-1}f^1, \dots, \mathcal{L}_{b,0}^{-1}f^d)$$

is well-defined and belongs to $MH_{p,\theta}^{\gamma+2} = H_{p,\theta-p}^{\gamma+2}$. Let $\tilde{u} = \mathcal{L}_{b,0}v$. By Remark 2.0.2, we have $\tilde{u} \in H_{p,\theta-p}^\gamma = MH_{p,\theta}^\gamma$. Furthermore, by the definition v satisfies $\Delta v - \lambda v = D_i \mathcal{L}_{b,0}^{-1}f^i$. We apply $\mathcal{L}_{b,0}$ to both parts of this equality to get

$$\Delta \tilde{u} - \lambda \tilde{u} = D_i \tilde{f}^i + \tilde{f}, \quad (48)$$

where

$$\tilde{f} = \Delta \mathcal{L}_{b,0}v - \mathcal{L}_{b,0}\Delta v + \mathcal{L}_{b,0}D_i \mathcal{L}_{b,0}^{-1}f^i - D_i f^i.$$

Next, we use

$$\begin{aligned} \mathcal{L}_{b,0}D_i \phi &= D_i \mathcal{L}_{b,0} \phi - 2\delta^{1i}M^{-1}\mathcal{L}_{b/2,0}\phi, \\ \mathcal{L}_{b,0}D_iD_j \phi &= D_iD_j \mathcal{L}_{b,0}\phi - 2\delta^{1j}D_iM^{-1}\mathcal{L}_{b/2,0}\phi - 2\delta^{1i}M^{-1}\mathcal{L}_{b/2,0}D_j\phi. \end{aligned}$$

We also use Remark 2.0.2 and Corollaries 2.0.5 and 2.0.13 and get

$$\tilde{f} = 2D_1M^{-1}\mathcal{L}_{b/2,0}M(M^{-1}v) + 2M^{-1}\mathcal{L}_{b/2,0}D_1v - 2M^{-1}\mathcal{L}_{b/2,0}\mathcal{L}_{b,0}^{-1}f^1,$$

$$\|M\tilde{f}\|_{\gamma-1,p,\theta} \leq N(\|M^{-1}v\|_{\gamma+2,p,\theta} + \|f^1\|_{\gamma-1,p,\theta}).$$

We see that $M\tilde{f} \in H_{p,\theta}^{\gamma-1}$, which by Remark 2.0.16 implies that $\tilde{f} = D_i \tilde{f}^i$ with $\|\tilde{f}^i\|_{\gamma,p,\theta} \leq N\|M\tilde{f}\|_{\gamma-1,p,\theta}$. Since $\gamma \geq 1$, by the result for $\gamma \geq 2$, the function $\bar{u} := \mathcal{R}_\gamma(\tilde{f}^1, \dots, \tilde{f}^d)$ is well-defined and belongs to $MH_{p,\theta}^{\gamma+1}$. To prove existence, it now remains only to notice that obviously $\bar{u} - \tilde{u}$ belongs to $MH_{p,\theta}^\gamma$ and solves eq. (38) because of eq. (48) and

$$\Delta \bar{u} - \lambda \bar{u} = D^i \tilde{f}^i = \tilde{f}.$$

To prove the estimate eq. (40), notice that by the definition of v and $\mathcal{R}_\lambda$,

$$\begin{aligned} \lambda \|M\bar{u}\|_{\gamma-2,p,\theta} + \|M^{-1}\bar{u}\|_{\gamma,p,\theta} &\leq \lambda N \|\bar{u}\|_{\gamma-2,p,\theta+p} + \|\bar{u}\|_{\gamma,p,\theta-p} \\ &= \lambda N \|\mathcal{L}_{b,0}v\|_{\gamma-2,p,\theta+p} + \|\mathcal{L}_{b,0}v\|_{\gamma,p,\theta-p} \\ &\leq \lambda N \|v\|_{\gamma,p,\theta+p} + \|v\|_{\gamma+2,p,\theta-p} \\ &\leq \lambda N \|Mv\|_{\gamma,p,\theta} + \|M^{-1}v\|_{\gamma+2,p,\theta} \\ &\leq N \sum_i \|\mathcal{L}_{b,0}^{-1}f^i\|_{\gamma+1,p,\theta} \end{aligned}$$

$$\leq N \sum_i \|f^i\|_{\gamma-1,p,\theta};$$

$$\begin{aligned} \lambda \|M\bar{u}\|_{\gamma-2,p,\theta} + \|M^{-1}\bar{u}\|_{\gamma,p,\theta} &\leq N \sum_i \|\bar{f}^i\|_{\gamma-1,p,\theta} \\ &\leq N \sum_i \|\bar{f}^i\|_{\gamma,p,\theta} \\ &\leq N \|M\bar{f}\|_{\gamma-1,p,\theta} \\ &\leq N (\|M^{-1}v\|_{\gamma+2,p,\theta} + \|f^1\|_{\gamma-1,p,\theta}) \\ &\leq N (\|f^1\|_{\gamma-1,p,\theta} + \sum_i \|\mathcal{L}_{b,0}^{-1}f^i\|_{\gamma+1,p,\theta}) \\ &\leq N \sum_i \|f^i\|_{\gamma-1,p,\theta}; \end{aligned}$$

$$\begin{aligned} \lambda \|Mu\|_{\gamma-2,p,\theta} + \|M^{-1}u\|_{\gamma+1,p,\theta} &\leq (\lambda \|M\bar{u}\|_{\gamma-2,p,\theta} + \|M^{-1}\bar{u}\|_{\gamma,p,\theta}) + (\lambda \|M\bar{u}\|_{\gamma-2,p,\theta} + \|M^{-1}\bar{u}\|_{\gamma,p,\theta}) \\ &\leq N \sum_i \|f^i\|_{\gamma-1,p,\theta}. \end{aligned}$$

By repeating the above argument, we consider the case $1 > \gamma \geq 0$, this time using the fact that $\gamma + 1 \geq 1$ and relying upon the result for $\gamma \geq 1$. One can continue in the same way and it only remains to prove the uniqueness of solutions in $MH_{p,\theta}^\gamma$.

It suffices to consider the case $f = 0$ (and $\gamma < 2$). In this case any solution $u \in MH_{p,\theta}^\gamma$ also belongs to $MH_{p,\theta}^2$ by Lemma 4.0.7 and its uniqueness follows from Lemma 4.0.8. The theorem is thus proved.

Remark 4.0.9 From the above derivation of Theorem 4.0.1 from Lemma 4.0.8 it is seen that for *any* fixed γ, p, θ satisfying the conditions of Theorem 4.0.1, if the assertion of Theorem 4.0.1 holds for these particular γ, p, θ , then it holds for any $\gamma \in \mathbb{R}$ with the same p, θ .

5 Parabolic Equations in $\mathbb{R}_+^d$

As in Section 4 we assume that condition eq. (37) is satisfied. In this section we are dealing with the equation

$$u_t = Lu + D_i f^i \quad t \in [0, T], \quad x^1 > 0, \quad (49)$$

where $T \in [0, \infty)$ is given,

$$Lu = a^{ij}(t)u x^i x^j,$$

and $a(t) = (a^{ij}(t))$ is a given symmetric $d \times d$ matrix valued function depending only on t . We assume that it is a Borel function satisfying, for each $t > 0$, condition eq. (43) with a fixed constant $\delta_0 > 0$.

We need some spaces of functions depending on t . Before introducing the space, we need another type of weighted Sobolev space.

Definition 5.0.1 Let $\zeta \in C_0^\infty(\mathbb{R}_+)$ be from Definition 1.0.1. For $\gamma, \theta \in \mathbb{R}$ and $p \in (1, \infty)$ we define a space $W_{p,\theta}^\gamma$ by all distributions u such that

$$\|u\|_{W_{p,\theta}^\gamma}^p := \sum_{n=-\infty}^{\infty} e^{n\theta} \|u(e^{n\cdot})\zeta\|_{B_{p,p}^\gamma}^p < \infty,$$

where $B_{p,p}^\gamma$ denotes the Besov space.

In the main paper, one uses $M^{1-2/p}H_{p,\theta}^{\gamma-2/p}$ for the initial function space. However, this works only if $p \geq 2$, and in addition, this is not optimal.

Definition 5.0.2 Let

$$\mathbb{H}_{p,\theta}^\gamma(T) = L_p((0, T), \mathfrak{B}(\mathbb{R}_+), H_{p,\theta}^\gamma), \quad \mathbb{L}_{p,\theta}(T) = \mathbb{H}_{p,\theta}^0(T).$$

For a function $u : [0, T] \rightarrow \mathcal{D}'(\mathbb{R}_+^d)^{(\text{xxv})}$ we write $u \in \mathfrak{H}_{p,\theta}^\gamma(T)$ if

(i) we have

$$M^{-1}u \in \bigcup_{\mu} \mathbb{H}_{p,\theta}^\mu(T), \quad u_x \in \mathbb{H}_{p,\theta}^{\gamma-1}(T), \quad u(0, \cdot) \in M^{1-2/p} W_{p,\theta}^{\gamma-2/p} =: U_{p,\theta}^\gamma; \quad (50)$$

(ii) there is a function $u_t = u_t(t)$ defined for $t \in (0, T)$ with values in the set of distributions of $\mathbb{R}_+^d$ such that

$$Mu_t \in \mathbb{H}_{p,\theta}^{\gamma-2}(T), \quad (u(t, \cdot), \phi) = (u(0, \cdot), \phi) + \int_0^t (u_t(s, \cdot), \phi) ds$$

for any $\phi \in C_0^\infty(\mathbb{R}_+^d)$ and $t \in [0, T]$.

Let

$$\|h\|_{U_{p,\theta}^\gamma} = \|M^{2/p-1}h\|_{\gamma-2/p,p,\theta}, \quad \|u\|_{\mathfrak{H}_{p,\theta}^\gamma(T)} = \|u_x\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)} + \|Mu_t\|_{\mathbb{H}_{p,\theta}^{\gamma-2}(T)} + \|u(0, \cdot)\|_{U_{p,\theta}^\gamma}. \quad (51)$$

Remark 5.0.3 Notice that such u_t is unique (t -a.e.). Indeed, if there exists f and g satisfies

$$(u(t, \cdot), \phi) = (u(0, \cdot), \phi) + \int_0^t (f(s, \cdot), \phi) ds = (u(0, \cdot), \phi) + \int_0^t (g(s, \cdot), \phi) ds$$

for every $\phi \in C_0^\infty(\mathbb{R}_+^d)$ and $t \in [0, T]$, where equations is meaningful, then for fixed $\phi \in C_0^\infty(\mathbb{R}_+^d)$,

$$\int_0^t (f(s, \cdot), \phi) ds = \int_0^t (g(s, \cdot), \phi) ds$$

holds for every $t \in [0, T]$, hence $(f(t, \cdot), \phi) = (g(t, \cdot), \phi)$ for almost every $t \in [0, T]$.^(xxvi) Since $C_0^\infty(\mathbb{R}_+^d)$ is a separable topological vector space, we obtain that $(f(t, \cdot), \phi) = (g(t, \cdot), \phi)$ for every $\phi \in C_0^\infty(\mathbb{R}_+^d)$ for almost every $t \in [0, T]$. This proves $f(t) = g(t)$ for almost every $t \in [0, T]$.

We will always understand eq. (49) in the sense of Definition 5.0.2 assuming that $f^i \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$.

Remark 5.0.4 By Theorem 2.0.6 and Corollary 2.0.13 the first two conditions in eq. (50) can be replaced with $M^{-1}u \in \mathbb{H}_{p,\theta}^\gamma(T)$. In addition,

$$\|M^{-1}u\|_{\mathbb{H}_{p,\theta}^\gamma(T)} \leq N\|u_x\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)} \leq N\|M^{-1}u\|_{\mathbb{H}_{p,\theta}^\gamma(T)}.$$

Remark 5.0.5 By Remark 2.0.2 it follows that $f^i \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$ only if $D_i f^i =: M^{-1}f$ with $f \in \mathbb{H}_{p,\theta}^{\gamma-2}(T)$. Therefore, $D_i f^i$ in eq. (49) may be replaced with $M^{-1}f$, where $f \in \mathbb{H}_{p,\theta}^{\gamma-2}(T)$.

On the other hand, by Remark 2.0.16 any $M^{-1}f$ with $f \in \mathbb{H}_{p,\theta}^{\gamma-2}(T)$ can be represented in a canonical way as $D_i f^i$ with $f^i \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$ and

$$\|f\|_{\mathbb{H}_{p,\theta}^{\gamma-2}(T)} \leq N \sum_i \|f^i\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)} \leq N\|f\|_{\mathbb{H}_{p,\theta}^{\gamma-2}(T)}.$$

^(xxv) Recall that $\mathcal{D}'(\mathbb{R}_+^d)$ is the set of all distributions on $\mathbb{R}_+^d$.

^(xxvi) We then have $\int_a^b (f(s, \cdot), \phi) ds = \int_a^b (g(s, \cdot), \phi) ds$ holds for every $0 \leq a \leq b \leq T$. This shows that $\int_0^T (f(t, \cdot), \phi) s(t) dt = \int_0^T (g(t, \cdot), \phi) s(t) dt$ for every simple Borel function s . Now take a sequence of simple functions $(s_n)_n$ such that $s_n(t) \rightarrow (f(t, \cdot) - g(t, \cdot), \phi)$ pointwise and $|s_n(t)| \leq |(f(t, \cdot) - g(t, \cdot), \phi)|$ for every t . Then apply the dominated convergence theorem.

Remark 5.0.6 Observe that for any $u \in \mathbb{H}_{p,\theta}^\gamma(T)$ and $f^i \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$ the right-hand side of eq. (49) belongs to $M^{-1}\mathbb{H}_{p,\theta}^{\gamma-2}(T)$, so that eq. (49) makes sense in $\mathfrak{H}_{p,\theta}^\gamma(T)$.^(xxvii) Indeed, the right-hand side of eq. (49) equals $(a^{ij}u_{x^j} + f^i)_{x^i}$ where $a^{ij}u_{x^j} + f^i \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$ by definitions and assumptions. It only remains to remember Remark 2.0.2.

This argument also shows that any $u \in \mathfrak{H}_{p,\theta}^\gamma(T)$ satisfies eq. (49) with $f^i := Q_i M(u_t - Lu)$,^(xxviii) where $M(u_t - Lu) \in \mathbb{H}_{p,\theta}^{\gamma-2}(T)$,^(xxix) $f^i \in \mathbb{H}_{p,\theta}^{\gamma-2}(T)$.

Furthermore, notice for the future that the operator $u \mapsto M(u_t - Lu)$ is a bounded operator from $\mathfrak{H}_{p,\theta}^\gamma(T)$ into $\mathbb{H}_{p,\theta}^{\gamma-2}(T)$. It follows from Remark 5.0.5 that to prove the unique solvability of eq. (49) for any $f^i \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$ it is necessary and sufficient to prove that the mapping $u \mapsto M(u_t - Lu)$ from $\mathfrak{H}_{p,\theta}^\gamma(T)$ to $\mathbb{H}_{p,\theta}^{\gamma-2}(T)$ is bijective.

Remark 5.0.7 It is known that $C_0^\infty(\mathbb{R}_+ \times \mathbb{R}_+^d)$ is dense in $\mathbb{H}_{p,\theta}^\gamma(T)$, and $C_0^\infty([0, T] \times \mathbb{R}_+^d)$ is dense in $\mathfrak{H}_{p,\theta}^\gamma(T)$. See Section 8.2 for the proof.

Here, notice that $\mathbb{R}_+ = (0, \infty)$. Thus support of functions in time is the type $[\epsilon, N]$ where $0 < \epsilon < N < \infty$. However, if $f \in C_0^\infty([0, T] \times \mathbb{R}_+^d)$, then the support of f may touch 0 or T . This means that we cannot use cutoff function to approximate $\mathfrak{H}_{p,\theta}^\gamma(T)$.

The actual reason why $C_0^\infty(\mathbb{R}_+ \times \mathbb{R}_+^d)$ is not dense in $\mathfrak{H}_{p,\theta}^\gamma(T)$ is that in the definition of the $\mathfrak{H}_{p,\theta}^\gamma(T)$ -norm, the initial value $u(0, \cdot)$ is introduced. Notice that for any $f \in C_0^\infty(\mathbb{R}_+ \times \mathbb{R}_+^d)$, $f(0, \cdot) = 0$, hence we cannot approximate with such functions.

The following is the main result of this section.

Theorem 5.0.8 For any $\gamma \in \mathbb{R}$, $u_0 \in U_{p,\theta}^\gamma$, and $f^i \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$ in the space $\mathfrak{H}_{p,\theta}^\gamma(T)$ there exists a unique solution of eq. (49) with initial value $u(0, \cdot) = u_0$. For this solution we have

$$\|u\|_{\mathfrak{H}_{p,\theta}^\gamma(T)} \leq N(d, \gamma, p, \theta, \delta_0) \left(\sum_i \|f^i\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)} + \|u_0\|_{U_{p,\theta}^\gamma} \right). \quad (52)$$

To prove this theorem we need two lemmas, the first of which is proved in next section.

Lemma 5.0.9 Theorem 5.0.8 holds for $\gamma = 2$.

Lemma 5.0.10 Let $\mu \leq \gamma$, $\theta_1 \in \mathbb{R}$, and $u \in \mathfrak{H}_{p,\theta_1}^\mu(T)$ be a solution of eq. (49) with initial value $u(0, \cdot) = u_0$. Assume that $M^{-1}u \in \mathbb{H}_{p,\theta}^\mu(T)$. Then $u \in \mathfrak{H}_{p,\theta}^\gamma(T)$ and

$$\|u\|_{\mathfrak{H}_{p,\theta}^\gamma(T)} \leq N(d, \gamma, \mu, \theta, p, \delta_0) \left(\sum_i \|f^i\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)} + \|M^{-1}u\|_{\mathbb{H}_{p,\theta}^\mu(T)} + \|u_0\|_{U_{p,\theta}^\gamma} \right).$$

proof. As in the proof of Lemma 4.0.7 it suffices to show the case when $\mu = \gamma - 1$. By virtue of Remark 5.0.4 and Remark 5.0.6, in order to prove the lemma, it suffices to show that

$$\|u_x\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)} \leq N(d, \gamma, \mu, \theta, p, \delta_0) \left(\sum_i \|f^i\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)} + \|M^{-1}u\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)} + \|u_0\|_{U_{p,\theta}^\gamma} \right). \quad (53)$$

Indeed, if $u_x \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$, then from eq. (49),

$$Mu_t = a^{ij}MD_iD_ju + MD_if^i \in \mathbb{H}_{p,\theta}^{\gamma-2}(T),$$

and eq. (53) is easily obtained from the definition of $\mathfrak{H}_{p,\theta}^\gamma(T)$ -norm.

^(xxvii) See eq. (50) that the space where u_t is located.

^(xxviii) Recall that $u = MD_iQ_iu$.

^(xxix) Since $u_x \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$, $MLu = a^{ij}M(u_{x^i})_{x^j} \in \mathbb{H}_{p,\theta}^{\gamma-2}(T)$.

Take a function $\zeta \in C_0^\infty(\mathbb{R}_+)$ satisfying condition eq. (1). Obviously the function $u_n(t, x) := u(e^{2n}t, e^n x)$ satisfy the equation

$$u_{nt} = L_n u_n + D_i f_n^i, \quad (54)$$

where

$$L_n v = a_n^{ij}(t) v_{x^i x^j} := a^{ij}(e^{2n}t) v_{x^i x^j}, \quad f_n^i(t, x) = e^n f^i(e^{2n}t, e^n x).$$

Hence by remembering that a is symmetric,

$$(\zeta u_n)_t = L_n(\zeta u_n) + \tilde{f}_n,$$

where $\tilde{f}_n = \zeta D_i f_n^i - 2a_n^{i1} \zeta' u_{nx^i} - a_n^{11} \zeta'' u_n$. Since $M^{-1}u \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$, for every $\eta \in C_0^\infty(\mathbb{R}_+)$ we have $\eta u_n \in \mathbb{H}_p^{\gamma-1}(e^{-2n}T)$ and $\eta u_{nx} \in \mathbb{H}_p^{\gamma-2}(e^{-2n}T)$. Indeed, $u_x \in \mathbb{H}_{p,\theta}^{\gamma-2}(T)$,

$$\eta(x) u_n(t, x) = e^n (M\eta)(x) (M^{-1}u)(e^{2n}t, e^n x), \quad \eta(x) u_{nx}(t, x) = e^n \eta(x) u_x(e^{2n}t, e^n x),$$

$$\begin{aligned} \|(M^{-1}u)(e^{2n}\cdot, e^n\cdot)\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(e^{-2n}T)}^p &= \int_0^{e^{-2n}T} \|(M^{-1}u)(e^{2n}t, e^n\cdot)\|_{\gamma-1,p,\theta}^p dt \\ &= e^{-n\theta p} \int_0^{e^{-2n}T} \|(M^{-1}u)(e^{2n}t, \cdot)\|_{\gamma-1,p,\theta}^p dt \\ &= e^{-n\theta p-2n} \int_0^T \|(M^{-1}u)(t, \cdot)\|_{\gamma-1,p,\theta}^p dt \\ &= e^{-n\theta p-2n} \|M^{-1}u\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)}^p. \end{aligned}$$

Similarly, as $f^i \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$, we have $\tilde{f}_n \in \mathbb{H}_p^{\gamma-2}(e^{-2n}T)$. Now following the proof of Lemma 3.6 of [KL99] almost gives the proof.^(xxx) Notice that we need a following inequality instead of the one in [KL99].

$$\begin{aligned} \sum_{n=-\infty}^{\infty} e^{(2-p+\theta)n} \|\zeta D_i f_n^i\|_{\mathbb{H}_p^{\gamma-2}(e^{-2n}T)}^p &= \sum_{n=-\infty}^{\infty} e^{(2-p+\theta)n} \|e^{2n} \zeta (D_i f^i)(e^{2n}\cdot, e^n\cdot)\|_{\mathbb{H}_p^{\gamma-2}(e^{-2n}T)}^p \\ &= \sum_{n=-\infty}^{\infty} e^{(2+\theta)n} \|e^n \zeta (D_i f^i)(e^{2n}\cdot, e^n\cdot)\|_{\mathbb{H}_p^{\gamma-2}(e^{-2n}T)}^p \\ &= \sum_{n=-\infty}^{\infty} e^{n\theta} \|e^n \zeta (D_i f^i)(\cdot, e^n\cdot)\|_{\mathbb{H}_p^{\gamma-2}(T)}^p \\ &= \sum_{n=-\infty}^{\infty} e^{n\theta} \|(MD_i f^i)(\cdot, e^n\cdot) M^{-1} \zeta\|_{\mathbb{H}_p^{\gamma-2}(T)}^p \\ &= \|MD_i f^i\|_{\mathbb{H}_{p,\theta}^{\gamma-2}(T)}^p \\ &\leq N \|f^i\|_{\mathbb{H}_{p,\theta}^{\gamma-1}(T)}^p, \end{aligned}$$

and similar for other terms. The lemma is proved.

proof. Theorem 5.0.8 Similar with the proof of Theorem 4.0.1, we split two cases to consider.

Case $\gamma \geq 2$. The uniqueness follows from Lemma 5.0.9 and the fact that $\mathfrak{H}_{p,\theta}^\gamma \subset \mathfrak{H}_{p,\theta}^2$, any two solutions coincide in $\mathfrak{H}_{p,\theta}^2$, and thus they are indistinguishable. For the existence of the solution, we use Lemma 5.0.9, Lemma 5.0.10 with $\mu = 2$, and $\|M^{-1}u\|_{\mathbb{H}_{p,\theta}^2(T)} \leq N \|u\|_{\mathfrak{H}_{p,\theta}^2(T)}$ by Remark 5.0.4.

Case $\gamma < 2$. As same as in the proof of Theorem 4.0.1, the case $1 \leq \gamma < 2$ is enough to consider.

Define an operator $\mathcal{R}$ maps $(u_0, f^1, \dots, f^d) \in U_{p,\theta}^\gamma \times (\mathbb{H}_{p,\theta}^{\gamma-1}(T))^d$ into the solution $u \in \mathfrak{H}_{p,\theta}^\gamma(T)$ of eq. (49) with initial value $u(0, \cdot) = u_0$. We already proved that such operator $\mathcal{R}$ is well-defined when $\gamma \geq 2$. Consider

$$v = \mathcal{R}(M^{1-2/p} \mathcal{L}_{2,0}^{-1} M^{2/p-1} u_0, \mathcal{L}_{2,0}^{-1} f^1, \dots, \mathcal{L}_{2,0}^{-1} f^d).$$

^(xxx) However, for $1 < p < 2$, we cannot apply all theorems introduced in [KL99] because the initial value space is not correct. For correct one, we should introduce Besov spaces.

Notice that by Theorem 2.0.17, $\mathcal{L}_{2,0}$ has a bounded inverse, so v is well-defined and belongs to $\mathfrak{H}_{p,\theta}^{\gamma+2}(T)$. Define $\tilde{u} = \mathcal{L}_{2,0}v$. It is not hard to show that $\tilde{u} \in \mathfrak{H}_{p,\theta}^{\gamma}(T)$. By definition v satisfies

$$v_t = a^{ij}v_{x^i x^j} + D^i \mathcal{L}_{2,0}^{-1} f^i$$

with initial value $v(0, \cdot) = M^{1-2/p} \mathcal{L}_{2,0}^{-1} M^{2/p-1} u_0$. Apply $\mathcal{L}_{2,0}$ at the both side we have^(xxxi)

$$\tilde{u}_t = a^{ij} \tilde{u}_{x^i x^j} + D^i f^i + \tilde{f}, \quad \tilde{u}(0, \cdot) = u_0 + \tilde{u}_0,$$

where

$$\tilde{f} = a^{ij} \mathcal{L}_{2,0} v_{x^i x^j} - a^{ij} (\mathcal{L}_{2,0} v)_{x^i x^j} + \mathcal{L}_{2,0} D^i \mathcal{L}_{2,0}^{-1} f^i - D^i f^i, \quad \tilde{u}_0 = \mathcal{L}_{2,0} M^{1-2/p} \mathcal{L}_{2,0}^{-1} M^{2/p-1} u_0 - u_0.$$

Formulas as in the proof of Theorem 4.0.1, and the formula

$$\mathcal{L}_{2,0} M^{\alpha} \phi = M^{\alpha} \mathcal{L}_{2,0} \phi + 2\alpha M^{\alpha+1} D^1 \phi + \alpha^2 M^{\alpha} \phi, \quad \forall \alpha \in \mathbb{R},$$

gives

$$\begin{aligned} \tilde{f} &= -2a^{i1} D^i M^{-1} \mathcal{L}_{1,0} v - 2a^{1j} M^{-1} \mathcal{L}_{1,0} D^j v - 2M^{-1} \mathcal{L}_{1,0} \mathcal{L}_{2,0}^{-1} f^1, \\ M^{2/p-1} \tilde{u}_0 &= (2 - 4/p) M D^1 \mathcal{L}_{2,0}^{-1} M^{2/p-1} u_0 + (1 - 2/p)^2 \mathcal{L}_{2,0}^{-1} M^{2/p-1} u_0. \end{aligned}$$

By Remark 2.0.2, Corollary 2.0.5, and Corollary 2.0.13, we find $\tilde{u}_0 \in U_{p,\theta}^{\gamma+1}$. In addition, as $v_x \in \mathbb{H}_{p,\theta}^{\gamma+1}(T)$ since $v \in \mathfrak{H}_{p,\theta}^{\gamma+2}(T)$, we have

$$M D^i M^{-1} \mathcal{L}_{1,0} v = M D^i \sum_j M D^j (D^j v) + M D^i D^1 v \in \mathbb{H}_{p,\theta}^{\gamma-1}(T).$$

Hence, $M \tilde{f} \in \mathbb{H}_{p,\theta}^{\gamma-1}(T)$. By Remark 2.0.16, one can take $\tilde{f}^i \in \mathbb{H}_{p,\theta}^{\gamma}(T)$ such that $\tilde{f} = M^{-1}(M \tilde{f}) = D^i \tilde{f}^i$. Since $\gamma \geq 1$, $\tilde{u} = \mathcal{R}(\tilde{u}_0, \tilde{f}^1, \dots, \tilde{f}^d)$ is well-defined and $\tilde{u} \in \mathfrak{H}_{p,\theta}^{\gamma+1}(T)$. Now $u = \tilde{u} - \tilde{u}$ belongs to $\mathfrak{H}_{p,\theta}^{\gamma}(T)$ and solves eq. (49). Estimate eq. (52) follows easily from the above construction and known estimates for $\mathcal{R}, \mathcal{L}_{2,0}, MD$.

For the uniqueness, it suffices to consider $f = 0, u_0 = 0$ (and $\gamma < 2$). In this case, any solutions $u \in \mathfrak{H}_{p,\theta}^{\gamma}(T)$ belongs to $\mathfrak{H}_{p,\theta}^2(T)$ by Lemma 5.0.10. Thus the uniqueness follows from Lemma 5.0.9. The theorem is now proved.

6 Proof of Lemma 5.0.9

First, we prove several auxiliary results.

Lemma 6.0.1 Let $u \in W_p^2(\mathbb{R}_+^d)$ and let c be a constant such that $1 - p < c < 1$. Assume that

$$\lim_{x^1 \downarrow 0} \frac{|u(x)|}{x^1} < \infty \quad \text{on} \quad \{x : x^1 = 0\} \text{ (a.e.)}. \quad (55)$$

Let ξ be a unit vector in $\mathbb{R}^d$. Then

$$\begin{aligned} - \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u u_{(\xi)(\xi)} dx &= c(1-c)p^{-1}(\xi^1)^2 \int_{\mathbb{R}_+^d} (x^1)^{c-2} |u|^p dx + (p-1) \int_{\mathbb{R}_+^d} x^c |u|^{p-2} (u_{(\xi)})^2 dx \\ &\geq (\xi^1)^2 \kappa \int_{\mathbb{R}_+^d} (x^1)^{c-2} |u|^p dx, \end{aligned} \quad (56)$$

where $\kappa = (1-c)(p-1+c)p^{-2}$, $u_{(\xi)} = u_{x^i} \xi^i$, and $u_{(\xi)(\xi)} = u_{x^i x^j} \xi^i \xi^j$.

^(xxxi) Notice that we understand $v_t = h$ by the distributions sense, which is

$$(v(t, \cdot), \phi) = (v(0, \cdot), \phi) + \int_0^t (h(s, \cdot), \phi) ds$$

for every $\phi \in C_0^\infty(\mathbb{R}_+^d)$. By applying $\mathcal{L}_{2,0}$ we mean that substitute $\mathcal{L}_{2,0}^* \phi$ in the above formula, where $\mathcal{L}_{2,0}^*$ is the formal adjoint of $\mathcal{L}_{2,0}$.

proof. If $\xi = (\xi^1, 0, \dots, 0)$, then eq. (56) reduces to the one-dimensional case by Fubini's theorem. In this case, we proceed as in the proof of Theorem 2.0.17 integrating by parts on the basis of Lemma 2.0.18. When we apply the integration by parts, we need to estimate boundary terms for $(x^1)^c |u|^{p-2} u u_{x^1}$ and $(x^1)^{c-1} |u|^p$. The boundary terms vanish for $x = 0$ due to assumption eq. (55) and the fact that u_x is bounded ($d = 1$). Recall that eq. (55) is equivalent to

$$\sup_{r>0} \inf_{0<x^1<r} \frac{|u(x)|}{x^1} < \infty \quad \text{on } \{x : x^1 = 0\} \text{ (a.e.)}.$$

Thus for every $r > 0$ one can take $0 < x^1 < r$ such that (also consider any x satisfies eq. (55)), since $c + p - 1 > 0$,

$$\begin{aligned} |(x^1)^c |u|^{p-2} u u_{x^1}| &\leq N(x^1)^{1-p} |u|^{p-1} (x^1)^{c+p-1} \leq N r^{c+p-1} \rightarrow 0, \\ |(x^1)^{c-1} |u|^p| &= (x^1)^{-p} |u|^p (x^1)^{c+p-1} \leq N r^{c+p-1} \rightarrow 0. \end{aligned}$$

Whereas for $x \rightarrow \infty$ they vanish since $c + 1 < 0$, and $|u|^p, |u|^{p-1} u_x \in L_1(\mathbb{R}_+)$, so that at least along a subsequence $x |u|^{p-1} u_x(x) \rightarrow 0$ and $|u|^p(x) \rightarrow 0$.^(xxxii) Hence we have the following calculation.

$$\begin{aligned} \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u u_{(\xi)(\xi)} dx &= \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u u_{x^1 x^1} dx \\ &= \int_{\mathbb{R}_+^d} (x^1)^c (|u|^{p-2} u u_{x^1})_{x^1} dx - (p-1) \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} (u_{x^1})^2 dx \\ &= -c \int_{\mathbb{R}_+^d} (x^1)^{c-1} |u|^{p-2} u u_{x^1} dx - (p-1) \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} (u_{x^1})^2 dx \\ &= -c \int_{\mathbb{R}_+^d} (x^1)^{c-1} (|u|^p)_{x^1} dx - (p-1) \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} (u_{x^1})^2 dx \\ &= c(c-1)p^{-1} \int_{\mathbb{R}_+^d} (x^1)^{c-2} |u|^p dx - (p-1) \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} (u_{x^1})^2 dx. \end{aligned}$$

To prove the inequality in eq. (56), define $v = |u|^{p/2}$ and observe that by Lemma 2.0.18 and Minkowski's inequality

$$\begin{aligned} \int_0^\infty x^{c-2} |u|^p dx &= \int_0^\infty x^{c-2} v^2 dx = \int_0^\infty x^c \left(\int_0^1 v'(yx) dy \right)^2 dx \\ &\leq \left(\int_0^1 \left(\int_0^\infty x^c (v'(yx))^2 dx \right)^{1/2} dy \right)^2 = \int_0^\infty x^c (v'(x))^2 dx \left(\int_0^1 y^b dy \right)^2, \end{aligned}$$

where $b = -1/2 - c/2 > -1$. By evaluating the last integral we get

$$\int_0^\infty x^{c-2} |u|^p dx \leq 4(1-c)^{-2} \int_{\mathbb{R}_+} x^c (v_x)^2 dx = p^2(1-c)^{-2} \int_{\mathbb{R}_+} x^c |u|^{p-2} (u_{x^1})^2 dx.$$

This yields the inequality in eq. (56),^(xxxiii) which is thus proved for $d = 1$ and for any d if $\xi = (\xi^1, 0, \dots, 0)$.

If $\xi^1 = 0$, by changing coordinates we can always assume $\xi = |\xi| e_2$, where e_2 is the second basis vector. Then integrating by parts we see that the left-hand side of eq. (56) equals

$$(p-1) \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u_{x^2}^2 dx \geq 0.$$

In the general case $\xi^1 \neq 0$, without loss of generality we assume that $\xi^1 > 0$ and make the change of coordinates $y^1 = x^1/\xi^1, y^i = x^i - \xi^i x^1/\xi^1$. Then

$$\left| \frac{\partial y}{\partial x} \right| = \frac{1}{\xi^1}, \quad u_{(\xi)(\xi)} = u_{y^i y^j} y_{(\xi)}^i y_{(\xi)}^j = u_{y^1 y^1}.$$

Hence, by the result for $\xi = (\xi^1, 0, \dots, 0)$ the left-hand side of eq. (56) equals

^(xxxii) Notice that for $f \in L_1(\mathbb{R}_+)$, we have $\lim_{x \rightarrow \infty} |f(x)| = 0$.

^(xxxiii) Here, as ξ is a unit vector, $\xi^1 = 1$.

$$-(\xi^1)^{1+c} \int_{\mathbb{R}_+^d} (y^1)^c |u|^{p-2} u u_{y^1 y^1} dy \geq (\xi^1)^{1+c} \kappa \int_{\mathbb{R}_+^d} (y^1)^{c-2} |u|^p dy = (\xi^1)^2 \kappa \int_{\mathbb{R}_+^d} (x^1)^{c-2} |u|^p dx.$$

This proves the lemma.

Corollary 6.0.2 If we are given a nonnegative symmetric $d \times d$ matrix a , then, under the assumptions of Lemma 6.0.1, we have

$$\begin{aligned} - \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u a^{ij} u_{x^i x^j} dx &= c(1-c) p^{-1} a^{11} \int_{\mathbb{R}_+^d} (x^1)^{c-2} |u|^p dx + (p-1) \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} a^{ij} u_{x^i} u_{x^j} dx \\ &\geq (\xi^1)^2 \kappa \int_{\mathbb{R}_+^d} (x^1)^{c-2} |u|^p dx, \end{aligned} \quad (57)$$

proof. Observe that $a^{ij} = \lambda^k \xi_k^i \xi_k^j$, where λ^k are the eigenvalues and ξ_k eigenvectors of a , which implies that the left-hand side of eq. (57) equals

$$-\lambda^k \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u u_{(\xi_k)(\xi_k)} dx.$$

The following is the basic a priori estimate.

Lemma 6.0.3 Let $u \in \mathfrak{H}_{p,\theta}^2(T)$ with $u(0, \cdot) = 0$. Then

$$\|M^{2/p-1} u(T, \cdot)\|_{0,p,\theta}^p + \|u\|_{\mathfrak{H}_{p,\theta}^2(T)}^p \leq N(d, p, \theta, \delta_0) \|M(u_t - Lu)\|_{\mathbb{L}_{p,\theta}(T)}.$$

proof. By Remark 5.0.6 and 5.0.7 we may assume that $u \in C_0^\infty([0, T] \times \mathbb{R}_+^d)$. By Lemma 5.0.10 we only need to prove^(xxxiv)

$$\|M^{2/p-1} u(T, \cdot)\|_{0,p,\theta}^p + \|M^{-1} u\|_{\mathbb{L}_{p,\theta}(T)}^p \leq N(d, p, \theta, \delta_0) \|M(u_t - Lu)\|_{\mathbb{L}_{p,\theta}(T)}. \quad (58)$$

Define $f = M(u_t - Lu)$, so that $u_t = Lu + M^{-1}f$, multiply the latter equation by $(x^1)^c |u|^{p-2} u$, where $c = 2 + \theta - d - p \in (1 - p, 1)$, and integrate by parts. Then by Hölder's inequality, we first obtain that

$$\int_0^\infty \int_{\mathbb{R}_+^d} (x^1)^{c-1} |u|^{p-2} u f dx dt \leq \|M^{-1} u\|_{\mathbb{L}_{p,\theta}(T)}^{p-1} \|f\|_{\mathbb{L}_{p,\theta}(T)}.$$

Observe that

$$\|M^{2/p-1} u(T, \cdot)\|_{0,p,\theta}^p = \int_{\mathbb{R}_+^d} (x^1)^{\theta-d+2-p} |u|^p(T, x) dx = \int_0^T \int_{\mathbb{R}_+^d} (x^1)^c (|u|^p)_t(t, x) dx dt$$

because of Lemma 2.0.18^(xxxv) and $u(0, \cdot) = 0$. Hence by Corollary 6.0.2 we find^(xxxvi)

$$\begin{aligned} p^{-1} \|M^{2/p-1} u(T, \cdot)\|_{0,p,\theta}^p &= \int_0^T \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u u_t dx dt \\ &= \int_0^T \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u (a^{ij} u_{x^i x^j} + M^{-1} f) dx dt \\ &= \int_0^T \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} u a^{ij} u_{x^i x^j} dx dt + \int_0^T \int_{\mathbb{R}_+^d} (x^1)^{c-1} |u|^{p-2} u f dx dt \\ &\leq -\delta_0 \kappa \|M^{-1} u\|_{\mathbb{L}_{p,\theta}(T)}^p + \|M^{-1} u\|_{\mathbb{L}_{p,\theta}(T)}^{p-1} \|f\|_{\mathbb{L}_{p,\theta}(T)}. \end{aligned}$$

This certainly implies eq. (58). The lemma is proved.

^(xxxiv) Since $u \in \mathfrak{H}_{p,\theta}^2(T)$, by Remark 5.0.6 one can take canonical f^i to apply Lemma 5.0.10.

^(xxxv) $|u|^p(t, x)$ is absolutely continuous in t and $(|u|^p)_t = p|u|^{p-2} u u_t$.

^(xxxvi) By the ellipticity, $-a^{11} \leq -\delta_0$. Also, remember that $c - 2 = \theta - d - p$.

Lemma 6.0.4 For any $u_0 \in U_{p,\theta}^2$ the equation $v_t = \Delta v$ with initial data $v(0, \cdot) = u_0$ has a solution $v \in \mathfrak{H}_{p,\theta}^2(T)$ such that

$$\|v\|_{\mathfrak{H}_{p,\theta}^2(T)} \leq N(p, d, \theta) \|u_0\|_{U_{p,\theta}^2}.$$

proof. The proof is rather long, so we will prove this in Appendix.

proof (Lemma 5.0.9). The uniqueness follows immediate from Lemma 6.0.3. For the existence, first of all, bearing in mind the method of continuity, we see that we can concentrate on the case $L = \Delta$. Then replacing u in eq. (49) with $u + v$, where v is taken from Lemma 6.0.4, we reduce the situation to the case $u_0 = 0$.

Indeed, let L be given and consider $L_\sigma = (1 - \sigma)\Delta + \sigma L$. Also, consider a set $\Sigma \subset [0, 1]$ by all collection of σ such that Lemma 5.0.9 is true by replacing L into L_σ . Later we will prove that $0 \in \Sigma$, thus for now, assume that $0 \in \Sigma$.

Fix $\sigma_0 \in \Sigma$. For $u \in \mathfrak{H}_{p,\theta}^2(T)$, define $v \in \mathfrak{H}_{p,\theta}^2(T)$ a solution to

$$v_t = L_{\sigma_0} v + D_i f^i + (L_{\sigma_0} - L_\sigma)u.$$

By Remark 5.0.6, $M(u_t - L_\sigma u) \in \mathbb{L}_{p,\theta}(T)$ for every $v \in \Sigma$. This shows that $M(L_{\sigma_0} - L_\sigma)u \in \mathbb{L}_{p,\theta}(T)$, so the given PDE makes sense, and by the definition of the set Σ , such v uniquely exists. So we can define a map $R : \mathfrak{H}_{p,\theta}^2(T) \rightarrow \mathfrak{H}_{p,\theta}^2(T)$ by $Ru = v$. By Remark 5.0.6, ^(xxxvii) and Lemma 6.0.3, for any $u_1, u_2 \in \mathfrak{H}_{p,\theta}^2(T)$,

$$\begin{aligned} \|Qu_1 - Qu_2\|_{\mathfrak{H}_{p,\theta}^2(T)} &\leq N \|M(\partial_t - L_{\sigma_0})(Qu_1 - Qu_2)\|_{\mathbb{L}_{p,\theta}(T)} \\ &= N \|M(L_{\sigma_0} - L_\sigma)(u_1 - u_2)\|_{\mathbb{L}_{p,\theta}(T)} \\ &= N |\sigma_0 - \sigma| \|M(L - \Delta)(u_1 - u_2)\|_{\mathbb{L}_{p,\theta}(T)} \\ &\leq N |\sigma_0 - \sigma| \|u_1 - u_2\|_{\mathfrak{H}_{p,\theta}^2(T)}, \end{aligned}$$

where N is a constant independent of u_1, u_2, σ_0 , and σ . Therefore, by taking σ close enough to σ_0 , one can make Q a contraction, and such distance between σ and σ_0 does not depends on σ and σ_0 . Hence, if $0 \in \Sigma$ is proved, we can obtain $1 \in \Sigma$ with a finitly many step, which proves Lemma 5.0.9 for the operator L .

Now by virtue of Lemma 6.0.3 and Remark 5.0.7, as in the proof of Lemma 4.0.8 we may assume that $f^i \in C_0^\infty(\mathbb{R}_+ \times \mathbb{R}_+^d)$ and we continue these functions on $\mathbb{R}_+ \times \mathbb{R}_+^d$ to be even function on x^1 if $i = 1$ and odd extension if $i \geq 2$. ^(xxxviii) Then, say by Theorem 4.10 of [Kry99a], eq. (49) considered for all $x \in \mathbb{R}^d$ with zero initial condition has a unique solution u such that $u \in \cap_\gamma C([0, T], H_p^\gamma)$. By Sobolev embedding theorem we get that $u \in \cap_{n=0}^\infty C([0, T], C^n(\mathbb{R}^d))$. Furthermore, by uniqueness, u is an odd function of x^1 so that $u(t, 0, x') = 0$ for $t \geq 0$. In particular, owing to the inequality $|M^{-1}u| \leq \sup_x |u_x|$, ^(xxxix) this implies that $M^{-1}u$ is bounded.

To finish the proof, we shall prove that $M^{-1}u \in \mathbb{H}_{p,\theta}^\gamma(T)$ for every $\gamma \geq 0$ and $\theta \in (d - 1, p + d - 1)$. By Corollary 2.0.3, it suffices to show that $M^{-1+|\alpha|} D^\alpha u \in \mathbb{L}_{p,\theta}(T)$.

Since $u(t, 0, x') = 0$, the fundamental theorem of calculus gives

$$\left| \frac{u(t, x)}{x^1} \right| \leq \int_0^1 |u_{x^1}(t, sx^1, x')| ds \leq \sup_{0 \leq y \leq 1} |u_x(t, y, x')|, \quad \forall x^1 \in (0, 1).$$

Hence, by Section 8.5,

$$\left| \frac{u(t, x)}{x^1} \right| \leq N \sup_{0 \leq x^1 \leq 1} e^{-\sqrt{2}|x|+8t} \leq N e^{-\sqrt{2}|x'|+8t}, \quad \forall x^1 \in (0, 1),$$

where $N = N(d, f)$. Using this estimate, we have

^(xxxvii) Since

$$M(L - \Delta) = M(\partial_t - \Delta) - M(\partial_t - L),$$

by Remark 5.0.6, the operator on the left-hand side is also bounded from $\mathfrak{H}_{p,\theta}^2(T)$ into $\mathbb{L}_{p,\theta}(T)$.

^(xxxviii) Then u a solution of eq. (49) should be an odd function because we are dealing with $D_1 f^1$, which is then an odd function of x^1 .

^(xxxix) See the proof of Lemma 4.0.8. Indeed, by Lemma 12.10.2 and Theorem 13.8.1 in [Kry08], $H_p^\gamma \subset L_\infty$ if $\gamma - d/p > 0$. From this, u_x is bounded.

$$\begin{aligned}
 \|M^{-1}u\|_{\mathbb{H}_{p,\theta}^0(T)}^p &= \int_0^T \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |M^{-1}u(t, x)|^p dx dt \\
 &= \int_0^T \int_{x^1 < 1} (x^1)^{\theta-d} |M^{-1}u(t, x)|^p dx dt + \int_0^T \int_{x^1 \geq 1} (x^1)^{\theta-d-p} |u(t, x)|^p dx dt \\
 &\leq N \int_0^T \int_{x^1 < 1} (x^1)^{\theta-d} e^{-p\sqrt{2}|x'|+8pt} dx dt + N \int_0^T \int_{x^1 \geq 1} (x^1)^{\theta-d-p} e^{-p\sqrt{2}|x'|+8pt} dx dt \\
 &= N \int_0^1 (x^1)^{\theta-d} dx^1 + N \int_1^\infty (x^1)^{\theta-d-p} dx^1 < \infty.
 \end{aligned}$$

Furthermore, for nonzero multiindex α we obtain

$$\begin{aligned}
 \|M^{-1+|\alpha|}D^\alpha u\|_{\mathbb{H}_{p,\theta}^0(T)}^p &= \int_0^T \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |M^{-1+|\alpha|}D^\alpha u(t, x)|^p dx dt \\
 &\leq N \int_0^T \int_{\mathbb{R}_+^d} (x^1)^{\theta-d-p+|\alpha|} e^{-p\sqrt{2}|x|+8pt} dx dt \\
 &= N \int_0^T \int_0^\infty \int_{\mathbb{R}^{d-1}} (x^1)^{\theta-d-p+|\alpha|} e^{-p(x^1)/\sqrt{2}} e^{-p|x'|/\sqrt{2}} e^{8pt} dx' dx^1 dt \\
 &= N \int_0^\infty (x^1)^{\theta-d-p+|\alpha|} e^{-p(x^1)/\sqrt{2}} dx^1.
 \end{aligned}$$

Since $|\alpha| \geq 1$, $\theta - d - p + |\alpha| > -1$ holds because $\theta - d > -1$. Therefore, the last integral is finite, proving that $M^{-1}u \in \mathbb{H}_{p,\theta}^\gamma$ for every $\gamma \geq 0$ provided that $\theta \in (d-1, p+d-1)$.

Finally, from eq. (49), it follows that $Mu_t \in \mathbb{H}_{p,\theta}^\gamma(T)$ for all γ . Thus $u \in \mathfrak{H}_{p,\theta}^2(T)$ and applying Lemma 6.0.3 we finish the proof of Lemma 5.0.9.

7 Some Additional Properties for Weighted Sobolev Space

7.1 Duality of $H_{p,\theta}^\gamma$

7.2 $H_{p,\theta}^\gamma$ as complex interpolation spaces

In this subsection, we are going to prove the following theorem by two different method.

Theorem 7.2.1 For $\kappa \in (0, 1)$, $p \in (1, \infty)$, $\gamma_i, \theta_i \in \mathbb{R}$, $i = 0, 1$,

$$\theta = \kappa\theta_1 + (1-\kappa)\theta_0, \quad \gamma = \kappa\gamma_1 + (1-\kappa)\gamma_0,$$

we have $[H_{p,\theta_0}^{\gamma_0}, H_{p,\theta_1}^{\gamma_1}]_\kappa = H_{p,\theta}^\gamma$, where $[\cdot, \cdot]_\kappa$ denotes the complex interpolation.

7.2.1 The direct method

Below proof is in [Kry99c]. In this paper, we fill some details where the main paper does not mention.

Before proving Theorem 7.2.1, we need a special holomorphic operator, $(1-\Delta)^z$. The following lemma states precisely.

Lemma 7.2.2 For any $f \in \mathcal{S}(\mathbb{R}^d)$, the function $g(z) = (1-\Delta)^z f$ is $L_2(\mathbb{R}^d)$ -valued entire function.

In order to prove this lemma, we need another lemma.

Lemma 7.2.3 Let X, Y be complex Banach spaces, f be a X -valued holomorphic function on a domain $\Omega \subset \mathbb{C}$, and $\Lambda \in L(X, Y)$. Then Λf is Y -valued holomorphic function on Ω and $(\Lambda f)' = \Lambda f'$.

proof. Fix $z_0 \in \Omega$. Then we have

$$\begin{aligned} \left\| \frac{(\Lambda f)(z_0 + h) - (\Lambda f)(z_0)}{h} - (\Lambda f')(z_0) \right\|_Y &= \left\| \Lambda \left(\frac{f(z_0 + h) - f(z_0)}{h} - f'(z_0) \right) \right\|_Y \\ &\leq \|\Lambda\|_{X \rightarrow Y} \left\| \frac{f(z_0 + h) - f(z_0)}{h} - f'(z_0) \right\|_X \\ &\rightarrow 0 \end{aligned}$$

as $|h| \rightarrow 0$. This proves the lemma

proof (Lemma 7.2.2). By Lemma 7.2.3, it suffices to show that $\hat{g}(z) = (1 + |\xi|^2)^z \hat{f}(\xi)$ is $L_2(\mathbb{R}^d)$ -valued entire function.

7.2.2 Using Interpolation Theories

8 Appendix

8.1 The space $l_p^\theta(\mathbb{Z}, H_p^\gamma)$

Lemma 8.1.1 Let $l_p^\theta(\mathbb{Z}, H_p^\gamma)$ be the space of all H_p^γ -valued sequences such that

$$\|f\|_{\gamma, p, \theta}^p = \sum_{n=-\infty}^{\infty} e^{n\theta} \|f_n\|_{\gamma, p}^p < \infty.$$

This space is a reflexive Banach space.

proof. Consider a measure μ_θ on $\mathbb{Z}$ by

$$\int_{\mathbb{Z}} f(n) \mu_\theta(dn) := \sum_{n=-\infty}^{\infty} e^{n\theta} f(n).$$

Then $l_p^\theta(\mathbb{Z}, H_p^\gamma) = L_p(\mathbb{Z}, \mu_\theta, H_p^\gamma)$, and the latter space is a Banach space by the Bochner integral theory.

8.2 The proof of Remark 5.0.7

Before proving the remark, we need an important lemma.

Lemma 8.2.1 Define $X = \bigcup_{\gamma, p, \theta} H_{p, \theta}^\gamma$. There exists operators $\Lambda_n, \Lambda_{m, n}$, and $\Lambda_{l, m, n}$ from X into X such that

(i) For any $u \in H_{p, \theta}^\gamma$, $\Lambda_{l, m, n} u \in C_0^\infty(\mathbb{R}_+^d)$;

(ii) We have

$$\sup_{l, m} (\|\Lambda_{m, n}\|_{H_{p, \theta}^\gamma \rightarrow H_{p, \theta}^\gamma} + \|\Lambda_{l, m, n}\|_{H_{p, \theta}^\gamma \rightarrow H_{p, \theta}^\gamma}) \leq N(n),$$

and

$$\sup_n \|\Lambda_n\|_{H_{p, \theta}^\gamma \rightarrow H_{p, \theta}^\gamma} \leq N.$$

(iii) $\lim_{l \rightarrow \infty} \Lambda_{l, m, n} u = \Lambda_{m, n} u$, $\lim_{m \rightarrow \infty} \Lambda_{m, n} u = \Lambda_n u$, and $\lim_{n \rightarrow \infty} \Lambda_n u = u$ in $H_{p, \theta}^\gamma$.

(iv) Let $u : \mathbb{R} \rightarrow H_{p,\theta}^\gamma$ such that u has a distributional derivative in t , that is if there exists f such that

$$(u(t, \cdot), \psi) = (u(0, \cdot), \psi) + \int_0^t (f(s, \cdot), \phi) ds$$

for every t and $\psi \in C_0^\infty(\mathbb{R}_+^d)$. In this case, we write $u_t = f$. Then

$$(\Lambda_n u)_t = \Lambda_n u_t, \quad (\Lambda_{m,n} u)_t = \Lambda_{m,n} u_t, \quad (\Lambda_{l,m,n} u)_t = \Lambda_{l,m,n} u_t.$$

proof. We first assume that γ be a nonnegative integer. Fix $\xi \in C_0^\infty(\mathbb{R}^d)$ such that $0 \leq \xi \leq 1$, $\xi \equiv 1$ on $|x| \leq 1$ and ξ vanishes on $|x| \geq 2$. Then put $\xi_R(x) = \xi(x/R)$ for $R \geq 1$. Then since γ is a nonnegative integer, we have for every $u \in H_p^\gamma$,

$$\begin{aligned} \|\xi_R u\|_{H_p^\gamma} &= \|\xi_R u\|_{W_p^\gamma} = \sum_{|\alpha| \leq \gamma} \|D^\alpha(\xi_R u)\|_{L_p} \\ &\leq \sum_{|\alpha| \leq \gamma} \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} R^{-|\alpha-\beta|} \|(D^{\alpha-\beta} \xi)(\cdot/R)\|_{L_\infty} \|D^\beta u\|_{L_p} \\ &\leq N(d, \gamma, \xi) \|u\|_{W_p^\gamma} = N(d, \gamma, \xi) \|u\|_{H_p^\gamma}; \end{aligned}$$

$$\begin{aligned} \|\xi_R u - u\|_{H_p^\gamma} &= \|\xi_R u - u\|_{W_p^\gamma} \\ &= \sum_{|\alpha| \leq \gamma} \|D^\alpha(\xi_R u) - D^\alpha u\|_{L_p} \\ &\leq \sum_{|\alpha| \leq \gamma} \sum_{\beta < \alpha} \binom{\alpha}{\beta} R^{-|\alpha-\beta|} \|(D^{\alpha-\beta} \xi)(\cdot/R)\|_{L_\infty} \|D^\beta u\|_{L_p} \\ &\leq N(d, \xi, \gamma) \sum_{|\alpha| \leq \gamma} \sum_{\beta < \alpha} R^{-|\alpha-\beta|} \|D^\beta u\|_{L_p} \\ &\leq N(d, \xi, \gamma) R^{-1} \|u\|_{H_p^\gamma}. \end{aligned}$$

Let us define a sequence of operator

$$T_n : \bigcup_{\gamma \in \mathbb{Z}_+, p} H_p^\gamma \rightarrow \bigcup_{\gamma \in \mathbb{Z}_+, p} H_p^\gamma$$

by $T_n u(x) = \xi_n(x) u(x) = \xi(x/n) u(x)$. What we shown in above is that

$$\|T_n\|_{H_p^\gamma \rightarrow H_p^\gamma} \leq N(d, \gamma, \xi), \quad \|T_n - I\|_{H_p^\gamma \rightarrow H_p^\gamma} \leq N(d, \gamma, \xi) n^{-1}. \quad (59)$$

Using the complex interpolation theorem (see [Tri83] for instance), eq. (59) also holds for every $\gamma \geq 0$ (need not to be an integer). Finally, if $\gamma < 0$, we use the duality. Since T_n is just an operator by multiplying smooth functions, $\langle T_n u, \phi \rangle = \langle u, T_n \phi \rangle$ holds for every $u \in H_p^\gamma$ and $\phi \in C_0^\infty(\mathbb{R}^d)$. Hence, we find

$$\begin{aligned} \|T_n u\|_{H_p^\gamma} &= \sup_{\phi \in C_0^\infty, \phi \neq 0} \frac{\langle T_n u, \phi \rangle}{\|\phi\|_{H_q^{-\gamma}}} = \sup_{\phi \in C_0^\infty, \phi \neq 0} \frac{\langle u, T_n \phi \rangle}{\|\phi\|_{H_q^{-\gamma}}} \\ &\leq \sup_{\phi \in C_0^\infty, \phi \neq 0} \frac{\|u\|_{H_p^\gamma} \|T_n \phi\|_{H_q^{-\gamma}}}{\|\phi\|_{H_q^{-\gamma}}} \leq N(d, \gamma, \xi) \|u\|_{H_p^\gamma}, \end{aligned}$$

and similarly, $\|T_n u - u\|_{H_p^\gamma} \leq N(d, \gamma, \xi) n^{-1} \|u\|_{H_p^\gamma}$, where $q = p/(p-1)$.

Take a sequence ζ_k from Corollary 1.0.22, and $\phi \in C_0^\infty(\mathbb{R}^d)$ such that $0 \leq \phi \leq 1$ and $\int \phi = 1$. Now we define

$$\Lambda_n u := \zeta_n u, \quad \Lambda_{m,n} u := T_m(\zeta_n u), \quad \Lambda_{l,m,n} u(x) = \phi_{1/l} * T_m(\zeta_n u).$$

where $\phi_\varepsilon(x) = \varepsilon^{-d} \phi(x/\varepsilon)$. Observe that we have the following inequality:

$$\|\Lambda_{l,m,n} u\|_{H_p^\gamma} \leq \|T_m(\zeta_n u)\|_{H_p^\gamma} \int \phi dx = \|T_m(\zeta_n u)\|_{H_p^\gamma}.$$

Because of Corollary 1.0.22, eq. (59), and theories about mollification, (i), (ii), and (iii) is now proved.

For (iv), since our construction of Λ_n , $\Lambda_{m,n}$, and $\Lambda_{l,m,n}$ are just a multiplication and mollification of smooth functions, one can easily check that (iv) holds. This proves the lemma.

Remark 8.2.2 Upper proof works similarly for $W_{p,\theta}^\gamma$.

To show that $C_0^\infty(\mathbb{R}_+ \times \mathbb{R}_+^d)$ is dense in $\mathbb{H}_{p,\theta}^\gamma(T)$, fix $u \in \mathbb{H}_{p,\theta}^\gamma(T)$ first. Notice that $u(t, \cdot) \in H_{p,\theta}^\gamma$ for almost every $t \in [0, T]$, thus there exists a modification $\bar{u}$ of u such that $\bar{u}(t, \cdot) \in H_{p,\theta}^\gamma$ for every $t \in [0, T]$. We will approximate $\bar{u}$ by smooth functions. After this, we write u instead of $\bar{u}$.

By Lemma 8.2.1, we have

$$\|\Lambda_{l,m,n}u(t, \cdot)\|_{\gamma,p,\theta} \leq N\|u(t, \cdot)\|_{\gamma,p,\theta},$$

and $\Lambda_{l,m,n}u(t, \cdot) \rightarrow u(t, \cdot)$ on $H_{p,\theta}^\gamma$ as $l, m, n \rightarrow \infty$ for every $t \in (0, T)$. Thus by the dominated convergence theorem, $\|\Lambda_{l,m,n}u - u\|_{\mathbb{H}_{p,\theta}^\gamma(T)} \rightarrow 0$ as $n \rightarrow \infty$. Furthermore, notice that $\Lambda_{l,m,n}u(t, \cdot)$ is $C_0^\infty(\mathbb{R}_+^d)$ for every n and for every $t \in (0, T)$. By taking subsequence, one can take a sequence u_n such that $u_n(t, \cdot)$ is $C_0^\infty(\mathbb{R}_+^d)$ for every $t \in (0, T)$, and $u_n \rightarrow u$ in $\mathbb{H}_{p,\theta}^\gamma(T)$ as $n \rightarrow \infty$.

Now consider $\eta_m \in C_0^\infty(\mathbb{R}_+)$ such that $0 \leq \eta_m \leq 1$, $\text{supp } \eta_m \in (1/m, T - 1/m)$, and $\eta_m = 1$ on $[2/m, T - 2/m]$. Clearly

$$\|\eta_m u_n\|_{\mathbb{H}_{p,\theta}^\gamma(T)}^p = \int_0^T \eta_m(t) \|u_n(t, \cdot)\|_{\gamma,p,\theta}^p dt \leq \|u_n\|_{\mathbb{H}_{p,\theta}^\gamma(T)}^p$$

holds for every m , and $\|1 - \eta_m(t)\| \|u_n(t, \cdot)\|_{\gamma,p,\theta} \rightarrow 0$ as $m \rightarrow \infty$ for every $t \in [0, T]$. The dominated convergence theorem gives $\eta_m u_n \rightarrow u_n$ in $\mathbb{H}_{p,\theta}^\gamma(T)$ as $m \rightarrow \infty$.

Finally consider $\phi \in C_0^\infty(\mathbb{R})$ such that $\text{supp } \phi \in (0, 1)$, $0 \leq \phi \leq 1$, and $\int \phi dt = 1$, and $\phi_\epsilon(t) = \epsilon^{-1} \phi(t/\epsilon)$. Then by the standard mollification argument gives that if we define

$$v_{\epsilon,m,n}(t, x) := \int_{\mathbb{R}} \phi_\epsilon(s) (\eta_m u_n)(t - s, x) ds,$$

then $v_{\epsilon,m,n} \rightarrow \eta_m u_n$ in $\mathbb{H}_{p,\theta}^\gamma(T)$ as $\epsilon \downarrow 0$. Finally observe that $v_{\epsilon,m,n} \in C_0^\infty(\mathbb{R}_+ \times \mathbb{R}_+^d)$.

Next, we will prove that $C_0^\infty([0, T] \times \mathbb{R}_+^d)$ is dense in $\mathfrak{H}_{p,\theta}^\gamma(T)$. Fix $u \in \mathfrak{H}_{p,\theta}^\gamma(T)$. Take a subsequence u_n of $\Lambda_{l,m,n}u$. Then Lemma 8.2.1 implies $u_{nt} = \Lambda_n u_t$, and thus

$$\begin{aligned} \|u_n - u\|_{\mathfrak{H}_{p,\theta}^\gamma(T)} &= \|M^{-1}(u_n - u)\|_{\mathbb{H}_{p,\theta}^\gamma(T)} + \|M(u_{nt} - u_t)\|_{\mathbb{H}_{p,\theta}^{\gamma-2}(T)} + \|M^{2/p-1}(u_n(0, \cdot) - u(0, \cdot))\|_{\gamma-2/p,p,\theta} \\ &= \|u_n - u\|_{\mathbb{H}_{p,\theta-p}^\gamma(T)} + \|\Lambda_n u_t - u_t\|_{\mathbb{H}_{p,\theta+p}^{\gamma-2}(T)} + \|u_n(0, \cdot) - u(0, \cdot)\|_{\gamma-2/p,p,\theta+2-p} \\ &\rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$.

To approximate u_n with smooth functions, together preserving value at $t = 0$, extend $u_n(t, \cdot) = u_n(0, \cdot)$ when $t \leq 0$ and $u_n(t, \cdot) = 0$ for $t > T$. After that, define

$$v_{\epsilon,n}(t, x) := (\phi_\epsilon *_t u_n)(t, x) := \int_{\mathbb{R}} \phi_\epsilon(s) u_n(t - s, x) ds.$$

Since the support of ϕ lie in $(0, 1)$ and our way to extend values of $u_n(t, \cdot)$ for negative t , we have

$$v_{\epsilon,n}(0, x) = \int_0^1 \phi_\epsilon(s) u_n(-s, x) ds = \int_0^1 \phi_\epsilon(s) u_n(0, x) ds = u_n(0, x).$$

In addition, for every $\psi \in C_0^\infty(\mathbb{R}_+^d)$ and $t \in [0, T]$, as $u_{nt} = \Lambda_n u_t \in M^{-1}\mathbb{H}_{p,\theta}^\gamma(T)$,

$$\begin{aligned} \int_{\mathbb{R}} \int_0^T \phi_\epsilon(s) |(u_{nt}(\sigma, \cdot), \psi)| d\sigma ds &\leq \int_{\mathbb{R}} \int_0^T \phi_\epsilon(s) \|u_{nt}(\sigma, \cdot)\|_{\gamma,p,\theta+p} \|\psi\|_{-\gamma,p',\theta'} d\sigma ds \\ &\leq \|\psi\|_{-\gamma,p',\theta'} \int_0^T \|u_{nt}(\sigma, \cdot)\|_{\gamma,p,\theta+p} d\sigma \\ &\leq \|\psi\|_{-\gamma,p',\theta'} T^{1/p'} \|u_{nt}\|_{\mathbb{H}_{p,\theta+p}^\gamma(T)} < \infty. \end{aligned}$$

Thus we can apply the Fubini's theorem to obtain

$$\begin{aligned}
 (v_{\epsilon,n}(t, \cdot), \psi) &= \int_{\mathbb{R}} \phi_{\epsilon}(s)(u_n(t-s, \cdot), \psi)ds \\
 &= \int_{\mathbb{R}} \phi_{\epsilon}(s) \left[(u_n(0, \cdot), \psi) + \int_0^{t-s} (u_{nt}(\sigma, \cdot), \psi)d\sigma \right] ds \\
 &= \int_{\mathbb{R}} \phi_{\epsilon}(s) \left[(u_n(-s, \cdot), \psi) + \int_{-s}^{t-s} (u_{nt}(\sigma, \cdot), \psi)d\sigma \right] ds \\
 &= \int_{\mathbb{R}} \phi_{\epsilon}(s)(u_n(-s, \cdot), \psi)ds + \int_{-s}^{t-s} \int_{\mathbb{R}} \phi_{\epsilon}(s)(u_{nt}(\sigma, \cdot), \psi)dsd\sigma \\
 &= \int_{\mathbb{R}} \phi_{\epsilon}(s)(u_n(-s, \cdot), \psi)ds + \int_0^t \int_{\mathbb{R}} \phi_{\epsilon}(s)(u_{nt}(\sigma-s, \cdot), \psi)dsd\sigma \\
 &= (v_{\epsilon,n}(0, \cdot), \psi) + \int_0^t ((\phi_{\epsilon} * u_{nt})(s, \cdot), \psi)ds.
 \end{aligned}$$

This, therefore, implies

$$\|M(\partial_t v_{\epsilon,n} - u_{nt})\|_{\mathbb{H}_{p,\theta}^{\gamma-2}(T)} = \|\phi_{\epsilon} * u_{nt} - u_{nt}\|_{\mathbb{H}_{p,\theta+p}^{\gamma-2}(T)} \rightarrow 0$$

as $\epsilon \downarrow 0$, and thus $\|v_{\epsilon,n} - u_n\|_{\mathfrak{S}_{p,\theta}^{\gamma}(T)} \rightarrow 0$ as $\epsilon \downarrow 0$. Finally, observe that $v_{\epsilon,n} \in C_0^{\infty}([0, T] \times \mathbb{R}_+^d)$.

8.3 The Poisson Kernel

For $t > 0$ and $x \in \mathbb{R}^d$, the Poisson kernel $P(t, x)$ is defined by

$$P(t, x) = \frac{2t}{\omega_{d+1}(|x|^2 + t^2)^{(d+1)/2}},$$

where $\omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ is the area of d dimensional sphere.

We first calculate $\|P(t, \cdot)\|_p$ and $\|\partial_t P(t, \cdot)\|_p$, respectively. For the first one, observe that $P(t, x) = t^{-d}P(1, t^{-1}x)$. Thus

$$\int_{\mathbb{R}^d} P^p(t, x)dx = t^{-dp} \int_{\mathbb{R}^d} P^p(1, t^{-1}x)dx = t^{-d(p-1)} \int_{\mathbb{R}^d} P^p(1, x)dx.$$

Especially, if $p = 1$, by using the polar coordinate,

$$\int_{\mathbb{R}^d} P(1, x)dx = \frac{2\omega_d}{\omega_{d+1}} \int_0^{\infty} \frac{r^{d-1}}{(r^2 + 1)^{(d+1)/2}} dr = \frac{2\Gamma((d+1)/2)}{\Gamma(d/2)\sqrt{\pi}} \int_0^{\infty} \frac{r^{d-1}}{(r^2 + 1)^{(d+1)/2}} dr.$$

By changing the variable $s = r^2/(r^2 + 1)$ (then $r^2 = s/(1-s)$) and recall the following formula

$$\int_0^1 t^{a-1}(1-t)^{b-1}dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

we have

$$\int_0^{\infty} \frac{r^{d-1}}{(r^2 + 1)^{(d+1)/2}} dr = \frac{1}{2} \int_0^1 s^{(d-2)/2}(1-s)^{-1/2}ds = \frac{\Gamma(d/2)\sqrt{\pi}}{2\Gamma((d+1)/2)}, \quad \int_{\mathbb{R}^d} P(1, x)dx = 1.$$

Thus for $1 \leq p < \infty$, we obtain $\|P(t, \cdot)\|_p = N(d, p)t^{-d(p-1)/p}$, where $N(d, 1) = 1$. In addition, it is easy to show that $\|P(t, \cdot)\|_{\infty} = N(d)t^{-d}$.

On the other hand, one can easily find that

$$\partial_t P(t, x) = t^{-1}P(t, x) - \frac{2(d+1)t^2}{\omega_{d+1}(|x|^2 + t^2)^{(d+3)/2}}, \quad \partial_t P(t, x) = t^{-d-1}Q(t^{-1}x),$$

where

$$Q(x) = P(1, x) - \frac{2(d+1)}{\omega_{d+1}(|x|^2 + 1)^{(d+3)/2}}.$$

Therefore, for $1 \leq p < \infty$, we have $\|\partial_t P(t, \cdot)\|_p = N(d, p)t^{-(p-1)(d+1)/p-p^{-1}}$, and $\|\partial_t P(t, \cdot)\|_\infty = N(d)t^{-d-1}$ by letting $p \rightarrow \infty$.

It is known that (see, for instance, [Fol95]) for any $g \in L_p(\mathbb{R}^d)$, where $1 \leq p \leq \infty$, $u(x) = g * P(x^1, \cdot)(x')$ is well-defined on $\mathbb{R}_+^{d+1}$ and is harmonic there. If $p < \infty$, then $u(x^1, \cdot) \rightarrow g$ in L_p as $x^1 \downarrow 0$.

In addition, let $1 \leq p, q, r \leq \infty$ such that $1 + r^{-1} = p^{-1} + q^{-1}$. Then by the Young's inequality, for $x^1 > 0$,

$$\|u(x^1, \cdot)\|_r \leq \|g\|_p \|P(x^1, \cdot)\|_q = N_1(d, q)(x^1)^{-d(q-1)/q} \|g\|_p \quad \left(\frac{\infty}{\infty} := 1\right);$$

$$\|u_{x^1}(x^1, \cdot)\|_r \leq \|g\|_p \|P_{x^1}(x^1, \cdot)\|_q = N_2(d, q)(x^1)^{-(q-1)(d+1)/q-q^{-1}} \|g\|_p \quad \left(\frac{\infty}{\infty} := 1, \frac{1}{\infty} := 0\right),$$

where $N_1(d, 1) = 1$. Especially, if $q = 1$,

$$\|u(x^1, \cdot)\|_p \leq \|g\|_p, \quad \|u_{x^1}(x^1, \cdot)\|_p \leq N(d)(x^1)^{-1} \|g\|_p.$$

8.4 Additional Informations about Lemma 4.0.8

In this section, we introduce the proof that if $f \in C_0^\infty(\mathbb{R}^d)$ and $u = (\lambda - \Delta)^{-1} f$, then u and all derivatives tends to zero exponentially as $|x| \rightarrow \infty$. More precisely, for every multiindex α ,

$$|D^\alpha u(x)| \leq N e^{-\mu|x|},$$

where $N = N(\alpha, d, \lambda, f) > 0$ and $\mu = \mu(\lambda) > 0$.

To prove this, observe that

$$G_\lambda(x) := \int_0^\infty e^{-\lambda t} p(t, x) dt, \quad p(t, x) := \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{|x|^2}{4t}\right).$$

is the Green's function to $(\lambda - \Delta)^{-1}$.

Let $\mu > 0$ be a constant specified in later. By Young's inequality,

$$\mu|x| \leq \frac{\epsilon}{8t}|x|^2 + 2t\mu^2, \quad e^{\mu|x|} p(t, x) \leq (4\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{8t}\right) e^{2t\mu^2}.$$

By integrating over x , we have^(xi)

$$\|e^{\mu|\cdot|} p(t, \cdot)\|_1 \leq 2^{d/2} e^{2t\mu^2}.$$

If we take μ such that $2\mu^2 - \lambda < 0$, we therefore obtain

$$\|e^{\mu|\cdot|} G_\lambda\|_1 \leq \int_0^\infty e^{-\lambda t} \|e^{\mu|\cdot|} p(t, \cdot)\|_1 dt \leq N(d)(\lambda - 2\mu^2).$$

Since G_λ is the Green's function, if $u = (\lambda - \Delta)^{-1} f$, we have $u = G_\lambda * f$. Therefore,

$$e^{\mu|x|} |u(x)| \leq \int_{\mathbb{R}^d} e^{\mu|x|} |G_\lambda(x-y)| |f(y)| dy \leq \int_{\mathbb{R}^d} (e^{\mu|x-y|} |G_\lambda(x-y)|) (e^{\mu|y|} |f(y)|) dy \leq \|e^{\mu|\cdot|} f\|_\infty \|e^{\mu|\cdot|} G_\lambda\|_1,$$

$$|u(x)| \leq N(d, \lambda, f) e^{-\mu|x|}.$$

Similarly, since $D^\alpha u = G_\lambda * D^\alpha f$, we can obtain the similar result. Here, the constant N also depends on α .

8.5 The heat Kernel and Boundedness of a Solution

Fix $T \in (0, \infty)$ and suppose that $f \in C_0^\infty(\mathbb{R}_+ \times \mathbb{R}^d)$. It is well-known that the function

$$u(t, x) = \int_0^t \int_{\mathbb{R}^d} p(t-s, x-y) f(s, y) dy ds, \tag{60}$$

satisfies $u_t = \Delta u + f$ with zero initial condition, where $p(t, x) = \mathbb{1}_{t>0} (4\pi t)^{-d/2} \exp(-|x|^2/(4t))$, a heat kernel. Since $\|p(t, \cdot)\|_1 = 1$ for every $t > 0$, we have

^(xi) It is easy to calculate $\int_{\mathbb{R}^d} e^{-a|x|^2} dx = (\pi/a)^{d/2}$ where $a > 0$.

$$\|u(t, \cdot)\|_p \leq \int_0^t \|p(t-s, \cdot) * f(s, \cdot)\|_p ds \leq \int_0^t \|p(t-s, \cdot)\|_1 \|f(s, \cdot)\|_p ds \leq \int_0^t \|f(s, \cdot)\|_p ds.$$

Notice that as f is infinitely differentiable in x , same argument shows that $u \in W_p^{1,2}(T)$. Hence, by the uniqueness on $W_p^{1,2}(T)$, every function satisfying $u_t = \Delta u + f$ can be represented by eq. (60) for $t \in [0, T]$ and $x \in \mathbb{R}^d$.

We are going to estimate u with the similar method as in Section 8.4. Fix constants $\mu > 0$, $\nu > 0$, and $\epsilon > 0$ will be specified later. By Young's inequality, we obtain

$$\mu|x| \leq \frac{\epsilon|x|^2}{2t} + \frac{\mu^2 t}{2\epsilon}, \quad e^{\mu|x|} p(t, x) \leq \mathbb{1}_{t>0} (4\pi t)^{-d/2} e^{\mu^2 t/(2\epsilon)} \exp\left(\frac{(2\epsilon-1)|x|^2}{4t}\right).$$

In order that the integral of the right hand side under x is finite, ϵ should be in range $(0, 1/2)$, and in that case, we have

$$\int_{\mathbb{R}^d} e^{\mu|x|} p(t, x) dx \leq N(\epsilon, d) e^{\mu^2 t/(2\epsilon)}.$$

Therefore,

$$\begin{aligned} e^{\mu|x|-\nu t} |u(t, x)| &\leq e^{\mu|x|-\nu t} \int_0^t \int_{\mathbb{R}^d} p(t-s, x-y) |f(s, y)| dy ds \\ &\leq \int_0^t \int_{\mathbb{R}^d} |e^{\mu|x-y|-\nu(t-s)} p(t-s, x-y)| |e^{\mu|y|-\nu s} f(s, y)| dy ds \\ &\leq \left(\sup_{s,y} e^{\mu|y|-\nu s} |f(s, y)| \right) \int_0^t \int_{\mathbb{R}^d} e^{\mu|y|-\nu s} p(s, y) dy ds \\ &\leq N \int_0^\infty \exp\left(-\nu s + \frac{\mu^2 s}{2\epsilon}\right) ds. \end{aligned}$$

Thus if $\mu^2 < 2\nu\epsilon$ and $0 < \epsilon < 1/2$, the integral is finite and the estimate is meaningful. Take $\mu = \sqrt{2}$, $\nu = 8$, and $\epsilon = 1/4$, for instance. Then we have the following inequality

$$|u(t, x)| \leq N e^{-\sqrt{2}|x|+8t}, \quad t \in [0, T], \quad x \in \mathbb{R}^d,$$

where $N = N(d, f)$.

In addition, since f is smooth in x , we have $D_x^\alpha u = p * D_x^\alpha f$ for every multiindex α . Therefore, we also obtain the following inequality:

$$|(D_x^\alpha u)(t, x)| \leq N e^{-\sqrt{2}|x|+8t}, \quad t \in [0, T], \quad x \in \mathbb{R}^d,$$

where $N = N(d, \alpha, f)$.

8.6 The space $C_0^\infty(\Omega)$ is separable topological space

Let Ω be an open set of $\mathbb{R}^d$. In this section, we prove that $C_0^\infty(\Omega)$ is a separable topological vector space. Before studying that space, we instead study $C_0^\infty(K)$ where K is a compact subset of $\mathbb{R}^d$.

First of all, clearly $C_0^\infty(K)$ is a vector space. We define a seminorm p_α for each multiindex α on $C_0^\infty(K)$ by

$$p_\alpha(\phi) := \sup_{x \in K} |D^\alpha \phi(x)|.$$

Consider $q_n(x) = c_n(1 - |x|^2)^n$ such that $\int_{|x| \leq 1} q_n(x) dx = 1$. Using the polar coordinate, we can obtain

$$1 = \int_{|x| \leq 1} c_n(1 - |x|^2)^n dx = \int_0^1 \int_{|x|=r} c_n(1 - r^2)^n d\sigma dr = N(d) \int_0^1 c_n(1 - r^2)^n r^{d-1} dr.$$

Here, if $d = 1$, we obtain $c_n \leq \sqrt{n}$, and if $d \geq 2$, then $c_n \leq Nn$ where $N = N(d)$ is a constant. So define $\kappa_1(n) = \sqrt{n}$ and $\kappa_d(n) = Nn$ for $d \geq 2$ to write simply.

For given $f \in C_0^\infty(K)$, define

$$f_n(x) := \int_{|y| \leq 1} q_n(y) f(x-y) dy.$$

One can easily notice that f_n is a polynomial, and since f is infinitely differentiable,

$$D^\alpha f_n(x) := \int_{|y| \leq 1} q_n(y) (D_x^\alpha f)(x - y) dy.$$

Now fix $\epsilon > 0$ and a positive integer k . Since $K + B[0, 1]$ is compact, we can take $\delta \in (0, 1)$ such that $|D^\alpha f(x) - D^\alpha f(x - y)| < \epsilon$ if $|y| < \delta$ for every $x \in K$, and multiindex α with $|\alpha| \leq k$. Then we obtain

$$\begin{aligned} \sup_{x \in K} |D^\alpha f(x) - D^\alpha f_n(x)| &\leq \sup_{x \in K} \int_{|y| \leq 1} q_n(y) |D^\alpha f(x - y) - D^\alpha f_n(x - y)| dy \\ &\leq 2 \left(\sup_{y \in K + B[0, 1]} |D^\alpha f(y)| \right) \int_{\delta \leq |y| \leq 1} q_n(y) dy + \epsilon \int_{|y| < \delta} q_n(y) dy \\ &\leq 2 \left(\sup_{y \in K + B[0, 1]} |D^\alpha f(y)| \right) \int_{\delta \leq |y| \leq 1} \kappa_d(n) (1 - \delta^2)^n dy + \epsilon \\ &= N(K, d, k, \delta, f) \kappa_d(n) (1 - \delta^2)^n + \epsilon \\ &\rightarrow \epsilon \end{aligned}$$

as $n \rightarrow \infty$ because $0 < 1 - \delta^2 < 1$ and $\kappa_d(n) = O(n)$. Notice that this argument holds for all k with same f_n . Hence, we proved that the set of all polynomials is dense in $C_0^\infty(K)$.

On $C_0^\infty(\Omega)$, consider a sequence of compact sets $(K_n)_n$ such that $K_n \uparrow \Omega$ as $n \rightarrow \infty$. The topology on $C_0^\infty(\Omega)$ consists of all seminorms $p_{n, \alpha}$ by

$$p_{n, \alpha}(\phi) = \sup_{x \in K_n} |D^\alpha \phi(x)|.$$

Hence, the set of all polynomials is also dense in $C_0^\infty(\Omega)$. Now one can obviously take a countable collection of polynomials which is still dense in $C_0^\infty(\Omega)$.

8.7 The proof of Lemma 1.0.11

Lemma 8.7.1 Let A be a $d \times d$ nonzero real matrix and $\gamma \in \mathbb{R}$. Consider

$$f_A(x) = \left(\frac{(Ax, x)}{|x|^2} \right)^\gamma, \quad x \in \mathbb{R}^d$$

Then for every multiindex α , we have

$$\sup_x |D^\alpha f_A(x)| \leq \|A_{\text{sym}}\|^\gamma |x|^{-|\alpha|} \max_{\|B\|=1} \max_{|x|=1} |D^\alpha f_B(x)| =: N(d, \alpha) \|A_{\text{sym}}\|^\gamma |x|^{-|\alpha|},$$

where $A_{\text{sym}} = (A + A^*)/2$.

proof. As we are dealing with real values, $(Ax, x) = (A_{\text{sym}}x, x)$ holds for every $x \in \mathbb{R}^d$. Thus it suffices to consider only for symmetric A .

Since f is homogeneous, we have

$$\sup_x |D^\alpha f_A(x)| = |x|^{-|\alpha|} \sup_{|x|=1} |D^\alpha f_A(x)|.$$

Here, observe that $D^\alpha f_A(x) = \|A\|^\gamma D^\alpha f_{A/\|A\|}(x)$, thus below inequality is obtained.

$$\sup_x |D^\alpha f_A(x)| \leq \|A_{\text{sym}}\|^\gamma |x|^{-|\alpha|} \sup_{\|B\|=1} \sup_{|x|=1} |D^\alpha f_B(x)| =: I \|A\|^\gamma |x|^{-|\alpha|}.$$

In order to show that I is finite, notice that $(x, B) \mapsto D^\alpha f_B(x)$ is a rational function on $\mathbb{R}^{d+d^2}$ if we identify a $d \times d$ matrix with an element of $\mathbb{R}^{d^2}$. Thus it is continuous on $\mathbb{R}^{d+d^2} \setminus \{0\}$. Since the set

$$\{(x, B) : |x| = 1, \|B\| = 1\}$$

is compact, finiteness of I is proved, and the lemma is also proved.

Lemma 8.7.2 Let σ be an invertible $d \times d$ matrix and $\gamma \in \mathbb{R}$. Consider

$$f(x) = \left(\frac{1 + |\sigma^* x|^2}{1 + |x|^2} \right)^\gamma.$$

Then there exists $\kappa > 0$ such that for every multiindex α ,

$$\sup_x |D^\alpha f(x)| \leq N(\lambda^{-\kappa}(\sigma) + \Lambda^\kappa(\sigma)) |x|^{-|\alpha|}$$

where N depends only on d and α .

proof. Consider

$$A = \begin{pmatrix} 1 & 0 \\ 0 & \sigma \sigma^* \end{pmatrix}.$$

Since $|\sigma^* x|^2 = (\sigma^* x, \sigma^* x) = (\sigma \sigma^* x, x)$, we have $f(x) = g(1, x)$, where

$$g(x) = \left(\frac{(Ax, x)}{|x|^2} \right)^\gamma, \quad x \in \mathbb{R}^{d+1}.$$

By Lemma 8.7.1, we find

$$\sup_{x \in \mathbb{R}^d} |D^\alpha f(x)| \leq \sup_{x \in \mathbb{R}^{d+1}} |D^\alpha g(x)| \leq N(d, \alpha) \|A\|^\gamma |x|^{-|\alpha|}.$$

One can easily check that $\|A\| = 1 \vee |\lambda(\sigma)| \vee |\Lambda(\sigma)|$.

proof (Proof of eq. (4)). By the change of variables in spaces H_p^γ , if u_n converges to u in H_p^γ , then $u_n(\sigma \cdot)$ converges to $u(\sigma \cdot)$ in H_p^γ . Thus it suffices to show eq. (4) for every $u \in \mathcal{S}$.

Let $\mathcal{F}, \mathcal{F}^{-1}$ denote the Fourier transform and Fourier inverse transform, respectively. Let $v = u(\sigma \cdot)$. Notice that

$$\mathcal{F}(v)(\xi) = |\det \sigma|^{-1} \mathcal{F}(u)(\sigma^{-*} \xi),$$

where $\sigma^{-*} = (\sigma^*)^{-1}$. From this, we have, by the definition of H_p^γ -norm and the changing of variables in spaces L_p and in Fourier transform,

$$\begin{aligned} \|u(\sigma \cdot)\|_{\gamma, p} &= \|v\|_{\gamma, p} \\ &= \|(1 - \Delta)^{\gamma/2} v\|_p \\ &= \left\| \mathcal{F}^{-1} [(1 + |\xi|^2)^{\gamma/2} \mathcal{F}(v)(\xi)] \right\|_p \\ &= \left\| \mathcal{F}^{-1} [(1 + |\xi|^2)^{\gamma/2} |\det \sigma|^{-1} \mathcal{F}(u)(\sigma^{-*} \xi)] \right\|_p \\ &= \left\| |\det \sigma^*| \mathcal{F}^{-1} [(1 + |\sigma^* \xi|^2)^{\gamma/2} |\det \sigma|^{-1} \mathcal{F}(u)(\xi)] (\sigma \cdot) \right\|_p \\ &= \left\| \mathcal{F}^{-1} [(1 + |\sigma^* \xi|^2)^{\gamma/2} \mathcal{F}(u)(\xi)] (\sigma \cdot) \right\|_p \\ &= |\det \sigma|^{-p} \left\| \mathcal{F}^{-1} [(1 + |\sigma^* \xi|^2)^{\gamma/2} \mathcal{F}(u)(\xi)] \right\|_p \\ &= |\det \sigma|^{-p} \left\| \mathcal{F}^{-1} \left[\frac{(1 + |\sigma^* \xi|^2)^{\gamma/2}}{(1 + |\xi|^2)^{\gamma/2}} (1 + |\xi|^2)^{\gamma/2} \mathcal{F}(u)(\xi) \right] \right\|_p \\ &=: |\det \sigma|^{-p} \left\| \mathcal{F}^{-1} [M(\xi)(1 + |\xi|^2)^{\gamma/2} \mathcal{F}(u)(\xi)] \right\|_p \\ &= |\det \sigma|^{-p} \left\| \mathcal{F}^{-1} [M(\xi) \mathcal{F}((1 - \Delta)^{\gamma/2} u)(\xi)] \right\|_p. \end{aligned}$$

First of all, we have

$$|\det \sigma| = |\det \sigma \sigma^*|^{1/2} \geq \lambda^{d/2}(\sigma).$$

In addition, by Theorem 2 in Appendix of [Mik65], if M is the constant as in the theorem in [Mik65], then we have

$$\left\| \mathcal{F}^{-1} [M(\xi) \mathcal{F}((1 - \Delta)^{r/2} u)(\xi)] \right\|_p \leq N(p, d) M \left\| (1 - \Delta)^{r/2} u \right\|_p = N(p, d) M \|u\|_{r,p}.$$

By Lemma 8.7.2, the inequality is now proved.

8.8 The proof of Lemma 2.0.18

Our assertion are nontrivial only if $p \in (1, 2)$, when one may encounter 0^{-1} in eq. (32).

Before proving the lemma, we should prove that u is bounded on $[a, b] \cap \mathbb{R}$. Since $u \in W_p^2(a, b)$, it is locally absolutely continuous on $[a, b] \cap \mathbb{R}$ and $u_x \in L_p(a, b)$. Since $p > 1$, $|u|^p$ is locally also absolutely continuous and $(|u|^p)_x = |u|^{p-2} u u_x$.

Hence, for fixed $t_0 \in [a, b] \cap \mathbb{R}$, we have

$$\sup_{t \in [t_0, b] \cap \mathbb{R}} |u(t)|^p \leq |u(t_0)|^p + \sup_{t \in [t_0, b]} \int_{t_0}^t |u(r)|^{p-1} |u'(r)| dr \leq |u(t_0)| + \int_a^b |u(r)|^{p-1} |u'(r)| dr.$$

Now Hölder's inequality shows that

$$\int_a^b |u(r)|^{p-1} |u'(r)| dr \leq \|u\|_{L_p(a,b)}^{p-1} \|u'\|_{L_p(a,b)}.$$

Such approach is same on $[a, t_0] \cap \mathbb{R}$. Hence, $\sup |u| < \infty$.

To prove (i), first assume that a and b are finite, take a sequence of smooth even positive functions $g_n(t)$ such that $g_n \rightarrow |t|^{p/2-1} =: g(t)$ in $L_2(-A, A)$ where $A = \sup |u| < \infty$. First of all, although g is undefined at the origin, since $1 < p < 2$, g lies in $L_2(-A, A)$. In addition, it is easy to show that

$$F(t) := \int_0^t g(s) ds = \frac{2}{p} |t|^{p/2-1} t, \quad G(t) := \int_0^t g^2(s) ds = \frac{1}{p-1} |t|^{p-2} t.$$

One can take $g_n(t) = (t^2 + n^{-2})^{p/4-1/2}$ for instance. Indeed, as $1 < p < 2$, we have $0 \leq g_n \leq g$, and $|g_n - g| \leq g$. Since $g \in L_2(-A, A)$, by the dominated convergence theorem, $\|g_n - g\|_2 \rightarrow 0$ as $n \rightarrow \infty$.

Now define

$$F_n(t) := \int_0^t g_n(s) ds \quad G_n(t) := \int_0^t g_n^2(s) ds.$$

By the Hölder's inequality (here $A < \infty$ is used),

$$\sup_{t \in [-A, A]} \left| F_n(t) - \frac{2}{p} |t|^{p/2-1} t \right| \leq \int_{-A}^A |g_n(s) - g(s)| ds \leq (2A)^{1/2} \|g_n - g\|_{L_2(-A, A)} \rightarrow 0;$$

$$\sup_{t \in [-A, A]} \left| G_n(t) - \frac{1}{p-1} |t|^{p-2} t \right| \leq \int_{-A}^A |g_n^2(s) - g^2(s)| ds \leq \|g_n + g\|_{L_2(-A, A)} \|g_n - g\|_{L_2(-A, A)} \rightarrow 0$$

as $n \rightarrow \infty$. Next, as $u \in W_p^2(a, b)$, both u and u_x are locally absolutely continuous. Thus there exists a Lebesgue measurable set $E \subset [a, b]$ such that $[a, b] \setminus E$ has measure zero and on E ,

$$(G_n(u) u_x)_x = G'_n(u) (u_x)^2 + G_n(u) u_{xx}.$$

Taking an integration on both sides, together with the fundamental theorem of calculus, we obtain

$$\int_a^b G_n(u) u_{xx} dx = G_n(u(b)) u_x(b) - G_n(u(a)) u_x(a) - I_n, \quad (61)$$

where

$$I_n = \int_a^b G'_n(u) (u_x)^2 dx = \int_a^b g_n^2(u) (u_x)^2 dx = \int_a^b (v_{nx})^2 dx, \quad v_n = F_n(u).$$

Since $[-A, A]$ is compact and G_n is smooth, $G_n(u)$ is bounded on $[a, b]$. From this, we have $G_n(u)u_{xx} \in L_1(a, b)$. Since $G_n \rightarrow G$ uniformly on $[-A, A]$, the dominated convergence theorem gives that $G(u)u_{xx} \in L_1(a, b)$.^(xli) On the other hand, clearly we have

$$\int_a^b v_n^2(x) dx = \int_a^b \left(\int_0^{u(x)} g_n(s) ds \right)^2 dx \leq \int_a^b \left(\int_{-A}^A g_n(s) ds \right)^2 dx \leq 2A(b-a) \|g_n\|_{L_2(-A, A)}.$$

By the dominated convergence theorem we have

$$\begin{aligned} \overline{\lim}_{n \rightarrow \infty} \|v_n\|_{W_2^1(a, b)}^2 &\leq N \overline{\lim}_{n \rightarrow \infty} (\|v_n\|_{L_2(a, b)}^2 + \|v_{nx}\|_{L_2(a, b)}^2) \\ &\leq N \overline{\lim}_{n \rightarrow \infty} (\|v_n\|_{L_2(a, b)}^2 + \|v_{nx}\|_{L_2(a, b)}^2) \\ &\leq N \overline{\lim}_{n \rightarrow \infty} \left(G_n(u(b))u_x(b) - G_n(u(a))u_x(a) - \int_a^b G_n(u)u_{xx} dx \right) + N \overline{\lim}_{n \rightarrow \infty} \|v_{nx}\|_{L_2(a, b)}^2 \\ &\leq N \left(G(u(b))u_x(b) - G(u(a))u_x(a) - \int_a^b G(u)u_{xx} dx \right) + NA(b-a) \|g\|_{L_2(-A, A)} \\ &< \infty. \end{aligned}$$

From this, the sequence v_n is bounded in $W_2^1(a, b)$, and thus there exists a subsequence while using same index and $v \in W_2^1(a, b)$ such that $v_n \rightarrow v$ weakly in $W_2^1(a, b)$, thus weakly in $L_2(a, b)$. On the other hand, since $v_n \rightarrow F(u)$ uniformly on $[a, b]$, it also converges in $L_2(a, b)$ by the dominated convergence theorem. Therefore, $F(u)$ and v are indistinguishable, proving that $F(u) = (2/p)|u|^{p/2-1}u \in W_2^1(a, b)$. Its absolute value has the same property, so that $|u|^{p/2} \in W_1^1(a, b)$, which implies $|u|^{p/2}$ is absolutely continuous.

To go further, it is useful to prove that $v_n \rightarrow \frac{2}{p}|u|^{p/2-1}u$ in $W_2^1(a, b)$. To this end it suffices to prove that

$$\lim_{n \rightarrow \infty} \int_a^b (v_{nx})^2 dx = \int_a^b |u|^{p-2}(u_x)^2 dx. \quad (62)$$

It follows from eq. (61) that the limit times $p-1$ equals (by the way, we use the dominated convergence theorem)

$$|u(b)|^{p-2}u(b)u_x(b) - |u(a)|^{p-2}u(a)u_x(a) - \int_a^b |u|^{p-2}uu_{xx} dx.$$

On the other hand let $(a, b) = \bigcup_i (a_i, b_i)$ be a decomposition of (a, b) into disjoint intervals on each of which u has the same sign and vanishes at end points except perhaps for the cases $a_i = a$ and $b_i = b$.^(xlii) Then, integrating by parts, we get

$$\begin{aligned} (p-1) \int_{a_i}^{b_i} |u|^{p-2}(u_x)^2 dx &= \int_{a_i}^{b_i} u_x d(|u|^{p-2}u) \\ &= u_x(b_i)|u(b_i)|^{p-2}u(b_i) - u_x(a_i)|u(a_i)|^{p-2}u(a_i) - \int_{a_i}^{b_i} |u|^{p-2}uu_{xx} dx. \end{aligned} \quad (63)$$

Indeed, for each interval (a_j, b_j) , the sign of u is fixed, say for instance, $u \geq 0$ on (a_j, b_j) . Consider any $\epsilon > 0$. Then on $(a_j + \epsilon, b_j - \epsilon)$, $|u|^{p-2}$ is absolutely continuous because there is no point which makes $u = 0$.^(xliii) by using the integration by parts, one have

$$\begin{aligned} (p-1) \int_{a_i+\epsilon}^{b_i-\epsilon} |u|^{p-2}(u_x)^2 dx &= (p-1) \int_{a_i+\epsilon}^{b_i-\epsilon} u^{p-2}(u_x)^2 dx \\ &= \int_{a_i+\epsilon}^{b_i-\epsilon} u_x d(u^{p-1}) \end{aligned}$$

^(xli) $\|G_n u_{xx} - G u_{xx}\|_1 \leq \|G_n - G\|_\infty \|u_{xx}\|_1 \rightarrow 0$ as $n \rightarrow \infty$.

^(xlii) Recall that every open sets in $\mathbb{R}$ can be expressed as the disjoint union of countable open intervals. Now as u is continuous (actually it is absolutely continuous), one can consider $\{u > 0\}$ and $\{u < 0\}$, respectively. However, here, we do not decompose (a, b) itself. Indeed (a, b) is connected, but countable disjoint union makes the set non-connected. Actually, we decompose a set $(a, b) \cap \{u \neq 0\}$.

^(xliii) Notice that $u(a_j) = 0$ unless $a_j = a$, and $u(b_j) = 0$ unless $b_j = b$.

$$\begin{aligned}
 &= u_x(b_i - \epsilon)u(b_i - \epsilon)^{p-1} - u_x(a_i + \epsilon)u(a_i + \epsilon)^{p-1} - \int_{a_i + \epsilon}^{b_i - \epsilon} u^{p-1} d(u_x) \\
 &= u_x(b_i - \epsilon)u(b_i - \epsilon)^{p-1} - u_x(a_i + \epsilon)u(a_i + \epsilon)^{p-1} - \int_{a_i + \epsilon}^{b_i - \epsilon} u^{p-1} u_{xx} dx \\
 &= u_x(b_i - \epsilon)|u(b_i - \epsilon)|^{p-2}u(b_i - \epsilon) \\
 &\quad - u_x(a_i + \epsilon)|u(a_i + \epsilon)|^{p-2}u(a_i + \epsilon) - \int_{a_i + \epsilon}^{b_i - \epsilon} |u|^{p-2}uu_{xx} dx.
 \end{aligned}$$

Since $u \in W_2^1(a, b)$, at least both u and u_x are continuous. Thus letting $\epsilon \downarrow 0$ gives eq. (63). Similarly, one can obtain eq. (63) when $u \leq 0$. Here, notice that our case is not a telescoping series because there is no way to reorder a_j , and b_j . However recall that $u(a_j) = 0$ unless $a_j = a$, and $u(b_j) = 0$ unless $b_j = b$. That means, when summing up eq. (63), all terms except one before integral in the right hand side is zero, and thus summing up eq. (63) gives eq. (62).

Now notice that $(G_n(u)u_x)_x = (v_{nx})^2 + G_n(u)u_{xx}$, where, by the above conclusions, the right-hand side converges in $L_1(a, b)$.^(xlv) It follows that the limit of $G_n(x)u_x$ is absolutely continuous. The limit being $\frac{1}{p-1}|u|^{p-2}uu_x$, the lemma is proved.

8.9 The proof of Lemma 6.0.4

If $u_0 \in C_0^\infty(\mathbb{R}_+^d)$, then there exists a unique function $u(t, x)$ which is bounded in $(0, \infty) \times \mathbb{R}_+^d$ together with all its derivatives and which is unique bounded solution of the heat equation $u_t = \Delta u$, $t > 0$ in $(0, \infty) \times \mathbb{R}_+^d$ with $u(0, \cdot) = u_0$ and boundary condition $u(t, x) = 0$ for $x^1 = 0$. Observe that if $p_t(x) = (4\pi t)^{-d/2} \exp(-|x|^2/(4t))\mathbb{1}_{t>0}$, $u(t, \cdot) = p_t * \bar{u}_0$ where $\bar{u}_0(x) = (\text{sgn } x^1)u_0(|x^1|, x')$, an odd extension with respect to x^1 . Since p_t is even and $\bar{u}_0$ is odd of x^1 , so is u .

By Lemma 2.0.18, Fubini's theorem, and integration by parts, we get the following where $c \in (1 - p, 1)$ is a constant which will be specified later.

$$\begin{aligned}
 \int_{\mathbb{R}_+^d} (x^1)^c |u(T, x)|^p dx - \int_{\mathbb{R}_+^d} (x^1)^c |u_0(x)|^p dx &= \int_{\mathbb{R}_+^d} (x^1)^c |u(T, x)|^p dx - \int_{\mathbb{R}_+^d} (x^1)^c |u(0, x)|^p dx \\
 &= \int_0^T \int_{\mathbb{R}_+^d} (x^1)^c p |u|^{p-2} u \Delta u dx dt \\
 &= - \int_0^T \int_{\mathbb{R}_+^d} pc(x^1)^{c-1} |u|^{p-2} uu_{x^1} dx dt - p(p-1)I \\
 &= - \int_0^T \int_{\mathbb{R}_+^d} c(x^1)^{c-1} (D_1 |u|^p) dx dt - p(p-1)I \\
 &= c(c-1)J - p(p-1)I,
 \end{aligned}$$

where

$$I = \int_0^T \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} (u_x)^2 dx dt, \quad J = \int_0^T \int_{\mathbb{R}_+^d} (x^1)^{c-2} |u|^p dx dt.$$

However in the above calculation, we should be careful to check the behavior at the boundary in order to integrate by parts.^(xlv) Since $\bar{u}_0 \in C_0^\infty(\mathbb{R}^d)$, $D^\alpha u = p_t * (D^\alpha \bar{u}_0)$. Take $R > 0$ such that $\bar{u}_0(x) = 0$ if $|x| > R$. Then

$$|D^\alpha u(t, x)| \leq (4\pi t)^{-d/2} \int_{|x| \leq R} e^{-|x-y|^2/(4t)} |D^\alpha \bar{u}_0|(y) dy \leq (4\pi t)^{-d/2} \exp\left(\frac{R^2}{4t}\right) \exp\left(-\frac{|x|^2}{16t}\right) \|D^\alpha \bar{u}_0\|_1.$$

Hence $D^\alpha u(t, x) \rightarrow 0$ exponentially fast as $|x| \rightarrow \infty$. Since $u(t, 0, x') = 0$ for every $t > 0$ and $x' \in \mathbb{R}^{d-1}$, by the fundamental theorem of calculus,

^(xlv) Actually, this can be obtained by using specific g_n , for instance, $g_n \uparrow |t|^{p/2-1}$.

^(xlv) Especially, if $1 < p < 2$, some steps are nonsense in the first place because 0^{-1} appears on the set where $u = 0$. However thanks to Lemma 2.0.18, we can just calculate like in above.

$$|u(t, x)| = |u(t, x^1, x') - u(t, 0, x')| \leq |x^1| \int_0^1 |D_1 u(t, x^1 a, x')| da.$$

Since $D_1 p_t(x) = -x^1/(2t)p_t(x)$, $|D_1 p_t(x)| \leq N(d)t^{-(d+1)/2}$, we have $|u(t, x)| \leq N(d, t)|x^1|$. Using them, we can see that boundary values vanishes as $x^1 \rightarrow 0$ and $|x| \rightarrow \infty$.

Now we further evaluate J . Denote $v = |u|^{p/2}$. By Lemma 2.0.18, v is then absolutely continuous on every closed interval of x^1 , and $v_{x^1} = (p/2)|u|^{p/2-1}u_{x^1}$. Use the Minkowski's inequality additionally,

$$\begin{aligned} \int_{\mathbb{R}_+^d} (x^1)^{c-2} |u|^p dx &= \int_{\mathbb{R}_+^d} (x^1)^{c-2} |v|^2 dx \\ &= \int_{\mathbb{R}_+^d} (x^1)^{c-2} \left| x^1 \int_0^1 v_{x^1}(tx^1, x') dt \right|^2 dx \\ &\leq \left(\int_0^1 \left(\int_{\mathbb{R}_+^d} (x^1)^c |v_{x^1}(tx^1, x')|^2 dx \right)^{1/2} dt \right)^2 \\ &= \int_{\mathbb{R}_+^d} (x^1)^c |v_{x^1}|^2 dx \left(\int_0^1 t^{-(c+1)/2} dt \right)^2 \\ &= 4(1-c)^{-2} \int_{\mathbb{R}_+^d} (x^1)^c |v_{x^1}|^2 dx \\ &= p^2(1-c)^{-2} \int_{\mathbb{R}_+^d} (x^1)^c |u|^{p-2} (u_{x^1})^2 dx. \end{aligned}$$

Thus integrating both side with respect to t on $[0, T]$, we have $J \leq p^2(1-c)^{-2}I$. What we have shown is that

$$\begin{aligned} \int_{\mathbb{R}_+^d} (x^1)^c [|u(T, x)|^p - |u_0(x)|^p] dx &= c(c-1)J - p(p-1)I \leq [c(c-1) - q^{-1}(1-c)^2]J; \\ [q^{-1}(1-c)^2 - c(c-1)]J &\leq \int_{\mathbb{R}_+^d} (x^1)^c |u_0(x)|^p dx, \end{aligned}$$

where $q = p/(p-1)$. In order that the scalar is positive, we have $c \in (1-p, 1)$.

Put $c = \theta - d + 2 - p$. Since $d-1 < \theta < p+d-1$, this c satisfies $c \in (1-p, 1)$. For that c , we have

$$\int_{\mathbb{R}_+^d} (x^1)^c |u_0(x)|^p dx = \|M^{2/p-1}u_0\|_{0,p,\theta}^p, \quad J = \int_0^T \|M^{-1}u(t, \cdot)\|_{0,p,\theta}^p dt.$$

Therefore, we obtain the following estimate

$$\int_0^T \|M^{-1}u(t, \cdot)\|_{0,p,\theta}^p dt \leq N \|M^{2/p-1}u_0\|_{0,p,\theta}^p.$$

We need another estimate related to this. Since $d-1 < \theta < p+d-1$, by Corollary 2.0.13 and as $MDu \in C_0^\infty(\mathbb{R}_+^d)$,

$$\begin{aligned} \int_0^T \|M^{-1}u(t, \cdot)\|_{1,p,\theta}^p &\leq N \int_0^T \|DMM^{-1}u(t, \cdot)\|_{0,p,\theta}^p \\ &= N \int_0^T \|M^{-1}MDu(t, \cdot)\|_{0,p,\theta}^p \\ &\leq N \|M^{2/p-1}MDu_0\|_{0,p,\theta}^p \\ &\leq N \sum_{i=1}^d \|M^{2/p-1}MD_i u_0\|_{0,p,\theta}^p \\ &\leq N \|M^{2/p-1}u\|_{0,p,\theta}^p + N \|DM^{2/p}u_0\|_{0,p,\theta}^p \\ &\leq N \|M^{2/p-1}u\|_{1,p,\theta}^p. \end{aligned}$$

Next, take a function $\zeta \in C_0^\infty(\mathbb{R}_+)$ satisfying condition eq. (1). For $u_n(t, x) := u(e^{2n}t, e^n x)$ we have

$$(\zeta u_n)_t = \Delta(\zeta u_n) - 2(\zeta' u_n)_{x^1} + \zeta'' u_n. \quad (64)$$

Hence by inequalities (IV.3.1) and (IV.3.2) of [LzSU68] (also see Section 2.11 in [Tri83]) for any n we obtain

$$\int_0^T \|(\zeta u_n)_{xx}(t, \cdot)\|_{-1,p}^p dt \leq N \|\zeta u_n(0, \cdot)\|_{W_p^{1-2/p}}^p + \int_0^T \|((2\zeta' u_n)_{x^1} - \zeta'' u_n)(t, \cdot)\|_{-1,p}^p dt. \quad (65)$$

As in the proof of Lemma 4.0.7 (and Remark 1.0.15),

$$N(\zeta, p) \|(\zeta u_n)_{xx}\|_{-1,p} \geq \|\zeta u_{nx}\|_{0,p} - \|\zeta' u_n\|_{0,p}.$$

Also we use

$$\|(2\zeta' u_n)_{x^1} - \zeta'' u_n\|_{-1,p} \leq \|(2\zeta' u_n)_{x^1}\|_{-1,p} + N \|\zeta'' u_n\|_{0,p} \leq N \|\eta u_n\|_{0,p},$$

where η is any $C_0^\infty(\mathbb{R}_+)$ function whose support covers that of ζ .

Therefore, from eq. (65) we get

$$\begin{aligned} & \int_0^T e^{np} \|\zeta u_x(e^{2n}t, e^n \cdot)\|_{0,p}^p dt \\ & \leq N \|\zeta u(0, e^n \cdot)\|_{W_p^{1-2/p}}^p + N \int_0^T e^{np} \|(M^{-1}u)(e^{2n}t, e^n \cdot) M \zeta'\|_{0,p}^p dt + N \int_0^T e^{np} \|(M^{-1}u)(e^{2n}t, e^n \cdot) M \eta\|_{0,p}^p dt. \end{aligned}$$

By changing variable in t , and multiply $e^{2n-np+n\theta}$, we find

$$\begin{aligned} & \int_0^T e^{n\theta} \|\zeta u_x(t, e^n \cdot)\|_{0,p}^p dt \\ & \leq N e^{2n-np+n\theta} \|\zeta u(0, e^n \cdot)\|_{W_p^{1-2/p}}^p + N \int_0^T e^{n\theta} \|(M^{-1}u)(t, e^n \cdot) M \zeta'\|_{0,p}^p dt + N \int_0^T e^{n\theta} \|(M^{-1}u)(t, e^n \cdot) M \eta\|_{0,p}^p dt \\ & \leq N e^{n\theta} \|\zeta(M^{2/p-1}u)(0, e^n \cdot)\|_{W_p^{1-2/p}}^p + N \int_0^T e^{n\theta} \|(M^{-1}u)(t, e^n \cdot) M \zeta'\|_{0,p}^p dt + N \int_0^T e^{n\theta} \|(M^{-1}u)(t, e^n \cdot) M \eta\|_{0,p}^p dt. \end{aligned}$$

Now sum the up and apply Lemma 1.0.6

$$\int_0^T \|u_x(t, \cdot)\|_{0,p,\theta}^p dt \leq N \|M^{2/p-1}u_0\|_{W_{p,\theta}^{1-2/p}}^p + N \int_0^T \|M^{-1}u(t, \cdot)\|_{0,p,\theta}^p dt.$$

Similarly from eq. (64) and inequalities (IV.3.1) and (IV.3.2) of [LzSU68] to write

$$\int_0^T \|(\zeta u_n)_{xx}(t, \cdot)\|_{0,p}^p dt \leq N \|\zeta u_n(0, \cdot)\|_{W_p^{2-2/p}}^p + \int_0^T \|((2\zeta' u_n)_{x^1} - \zeta'' u_n)(t, \cdot)\|_{0,p}^p dt. \quad (66)$$

For the same reason as in above, if η_1 and η_2 are functions of class $C_0^\infty(\mathbb{R}_+)$ with supports covering that of ζ , we obtain

$$\int_0^T \|\zeta u_{nxx}(t, \cdot)\|_{0,p}^p dt \leq N \|\zeta u_n(0, \cdot)\|_{W_p^{2-2/p}}^p + \int_0^T \|\eta_1 u_{nx}\|_{0,p}^p dt + \int_0^T \|\eta_2 u_n\|_{0,p}^p dt,$$

and

$$\int_0^T \|Mu_{xx}(t, \cdot)\|_{0,p,\theta}^p dt \leq N \|M^{2/p-1}u_0\|_{W_{p,\theta}^{2-2/p}}^p + \int_0^T \|u_x(t, \cdot)\|_{0,p,\theta}^p dt + \int_0^T \|M^{-1}u(t, \cdot)\|_{0,p,\theta}^p dt.$$

Furthermore, we actually have

$$\int_0^T \|u_x(t, \cdot)\|_{1,p,\theta}^p dt \leq N \|M^{2/p-1}u_0\|_{W_{p,\theta}^{2-2/p}}^p + N \int_0^T \|M^{-1}u(t, \cdot)\|_{1,p,\theta}^p dt.$$

Therefore, combining all inequalities in above together with the equation $u_t = M^{-1}M\Delta u$ and $\|\Delta u\|_{0,p,\theta} \leq N \|u_{xx}\|_{0,p,\theta}$, we have

$$\|u\|_{\mathcal{H}_{p,\theta}^2(T)}^p \leq N \|u_x\|_{\mathbb{H}_{p,\theta}^1(T)}^p + N \|Mu_t\|_{\mathbb{L}_{p,\theta}(T)}^p + N \|u(0, \cdot)\|_{U_{p,\theta}^2}^p$$

$$\begin{aligned}
 &\leq N\|u_x\|_{\mathbb{H}_{p,\theta}^1(T)}^p + N\|Mu_{xx}\|_{\mathbb{L}_{p,\theta}(T)}^p + N\|u(0, \cdot)\|_{U_{p,\theta}^2}^p \\
 &\leq N\|u(0, \cdot)\|_{U_{p,\theta}^2}^p + N\|u_x\|_{\mathbb{L}_{p,\theta}(T)}^p + N\|M^{-1}u\|_{\mathbb{L}_{p,\theta}(T)}^p + N\|M^{-1}u\|_{\mathbb{H}_{p,\theta}^1(T)}^p \\
 &\leq N\|u(0, \cdot)\|_{U_{p,\theta}^2}^p + N\|M^{-1}u\|_{\mathbb{L}_{p,\theta}(T)}^p + N\|M^{-1}u\|_{\mathbb{H}_{p,\theta}^1(T)}^p \\
 &\leq N\|u(0, \cdot)\|_{U_{p,\theta}^2}^p.
 \end{aligned}$$

This proves Lemma 6.0.4 for $u_0 \in C_0^\infty(\mathbb{R}_+^d)$. Actually, above we have constructed a mapping

$$\Pi : C_0^\infty(\mathbb{R}_+^d) \ni u_0 \mapsto u \in \mathfrak{H}_{p,\theta}^2(T),$$

and

$$\|\Pi u_0\|_{\mathfrak{H}_{p,\theta}^2(T)} \leq N(p, \theta, d)\|u_0\|_{U_{p,\theta}^2}. \quad (67)$$

Since $\mathfrak{H}_{p,\theta}^2(T)$ is a Banach space and $C_0^\infty(\mathbb{R}_+^d)$ is dense in $U_{p,\theta}^2$, we can extend Π into whole space $U_{p,\theta}^2$ in a way that $(\Pi u_0)_t = \Delta \Pi u_0$, $\Pi u_0|_{t=0} = u_0$, and eq. (67) holds. This proves the lemma.

8.10 Another proof of the regularity of $M^{-1}u$ in the proof of Lemma 5.0.9

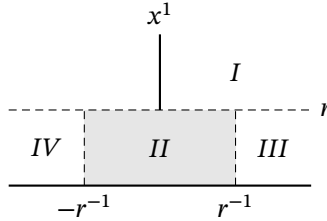
in the same way as in the proof of Lemma 4.2 of [KL99], using boundedness of all derivatives of u with respect to x , we will prove that

$$M^{-1}u \in \mathbb{H}_{p,\theta}^\gamma(T) \quad \forall \gamma \geq 0, \theta \in (d-1, p+d-1).$$

Since f has a compact support, there exists $r > 0$ such that the support of f lies in the domain

$$\Omega := (2r, r^{-1}) \times \{x : |x'| < r^{-1}\}.$$

First of all, we want to evaluate $\|M^{-1}u\|_{\mathbb{L}_{p,\theta}}$ on the following subregions (actually *III* and *IV* are symmetric).



Notice that then $u_t = \Delta u$ outside of Ω with zero initial and zero boundary value for $x^1 = 0$. If we set $u(t, x) = 0$ for $t < 0$, then it satisfies the heat equation for all $t \leq T$ and $x \in \mathbb{R}_+^d \setminus \Omega$. Similar method in the proof of Lemma 4.2 of [KL99] gives that for any multiindex α ,

$$\sup_{0 < x^1 < r, |x'| < r^{-1}, t \leq T} |D^\alpha u(t, x)| \leq \frac{N(\alpha)}{r^{|\alpha|}} \sup_{x^1 = 2r, |x'| = 2r, t \leq T} |u(t, x)|. \quad (68)$$

By eq. (68),^(xlv)

$$\begin{aligned}
 \int_0^T \int_{(0,r) \times \{|x'| < r\}} (x^1)^{\theta-d} |M^{-1}u(t, x)|^p dx dt &\leq N \sup_{x^1 = 2r, |x'| = 2r, t \leq T} |u(t, x)|^p \int_0^T \int_{(0,r) \times \{|x'| < r\}} (x^1)^{\theta-d} dx dt \\
 &\leq N T r^d \sup_{x^1 = 2r, |x'| = 2r, t \leq T} |u(t, x)|^p \int_0^1 (x^1)^{\theta-d} dx^1.
 \end{aligned}$$

Since $\theta > d-1$, the right-hand side integral is finite. Upon translating with respect to x' , applying same argument, and sum them up shows, therefore, that

$$\int_0^T \int_{II} (x^1)^{\theta-d} |M^{-1}u(t, x)|^p dx dt \leq N T r^d \sup_{x \in II', t \leq T} |u(t, x)| \int_0^1 (x^1)^{\theta-d} dx^1 \leq N(d, \theta, T, f) < \infty$$

^(xlv)In addition, $|M^{-1}u| \leq \sup |u_x|$.

where II' is some subset of $\mathbb{R}_+^d$ containing II and disjoint with I . In addition, as $(x^1)^{\theta-d-p} \leq 1$ if $x^1 \geq 1$, we have

$$\int_0^T \int_I (x^1)^{\theta-d} |M^{-1}u(t, x)|^p dt dx \leq \int_0^T \int_I |u(t, x)|^p dt dx \leq T \sup_{t \leq T} \|u(t, \cdot)\|_{L_p}^p < \infty,$$

because $u \in C([0, T], L_p)$.

The tricky part is to estimate III . By the translation with respect to x' , we can obtain the same inequality eq. (68) with r replaced by nr^{-1} for $n = 1, 2, \dots$. Since u is bounded because of the Sobolev embedding theorem, for $n = 1, 2, \dots$ that

$$\begin{aligned} \int_0^T \int_{(0,r) \times \{nr^{-1} \leq |x'| \leq (n+1)r^{-1}\}} (x^1)^{\theta-d} |M^{-1}u(t, x)|^p dx dt &\leq \frac{N}{n^p} \sup_{t,x} |u|^p \int_0^T \int_{(0,r) \times \{nr^{-1} \leq |x'| \leq (n+1)r^{-1}\}} (x^1)^{\theta-d} dx dt \\ &\leq \frac{N}{n^p} \sup_{t,x} |u|^p \int_0^1 (x^1)^{\theta-d} dx^1. \end{aligned}$$

Since $p > 1$, summing up with respect to n gives a finite estimate on III . Notice that IV is same, therefore, $M^{-1}u \in \mathbb{L}_{p,\theta}(T)$. Similarly, one can prove that for every multiindex α , (by the induction over $|\alpha|$)

$$\int_0^T \int_{\mathbb{R}_+^d} (x^1)^{\theta-d} |M^{|\alpha|-1} D^\alpha u| dx dt < \infty,$$

which proves $M^{-1}u \in \mathbb{H}_{p,\theta}^\gamma(T)$ for every nonnegative γ by Corollary 2.0.3.

Remark 5.0.4 shows that the same inclusions are true for u_x and by Remark 2.0.2 they also hold for Mu_{xx} . Finally, from eq. (49), it follows that $Mu_t \in \mathbb{H}_{p,\theta}^\gamma(T)$ for all γ . Thus $u \in \mathfrak{H}_{p,\theta}^2(T)$ and applying Lemma 6.0.3 we finish the proof of Lemma 5.0.9.

References

- [Fol95] Gerald B. Folland, *Introduction to partial differential equations*, second ed., Princeton University Press, Princeton, NJ, 1995. MR 1357411
- [GT01] David Gilbarg and Neil S. Trudinger, *Elliptic partial differential equations of second order*, Classics in Mathematics, Springer-Verlag, Berlin, 2001, Reprint of the 1998 edition. MR 1814364
- [KL99] N. V. Krylov and S. V. Lototsky, *A sobolev space theory of spdes with constant coefficients in a half line*, SIAM Journal on Mathematical Analysis **31** (1999), no. 1, 19–33.
- [Kry94] N. V. Krylov, *A generalization of the Littlewood-Paley inequality and some other results related to stochastic partial differential equations*, Ulam Quart. **2** (1994), no. 4, 16 ff., approx. 11 pp. MR 1317805
- [Kry95] ———, *Introduction to the theory of diffusion processes*, Translations of Mathematical Monographs, vol. 142, American Mathematical Society, Providence, RI, 1995, Translated from the Russian manuscript by Valim Khidkel and Gennady Pasechnik. MR 1311478
- [Kry99a] ———, *An analytic approach to SPDEs*, Stochastic partial differential equations: six perspectives, Math. Surveys Monogr., vol. 64, Amer. Math. Soc., Providence, RI, 1999, pp. 185–242. MR 1661766
- [Kry99b] ———, *Weighted Sobolev spaces and Laplace's equation and the heat equations in a half space*, Comm. Partial Differential Equations **24** (1999), no. 9-10, 1611–1653. MR 1708104
- [Kry99c] Nicolai V. Krylov, *Some properties of weighted Sobolev spaces in $\mathbb{R}_+^d$* , Ann. Scuola Norm. Sup. Pisa Cl. Sci. (4) **28** (1999), no. 4, 675–693. MR 1760536
- [Kry08] N. V. Krylov, *Lectures on elliptic and parabolic equations in Sobolev spaces*, Graduate Studies in Mathematics, vol. 96, American Mathematical Society, Providence, RI, 2008. MR 2435520
- [KZiPSi76] M. A. Krasnoselskii, P. P. Zabreiko, E. I. Pustynnik, and P. E. Sobolevskii, *Integral operators in spaces of summable functions*, Monographs and Textbooks on Mechanics of Solids and Fluids, Mechanics: Analysis, Noordhoff International Publishing, Leiden, 1976, Translated from the Russian by T. Ando. MR 385645

- [LzSU68] O. A. Ladyženskaja, V. A. Solonnikov, and N. N. Uraltceva, *Linear and quasilinear equations of parabolic type*, Translations of Mathematical Monographs, vol. Vol. 23, American Mathematical Society, Providence, RI, 1968, Translated from the Russian by S. Smith. MR 241822
- [Mik65] S. G. Mikhlin, *Multidimensional singular integrals and integral equations*, Pergamon Press, Oxford-New York-Paris, 1965, Translated from the Russian by W. J. A. Whyte. MR 185399
- [Tri83] Hans Triebel, *Theory of function spaces*, Monographs in Mathematics, vol. 78, Birkhäuser Verlag, Basel, 1983. MR 781540
- [Tri92] ———, *Theory of function spaces. II*, Monographs in Mathematics, vol. 84, Birkhäuser Verlag, Basel, 1992. MR 1163193